

Improving the ecological validity of synthetic datasets for deep neural network-based audio processing

Enric Gusó Muñoz

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Directors de la tesi

Dr. Xavier Serra Casals
Department of Engineering
Universitat Pompeu Fabra, Barcelona

Dr. Umut Sayın Saraç
Multimedia Technologies Unit
Eurecat - Centre Tecnològic de Catalunya, Barcelona

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Dr. Xavier Serra

Thesis supervisor

Universitat Pompeu Fabra, Barcelona

Dr. Umut Sayın

Thesis co-supervisor

Eurecat, Centre Tecnològic de Catalunya, Barcelona

Dr. Ishwarya Ananthabhotla

Thesis committee president

Meta Reality Labs, USA

Dr. Oriol Nieto

Thesis committee secretary

Adobe Research, USA

Dr. Frederic Font

Thesis committee member

Universitat Pompeu Fabra, Barcelona

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Abstract

Deep Neural Network (DNN) based audio processing models often fail to generalize from their test sets to real-world conditions, where performance can degrade substantially. This thesis pursues to reduce this gap by improving the ecological validity of the (often synthetic) data. We develop this idea across three audio processing tasks.

- *(i)* For binaural Speech Enhancement (SE) in Hearing Aids (HAs) we propose an evaluation methodology. Instead of relying on test signals under lab conditions, we propose to leverage spatial audio for generating realistic speech-in-noise 3D test scenes, which are played back through a loudspeaker array and recorded with a KU100 dummy head, both wearing and not wearing HAs. The recordings are then used to compare the traditional HAs algorithms with offline DNN-based SE models, showing that the latter demonstrate higher consistency.
- *(ii)* For normal-hearing monaural speech enhancement, we introduce the MB-RIRs dataset which examines three strategies for raising the ecological validity of synthetic Room Impulse Response (RIR) datasets — multiband absorption, source directivity and receiver directivity — and show that frequency-dependent absorption coefficients yield measurable gains when using real RIRs in the evaluation.
- *(iii)* For Music Source Separation (MSS) of live recordings from the audience, where the performance of studio-trained models degrades because of ignoring the venue acoustics, the speaker system response and audience noise, we introduce CrowdioSet (real ambient tracks plus synthetic sing-alongs, generated from zero-shot singing voice conversions) and PaRIR-set (stereo impulse responses captured across 40 real professional concert venues), and show that training on them improves separation under live conditions, both objectively and subjectively.

In each of the three tasks, we obtain performance gains without adding complexity at inference, investing it upstream in building more ecologically valid data instead.

Resum

Els models de processament d'àudio basats en xarxes neuronals profundes (Deep Neural Networks, DNN) sovint no generalitzen bé en condicions reals, on el rendiment es pot degradar substancialment. Aquesta tesi pretén reduir aquesta bretxa millorant la validesa ecològica de les dades (sovint sintètiques), en tres tasques de processament d'àudio diferents:

- *(i)* Per a la millora de la parla (Speech Enhancement, SE) binaural en audiòfons (Hearing Aids, HAs) proposem una metodologia d'avaluació que, enlloc de senyals de test al laboratori, utilitza tècniques d'àudio espacial per generar escenes 3D realistes de parla amb soroll, les quals reproduïm amb un conjunt d'altaveus i re-enregistrem amb un cap de maniquí KU100, tant portant com sense portar audiòfons. Comparant els algorismes tradicionals dels audiòfons amb models prototip basats en DNN, mostrem que aquests últims són més consistents.
- *(ii)* Per a la millora de la parla monoaural per oients normals, introduïm el conjunt de dades MB-RIRs per estudiar tres estratègies que augmentin la validesa ecològica de conjunts sintètics de respostes impulsional de sala (Room Impulse Responses, RIRs) —absorció multibanda, directivitat de la font i directivitat del receptor— i mostrem que emprar coeficients d'absorció dependents de la freqüència proporciona guanys mesurables quan s'avalua amb RIRs reals.
- *(iii)* Per a la separació de fonts musicals (Musical Source Separation, MSS) d'enregistraments en directe des del públic, on els models entrenats amb gravacions d'estudi es degraden al ignorar l'acústica del recinte, la resposta del sistema de d'altaveus i el soroll del públic, introduïm CrowdioSet (pistes d'ambients reals combinats amb cors sintètics, generats a partir de conversions de veu cantada) i PaRIRset (respostes impulsional estereo captades en 40 recintes de concerts professionals reals), i mostrem que entrenar amb aquestes dades millora la separació en aquest cas d'ús.

En cadascuna de les tasques, el nostre enfoc es mostra efectiu i trasllada la complexitat de la inferència a la generació de les dades.

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Glossary

ASR Automatic Speech Recognition. 3, 4, 39, 43, 66

BiMagLS Bilateral Magnitude Least Squares. 33, 52, 55, 56, 63, 72, 106, 121

BTE Behind-The-Ear. 45, 52

CNN Convolutional Neural Network. xix, xx, 15, 16, 20, 21

DNN Deep Neural Network. xi, xiii, xxi, xxii, xxv, 4, 5, 11, 12, 13, 15, 25, 33, 39, 46, 47, 48, 51, 53, 54, 55, 56, 58, 59, 61, 62, 63, 65, 105, 106, 108

DNS Deep Noise Suppression. xxvi, 65, 66, 67, 68, 69, 72, 75, 76, 78, 79, 80

FOA First Order Ambisonics. xxi, 29, 30, 32, 39, 40

FOH Front Of House. xxiii, 91, 92, 93

GAN Generative Adversarial Network. 23, 24

GMACs Giga Multiply-Add Operations. xix, 5

GPU Graphics Processing Unit. 9, 20, 21, 75, 96

GRU Gated Recurrent Unit. xx, 19, 20

HA Hearing Aid. xi, xiii, xxi, xxii, xxv, 2, 3, 4, 10, 11, 13, 26, 34, 41, 45, 46, 47, 48, 50, 51, 52, 53, 55, 56, 58, 59, 60, 61, 62, 63, 65, 86, 105, 106, 108, 117, 121

HASPI Hearing-Aid Speech Perception Index. 54, 59, 60, 108

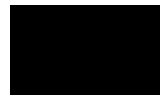
HASQI Hearing-Aid Speech Quality Index. 54, 59, 60, 108

HOA Higher Order Ambisonics. xxi, 29, 30, 32, 33

HRIR Head-Related Impulse Response. 32, 33, 52

- HRTF** Head-Related Transfer Function. xxii, 27, 32, 33, 34, 52, 53, 58, 63, 67, 72, 106
- ICASSP** International Conference on Acoustics, Speech and Signal Processing. 110
- ILD** Interaural Level Difference. 26, 29
- ISM** Image Source Method. xxi, xxii, xxvi, 36, 37, 38, 39, 41, 43, 49, 65, 66, 67, 68, 69, 70, 72, 73, 75, 81, 107, 109, 110, 113, 117
- ITD** Interaural Time Difference. 26, 29, 33
- KU100** Neumann Dummy Head. xxi, 47, 49, 50, 51, 52, 58, 63, 73, 121
- LFE** Low Frequency Extension. 33, 52
- LSTM** Long Short-Term Memory. xx, 19, 20
- MagLS** Magnitude Least Squares. 33, 121
- MASP** Multichannel Acoustic Signal Processing. xxii, 49, 52, 56, 57, 69, 72, 117
- MBSTOI** Modified Binaural Short-Time Objective Intelligibility. 54, 59, 60, 62
- MIR** Music Information Retrieval. 1, 2, 3, 84, 105
- MLS** Multilingual LibriSpeech Studio. 55, 56
- MS** Mid-Side. 29, 31
- MSS** Music Source Separation. xi, xiii, xxiii, 2, 11, 13, 34, 83, 84, 85, 86, 95, 96, 100, 102, 107, 110
- MTG** Music Technology Group. 83, 107
- MUSHRA** Multiple Stimuli with Hidden Reference and Anchor. xxii, 62, 75, 78
- PA** Public Address. xxiii, xxvi, 83, 84, 85, 90, 91, 92, 102, 111, 112, 123, 125
- PESQ** Perceptual Evaluation of Speech Quality. 76, 77

- RIC** Receiver-In-Cannal. xxi, xxv, 45, 46, 50, 58, 61, 63, 121
- RIR** Room Impulse Response. xi, xiii, xxii, xxiii, xxv, xxvi, 3, 11, 12, 13, 25, 34, 35, 36, 37, 39, 41, 43, 65, 66, 67, 68, 69, 70, 72, 73, 75, 76, 77, 78, 79, 80, 81, 84, 85, 86, 90, 91, 92, 93, 94, 95, 97, 102, 106, 107, 108, 109, 110, 112, 113, 118, 121
- RNN** Recurrent Neural Network. xx, 19, 20
- SDR** Signal-to-Distortion Ratio. xxiii, xxvi, 76, 94, 96, 97, 98, 99, 100, 101
- SE** Speech Enhancement. xi, xiii, xxv, 1, 2, 3, 6, 10, 11, 13, 26, 34, 39, 41, 43, 46, 47, 55, 56, 58, 63, 65, 66, 67, 73, 75, 76, 78, 80, 81, 90, 94, 102, 105, 106, 107, 108, 118, 119
- SH** Spherical Harmonics. 27, 29, 32, 33
- SI-SDR** Scale-Invariant Signal-to-Distortion Ratio. xix, 5, 6, 54, 55, 59, 60, 62, 76, 77, 78
- SNR** Signal to Noise Ratio. xxv, 35, 46, 53, 57, 58
- SOFA** Spatially Oriented Format for Acoustics. 52, 117, 121
- SOTA** State-of-the-Art. xix, 5, 6, 11, 65, 67, 105, 108
- SPL** Sound Pressure Level. 50, 54
- STOI** Short-Time Objective Intelligibility. 76, 77
- VBAP** Vector-Based Amplitude Panning. xxi, 28, 31, 32, 33, 106



Introduction

Audio processing is the subfield of signal processing concerned with the analysis, modification, and generation of sound signals. Historically, its applications have grown around several research communities, some of which this thesis touches upon: the *speech* field, devoted to understanding, transcribing and enhancing spoken language; the *machine listening* field, which studies how to extract information about the acoustic environment; the *computational audiology* field, which studies how to compensate for hearing loss and improve listening in adverse conditions; the *music* field, where the retrieval and manipulation of musical content is studied under the umbrella of Music Information Retrieval (MIR); and the *acoustic signal processing* field, which encompasses the full range of audio tasks, among others. The work in this thesis sits at the intersection of these communities, several of whose most representative tasks are listed in Table 1.1.

Until the last decade, these communities have relied on analytical, hand-engineered solutions grounded in acoustics, psychoacoustics and statistical signal processing. However, the advent of deep learning (LeCun et al., 2015) has reshaped all of them, shifting the dominant paradigm from explicit modelling to data-driven learning, bringing unprecedented performance, especially in tasks where analytical approaches are limited. This has led to significant advances in tasks such as retrieving specific elements from an audio mixture (source separation) and removing noise (denoising) or reverberation (dereverberation) to improve quality and/or intelligibility of signals, and has also given rise to new tasks and interfaces previously unfeasible, such as text-to-audio.

These tasks (listed in Table 1.1) can be grouped into three paradigms: those that *categorize* audio content—such as tagging or key detection; those that *infer* a structured output from the signal—such as source separation or SE;

and those that *generate* new audio—such as speech synthesis or text-to-music. These paradigms correspond, respectively, to the machine learning problems of *classification*, *regression*, and more recently to the *sampling* of generative models. Although the contributions of this thesis may prove useful across all three paradigms, it particularly focuses on a subset of the regression-oriented family of tasks, which we refer to as *audio processing* tasks, in the sense that both the model’s inputs and outputs are audio signals and to differentiate them from *audio analysis* and *audio generation* models.

In the present work, we focus on three of these representative tasks, the choice of which is also motivated by the industrial partners involved in the thesis, and each one from a distinct audio research field: *(i) Speech Enhancement (SE)* from the speech field, *(ii) Music Source Separation (MSS)* from the Music Information Retrieval (MIR) field, and *(iii) Hearing Aids (HAs)* processing from the computational audiology field.

1.1 Context

To situate the contributions of this thesis within the context of audio processing, we first trace the transition from analytical to data-driven methods, which has unfolded along two axes. The first concerns the signal representations. Models that used to operate on hand-crafted descriptors built from expert domain knowledge have progressively given way to architectures that learn their own representations directly from time–frequency transforms or even from the raw waveform (Pons Puig, 2019). These learned representations have themselves grown more abstract: continuous, high-dimensional feature maps have transitioned to discrete latent codes, learned end-to-end and quantised into a finite vocabulary. Neural audio codecs epitomise this shift, compressing the waveform into a handful of discrete tokens that preserve perceptual fidelity (Zeghidour et al., 2021; Défossez et al., 2023) while exposing a tokenised substrate over which language-model architectures can operate directly (Borsos et al., 2023). Continuous latent spaces have remained equally productive, most notably the encoder-decoder structure of the Variational Auto Encoder (VAE) that underpins latent-diffusion generative models such as Evans et al. (2025), where a compressed continuous latent serves as the domain over which the generative process is defined. However, these models work at a latent rate in the order of ~ 10 to 20 Hz, so the compactness of the representations intrinsically makes them hard to deploy in real time due to their latency constraints (with some exceptions such as Caillon et al., 2025; Novack et al., 2025).

Field	Venues	Representative tasks
Acoustics	ICASSP WASPAA EUSIPCO	DoA estimation Beamforming Echo cancellation RIR estimation
Speech	Interspeech ASRU ICASSP WASPAA	ASR Speaker verification SE Text-to-Speech Voice conversion
Machine listening	DCASE ICASSP WASPAA	Acoustic scene classification Sound event detection Audio moment retrieval Sound localization
Computational audiology	VCCA AES EMBC ICASSP WASPAA	HAs processing Cochlear implant processing Noise reduction Speech intelligibility prediction Hearing loss simulation
MIR	ISMIR SMC ICASSP WASPAA	Beat tracking Chord recognition Source separation Transcription Text-to-song

Table 1.1: Some representative audio processing tasks for each research community and the main venues where they are traditionally presented. This selection is merely illustrative and intended to contextualize the thesis within the broader audio processing landscape; it is by no means exhaustive, and the tasks are not mutually exclusive — for example, source separation is relevant to both the acoustics and music fields. A more comprehensive taxonomy can be found in Huang et al. (2025).

The second axis concerns another recent inflection, visible across machine learning as a whole, which is the move from purely supervised training towards large-scale self- and semi-supervised *pre-training*. Instead of training a model from scratch for a single task, the encoder-decoder model is first trained on vast amounts of unlabelled or weakly labelled data to learn general-purpose representations by contrastive or masking approaches, and only then is adapted to specific downstream tasks, usually by adding task-specific layers. The resulting models, often termed *foundation models*, have delivered remarkable performance in Natural Language Processing (NLP) (Devlin et al., 2019; Touvron et al., 2023) and are increasingly replicated in the audio and speech domains (Baevski et al., 2020; Mohamed et al., 2022), but they are usually

bound to large parameter counts and inference budgets that also make them difficult to deploy in real time and on consumer-grade hardware, and that are particularly problematic for wearable applications such as Hearing Aids (HAs). Likewise, the trajectory from analytical methods to learned, increasingly end-to-end models and with weaker supervision has been observed across the three fields we address.

*(i) In the **speech** field*, classical Automatic Speech Recognition (ASR) used Mel-Frequency Cepstral Coefficients (Davis & Mermelstein, 1980) feeding statistical acoustic models (Rabiner, 1989), an approach later overtaken by DNN acoustic modelling (Hinton et al., 2012) and, more recently, by large end-to-end models such as Whisper, trained on hundreds of thousands of hours of weakly supervised audio (Radford et al., 2023).

*(ii) In the **music** field*, source separation was long dominated by analytical decompositions such as non-negative matrix factorization (Virtanen, 2007), before being superseded by DNN-based systems exemplified by open, reproducible baselines such as Open-Unmix (Stöter et al., 2019).

*(iii) In the **computational audiology** field*, traditional methods of HAs improve the signal-to-noise ratio through directional microphones and adaptive beamforming (Ricketts & Dhar, 1999; Gode & Doclo, 2022), techniques rooted in array signal processing and typically validated under controlled laboratory conditions. More recently, Deep Neural Networks (DNNs) have been proposed as enhancement back-ends for HAs for binaural separation (Han et al., 2020), or as evaluation metrics (Chiang et al., 2024), with recent interest in distillation to smaller models that are compatible with wearable constraints (Wang et al., 2023b; Andersen et al., 2021).

Both axes and examples point in the same direction: towards models that are more powerful and capable of tasks unthinkable a decade ago, but at a steep computational cost. The representations are richer and the pre-trained backbones larger, yet this capability is increasingly bought with parameter counts and inference budgets that place these models out of reach for local or consumer-grade hardware, and that make real-time deployment difficult. This thesis explores an alternative direction. Rather than chasing capability through ever-larger models trained on ever-more data, can we displace part of the complexity onto the selection and construction of the training data, while keeping model size fixed? Put differently: **can *better* data — more representative of the conditions a model will actually face — improve performance without resorting to more data and larger models?**

1.2 Motivation: the importance of ecological validity

In this section, we lay out two examples that have motivated the data-centric direction this thesis pursues. We have mentioned that DNN-based audio processing performs extremely well on many tasks. This performance is traditionally measured on benchmark datasets the community has standardized around, and progress on these benchmarks is pursued so relentlessly that successive systems have *saturated* them. By saturation we mean that performance gains have largely plateaued, so that further improvements demand disproportionately more compute and parameters for diminishing returns. We illustrate this for the two-speaker speech separation task and the progress of the State-of-the-Art (SOTA) across the last decade, which we depict in Figure 1.1.

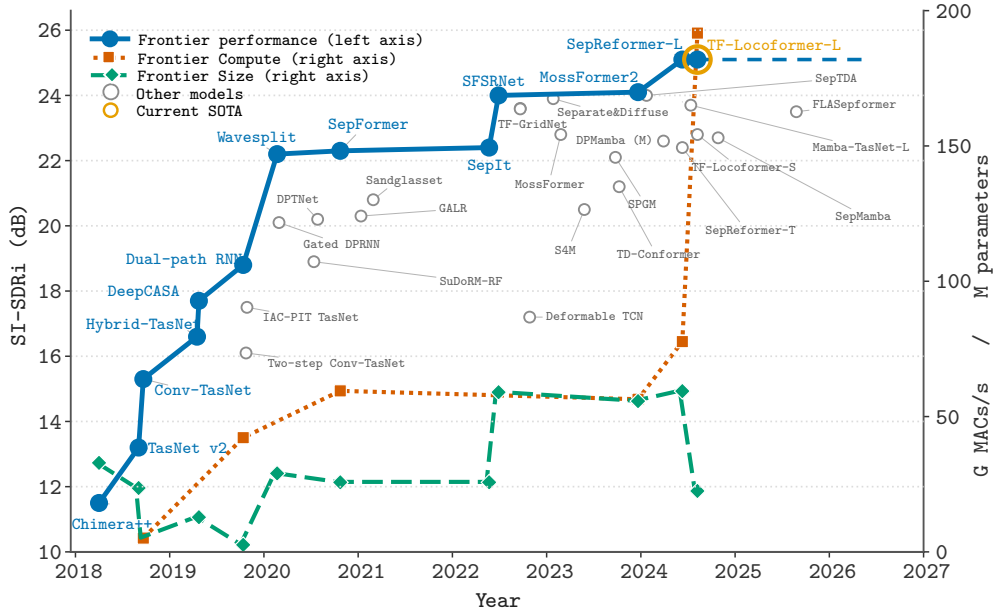


Figure 1.1: Progress on clean two-speaker speech separation over the last decade, evaluated on the WSJ0-2mix test set. Each marker is a published model: grey markers are individual systems, and blue markers trace the *frontier* of performance — the SOTA SI-SDRi at each point in time (left y-axis), with the current SOTA highlighted. The orange and green lines (right y-axis) report the computational cost and model size of the frontier models for which such data was available, expressed in GMACs/s and millions of parameters, respectively. While performance has plateaued around 22 dB, compute and size have continued to grow, illustrating the saturation discussed in the text. Computational figures are taken from Wang et al. (2025).

Speech separation performance is commonly measured by the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) proposed by Le Roux et al. (2019), which quantifies the similarity between the separation estimate and a reference ground truth source signal, usually reported as improvement upon the mixture (denoted with the i in SI-SDR i). Traditionally, the speech community has used the WSJ0-2mix dataset (Hershey et al., 2016) to assess progress on clean speech separation and to compare with previous existing literature. SOTA performance over time shows that progress is increasingly hard, and although the saturation seemed more pronounced at the beginning of this thesis (around 2022, where performance progress had largely stalled), the performance leaps that have been made in Rixen & Renz (2022) and Shin et al. (2024) grow smaller every time. Note also how the computational cost, here measured in terms of the number of matrix multiplications per second, is also increasing. As a rule of thumb, an SI-SDR i >20 dB is often taken for near-perfect separation, which was already achieved in 2022.

However, this apparent progress can be misleading: when the same models are evaluated across datasets (cross-dataset evaluation), performance often degrades sharply, revealing that they have, to a large extent, overfit to the idiosyncrasies of a particular dataset rather than generalised to unseen conditions. Continuing with the two-speaker clean speech separation example, Kadioğlu et al. (2020) proposed cross-dataset evaluation and showed that a Conv-TasNet trained on WSJ0-2mix degrades sharply on LibriTTS and vice-versa, as we depict in Figure 1.2. A similar pattern holds for tasks other than separation: SE models trained on one dataset degrade on others, in their case due to channel and recording-condition mismatch (Pandey & Wang, 2020).

We frame this gap through the notion of *ecological validity*. The term was originally introduced in psychology by Egon Brunswik (Brunswik, 1955) with a narrow, technical meaning: within his lens model of perception, it denoted the statistical reliability of a perceptual cue. In its contemporary and considerably broader usage, however, the term has come to mean something Brunswik did not intend: the extent to which findings obtained under controlled laboratory conditions generalize to real-life settings (Leglaive et al., 2023). We adopt this modern sense throughout this thesis.

Transposed to supervised audio processing, ecological validity concerns the degree to which the training and evaluation data reproduce the acoustic conditions of the intended deployment scenario: realistic noise and reverberation, source and receiver directivity, room and venue acoustics, transducer responses,

1.2. MOTIVATION: THE IMPORTANCE OF ECOLOGICAL VALIDITY 7

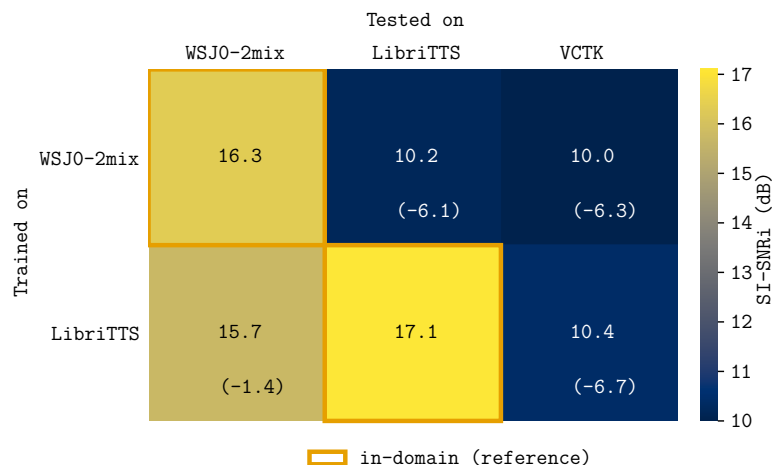


Figure 1.2: Cross-dataset generalization of Conv-TasNet on two-speaker speech separation, reported in SI-SNR_i (dB) in the original study. Each row corresponds to the training dataset and each column to the test set; diagonal cells, outlined in orange, are in-domain (matched train/test) references. Off-diagonal cells report out-of-domain performance, with the drop relative to the corresponding in-domain reference shown in parentheses. Models trained on one corpus lose roughly 6 dB when evaluated on a different one (e.g. -6.1 and -6.3 dB for the WSJ0-2mix model on LibriTTS and VCTK), illustrating that high in-domain scores do not transfer across datasets. Values taken from Kadioğlu et al. (2020).

and so on. From this perspective, saturating a benchmark is a perfectly legitimate measure of progress, but it must be complemented by evaluations that are themselves ecologically valid, so that gains translate into the real world.

A complementary, more quantitative view of the same gap comes from the machine learning literature, where it is studied under the notion of *domain mismatch* (or *distribution shift*). Supervised models are trained under the implicit assumption that training and real data are drawn from the same underlying distribution; when this assumption is violated performance degrades, often severely (Quiñonero-Candela et al., 2008). In audio processing, this mismatch is rarely a single factor but the compound effect of many: the recording channel and microphone response, room acoustics and reverberation, the noise types and signal-to-noise ratios present, speaker or instrument characteristics, and the way mixtures are constructed in synthetic benchmarks. The cross-dataset degradation reported above (Figure 1.2 and the channel-mismatch effects of Pandey & Wang, 2020) are concrete examples of this shift. Reducing domain mismatch and improving ecological validity are thus two framings of a single objective, and the one this thesis pursues: aligning the data a model is trained and evaluated on with the conditions of its intended use.

This intuition can be made precise. Let $\mathcal{X}$ denote the space of input signals and $\mathcal{Y}$ the space of targets (separated sources/instruments, enhanced speech, etc.), and let a model $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ be trained to minimize the expected loss $\mathcal{L}$ over a *source* domain $p_S(x, y)$:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(x,y) \sim p_S} [\mathcal{L}(f_\theta(x), y)]. \quad (1.1)$$

At deployment, however, the model is evaluated on a *target* distribution $p_T(x, y)$, and the quantity we actually care about is the target risk $\mathbb{E}_{(x,y) \sim p_T} [\mathcal{L}(f_{\theta^*}(x), y)]$. Empirical risk minimization offers guarantees only when $p_S = p_T$; *domain mismatch* is precisely the condition

$$p_S(x, y) \neq p_T(x, y), \quad (1.2)$$

under which a model optimal on the source domain need not be optimal—or even adequate—on the target domain.

We can factorize the joint distribution as $p(x, y) = p(x)p(y | x)$. In the audio processing tasks we consider, the conditional input–output relationship is largely physical and stable across domains (e.g. the mapping from a noisy mixture to its clean elements is governed by the same acoustics everywhere), so $p_S(y | x) \approx p_T(y | x)$, and the dominant discrepancy lies in the marginal over inputs, $p_S(x) \neq p_T(x)$. Each benchmark fixes a particular $p(x)$ —a specific distribution of channels, rooms, noises and speakers—and the cross-dataset degradation reported above is the gap between performance under $p_S(x)$ and under a different $p_T(x)$. The contributions of this thesis can be read as efforts to bring the training input distribution $p_S(x)$ closer to the deployment distribution $p_T(x)$ through the selection and construction of training data. Several broad strategies exist to mitigate this mismatch.

A first family acts on the model: *domain adaptation* techniques adjust a trained model towards the target domain, often using small amounts of unlabelled target data (Hsu et al., 2017). A second, radically different response to domain mismatch is simply to enlarge the training distribution until it subsumes the deployment one. Rather than adapting to a specific target, one trains a single large model on a corpus so vast and heterogeneous that few conditions remain genuinely out-of-distribution; the target $p_T(x)$ then falls within the support of an enormous $p_S(x)$. This is the logic behind large-scale weak supervision in speech: Radford et al. (2023), for instance, train on hundreds of thousands of hours of diverse web audio and obtain strong *zero-shot* robustness across datasets, attributing the gain to data diversity rather than to any adaptation

mechanism. From the perspective of the previous paragraphs, this is a brute-force way of minimising the gap between $p_S(x)$ and $p_T(x)$ —not by carefully shaping $p_S(x)$, but by making it large enough to cover everything.

This strategy is remarkably effective, but it is precisely the one that the constraints of this thesis rule out; the parameter counts and inference budgets inherent to web-scale training are incompatible with the real-time, on-device deployment targets that motivate our work. When scaling the model and the data is not an option, the quality and representativeness of a fixed-size training set become the primary lever — which returns us to ecological validity. This constraint also shapes the very *choice* of problems studied in this thesis. Every experiment reported here was run on a single Graphics Processing Unit (GPU), so we have deliberately favoured tasks where progress comes from improving the *data* rather than from scaling the model: the design of ecologically valid evaluations and the construction of better training datasets, where a small group can still make a measurable difference.

1.3 Industrial PhD

The choice of the specific tasks in which we investigate towards improving ecological validity has been motivated by the partners involved in this PhD. This dissertation has been carried out as an *industrial PhD*, a doctoral modality designed to bridge academia and industry by embedding the doctoral candidate within a company or organisation while maintaining a formal partnership with a university or research centre. Unlike a traditional academic PhD — which primarily advances knowledge within a scientific discipline — an industrial PhD aligns research objectives with the strategic needs of the industrial partner, targeting outcomes such as technological innovation, process improvement, or competitive advantage. The candidate operates across both environments simultaneously, contributing to the partner’s goals while producing original research that culminates in a doctoral thesis.

From the academic side, the Music Technology Group (MTG) at the Universitat Pompeu Fabra (UPF) in Barcelona is one of the leading research groups in sound and music computing worldwide. Founded in 1994 within UPF’s Department of Information and Communication Technologies, and now part of the Department of Engineering, the MTG conducts research spanning signal processing, machine learning, music perception, computational musicology, and human-computer interaction, with applications ranging from music creation

and performance to education and cultural heritage. A defining characteristic of the group is its commitment to open science: flagship initiatives such as Freesound, a collaborative repository of Creative Commons audio samples, and Essentia, an open-source library for audio analysis and music description, have become widely adopted tools in both the research community and industry. The MTG has grown into an interdisciplinary team that combines scientific rigour with artistic and societal relevance, and maintains active collaborations with musicians, educators, and industry partners.

From the industrial side, Eurecat is a foundation whose mission is to promote research, technological development, and knowledge transfer in order to improve the competitiveness of companies and the well-being of society. It was created in 2015 through the consolidation of several industrial research units from across Catalonia. The audio research group at Eurecat traces its origins to Barcelona Media, a foundation for innovation and interdisciplinary research that operated from 2002 and developed solutions across a broad range of technological domains, including image, audio, and social media analysis. Among its most notable achievements, the Barcelona Media audio team developed the immersive 3D audio technology behind Imm Sound, a cinema sound solution acquired by Dolby Laboratories in 2012. The researchers who remained at Barcelona Media after that acquisition formed the core of what would become Eurecat’s audio group when the centre was established in 2015.

Three Eurecat projects underpin the work presented in this thesis.

- **Eurecat-Microson** collaboration project, focusing on the development of a new benchmark for evaluating HAs and speech denoising algorithms in realistic conditions.
- **GuestXR** (Horizon 2020) project, aimed at creating an immersive on-line social space through extended reality, centred on a machine learning agent called “The Guest” designed to facilitate interaction between participants, mitigate conflicts and cyberbullying. Because supporting participants with and without hearing difficulties requires intelligible speech, our contribution addresses SE.
- **Traça IAGen**, an internal Eurecat initiative funded by the Catalan Government with the goal of driving the adoption of generative AI across the centre’s different research units.

1.4 Research objectives and thesis outline

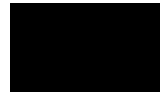
In light of the above, this thesis investigates whether improving the ecological validity of (often synthetic) training and evaluation data is an effective lever for closing the generalization gap, across three audio processing tasks — one per partner and field. The remainder of the manuscript tackles the following Research Objectives (RO) and is organized as follows.

- **Chapter 2** introduces the scientific background used broadly throughout the thesis: deep-learning for audio, spatial-audio techniques and virtual-acoustics on which the contributions build, together with a summary of their corresponding publicly available datasets.
- **Chapter 3** addresses the HAs field and is the result of the collaboration with Microson and within the GuestXR project: the development of an ecologically-valid *evaluation* for HAs (**RO1**), and whether DNNs are a promising route to generalise better than traditional HAs algorithms (**RO2**). Beyond their technical interest, both objectives foster accessibility, and thus inclusivity, for people with hearing loss.
- **Chapter 4** carries the idea to more general models, again part of the GuestXR project. It investigates whether Room Impulse Response (RIR)s generated with frequency-dependent absorption coefficients and source or receiver directivity improve the generalisation of monaural Speech Enhancement (SE) models to real data (**RO3**), and, in pursuit of the same ecological validity, whether using fewer but more realistic mesh-based RIRs improves performance (**RO4**). While large, realistic RIR datasets already exist privately within industry — where major labs maintain their own simulation tools — replicating them for open science and releasing them freely to every research lab carries value in itself.
- **Chapter 5** brings the same idea to Music Source Separation (MSS), in line with the research interests of the MTG (UPF). We identify that State-of-the-Art (SOTA) models designed for studio music do not work well on recordings captured by attendees (mobile-phone recordings from the audience). This raises several questions: can we generate synthetic data that transforms studio datasets into live datasets? Can a generative singing-voice conversion model produce synthetic audiences that sing along to the vocals of the studio songs (**RO5**)? Can we curate additional audience-like noise from internet audio archives (**RO6**)? And what

about acoustics — does adding impulse responses specifically measured in real, modern music venues improve upon using existing non-concert RIRs (**RO7**)?

- **Chapter 6** closes the dissertation with some additional discussion, general conclusions and perspectives for future work.
- We provide a list of publications (Appendix B) and research materials (Appendix C).

Having grounded the concepts of DNN-based *audio processing* and *ecological validity*, the title of this thesis can now be read in full: our aim is to improve the ecological validity of datasets as a lightweight route to better-generalizing, deployable audio processing models—one per project of this Industrial PhD.



Scientific Background

The contributions of this thesis draw on concepts and tools from different fields, and understanding them requires a working familiarity with a number of recurring building blocks: the DNNs that drive modern audio processing, the spatial-audio techniques used to reproduce sound fields, the virtual acoustics that let us simulate reverberation, and the state of the art of the three tasks we address — SE, HAs processing, and MSS. This chapter introduces that background in the order in which the remainder of the manuscript relies on it. We begin with an overview of DNN-based audio processing (Section 2.1). We then turn to spatial audio and Ambisonics (Section 2.2), followed by the measurement and simulation of RIRs (Section 2.3).

Note that in the chapters that follow we also review additional scientific background specific to each of the three tasks: Hearing Aids (HAs) in Chapter 3, Speech Enhancement (SE) in Chapter 4, and Music Source Separation (MSS) in Section 5.1.

2.1 Deep learning for audio

This section is intended as a non-exhaustive, deliberately synthetic recap of the DNN concepts needed to follow the rest of the thesis, rather than a thorough survey. It is written primarily for the neophyte, or for the audio researcher whose expertise lies in a neighbouring field^{2.1}.

^{2.1}This section grew out of a series of internal seminars held within the Multimedia Unit at Eurecat. Accordingly, the presentation prioritises intuition over rigour; the interested reader is referred to standard texts for a complete treatment.

2.1.1 The advent of DNNs (2014–2024)

The Perceptron — The basic unit of a neural network is the *perceptron* (Rosenblatt, 1958), depicted in Figure 2.1. Given an input $x \in \mathcal{X}$ with components $x_1, \dots, x_n$, the perceptron forms a weighted sum of its inputs, adds a bias, and passes the result through a non-linear *activation* function f to produce an output o :

$$o = f_{\theta}(x) = f\left(\sum_{k=1}^n w_k x_k + b\right), \quad \theta = \{w_1, \dots, w_n, b\}. \quad (2.1)$$

The parameters θ — the weights w_k and the bias b — are exactly the θ of Equation 1.1: the perceptron is the simplest instance of the parametric map $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ we wish to fit.

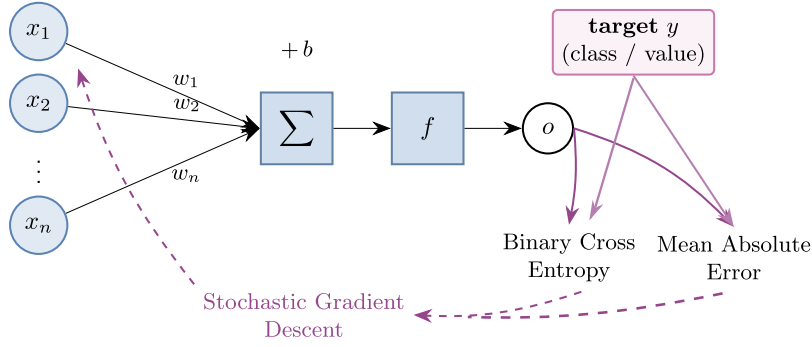


Figure 2.1: The Perceptron, the basic unit of a neural network.

Training begins from *randomly initialised* weights and proceeds by making the output o progressively closer to a target y by a *loss* $\mathcal{L}(o, y)$ whose form depends on the task. For a *classification* task the target y is a class label and a common choice is the binary cross-entropy, $\mathcal{L}_{\text{BCE}}(o, y) = -[y \log o + (1 - y) \log(1 - o)]$, whereas for a *regression* task — the family this thesis focuses on, where y is itself a signal — the loss measures a numerical distance, such as the mean absolute error $\mathcal{L}_{\text{MAE}}(o, y) = |o - y|$.

The weights are then adjusted to reduce this loss by *back-propagation*, which computes the gradient of the loss with respect to every parameter, followed by Stochastic Gradient Descent (SGD), which takes a small step against that gradient: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(o, y)$, where η is the learning rate. Because evaluating the expectation in Equation 1.1 over the whole dataset at once is impractical, in practice SGD approximates it on small random subsets of the data, the *mini-batches*: each iteration of SGD is computed on a fresh mini-batch, and

the loop is repeated until the loss converges. This cycle — forward pass, loss, back-propagation, parameter update — is the engine behind every model in this thesis. Crucially, all of the above presupposes that for every input x we have access to its corresponding target y : it is the pairing (x, y) that makes the loss in Equation 1.1 computable. This is known as *supervised learning*.

The multi-layer perceptron (Rumelhart et al., 1986) stacks many perceptron units into layers and composing them yields a DNN, but the learning principle is already visible in a single unit (the perceptron).

The Convolutional Neural Network (CNN) (LeCun et al., 1998) is a pivotal refinement: instead of learning an independent weight for every input–unit connection, we learn a set of *filters* (or *kernels*) that are slid across the input, as illustrated in Figure 2.2. Each is applied at every spatial location, so the same weights are reused throughout the input — a form of weight sharing that drastically reduces the number of parameters and builds in translation equivariance. CNNs were first popularised for handwritten-digit recognition and, more than a decade later, became the workhorse of computer vision.

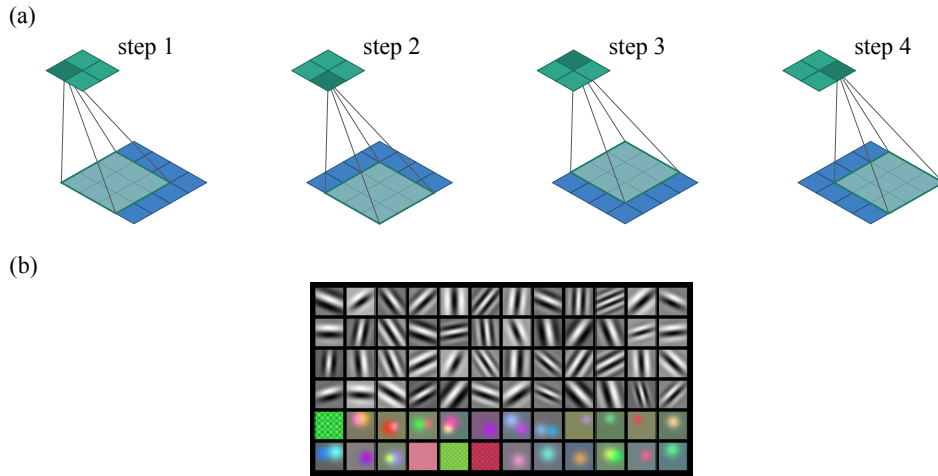


Figure 2.2: Convolutional filters in a CNN. **(a)** A 3×3 filter (kernel) slid across a 4×4 input image or feature map (blue) to produce a 2×2 output feature map (green); at each of the four positions the filter covers a local region of the input and produces a single output value. **(b)** Illustrative examples of filters learned: the grayscale filters (top) resemble oriented Gabor functions acting as edge detectors, while the colour filters (bottom) capture colour or other patterns.

In audio, the two-dimensional input is not an image but a spatio-temporal representation of the signal, most commonly the Short-Time Fourier Transform. Early systems operated on the magnitude alone, discarding the phase,

although recent models process the complex spectrogram (Tong et al., 2024; Schröter et al., 2023) — real and imaginary parts jointly.

Sliding a filter over the input yields a smaller *feature map*, one per filter; stacking many filters per layer and many layers produces a hierarchy of feature maps of increasing abstraction. While the filters learned in the first layers of an image CNN resemble Gabor functions tuned to particularly oriented edge detectors (Figure 2.2); in music, the analogous filters tend to specialise instead on timbre or rhythm. Deeper layers compose these primitives into progressively complex detectors for more concrete concepts — particular colours or shapes in vision, or higher-level acoustic patterns in audio.

Formally, writing $\mathbf{H}^{(0)} = x$ for the input and $*$ for the convolution, the k -th feature map of layer l is

$$\mathbf{H}_k^{(l)} = f \left(\sum_{c=1}^{C_{l-1}} \mathbf{W}_{k,c}^{(l)} * \mathbf{H}_c^{(l-1)} + b_k^{(l)} \right), \quad (2.2)$$

where the learned filters $\mathbf{W}_{k,c}^{(l)}$ and biases $b_k^{(l)}$ now play the role of the parameters θ of Equation 1.1, and f is the (non-linear) activation, typically a rectified linear unit. After L such layers, interleaved with pooling operations that progressively reduce the spatial resolution, the final feature maps are flattened into a single vector $z = \text{vec}(\mathbf{H}^{(L)}) \in \mathbb{R}^d$ known as the *embedding*: a compact, learned representation of the input that concentrates the information relevant to the task. The output is then a simple — often linear — read-out of this embedding, $o = f_\theta(x) = g(\mathbf{W}_o z + \mathbf{b}_o)$ with g a softmax over classes for a classification task, sometimes called Fully-Connected (FC) as in Figure 2.3.

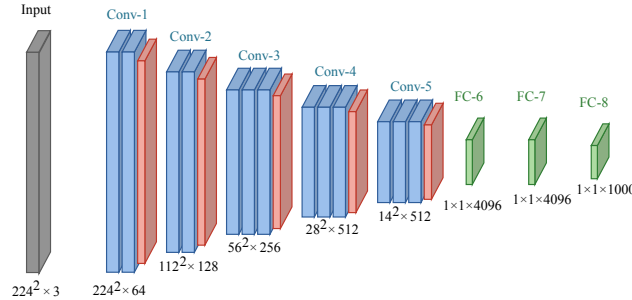


Figure 2.3: The VGG-16 architecture (Simonyan & Zisserman, 2015), a canonical CNN. Stacks of 3×3 convolutions (blue) followed by max-pooling (red) progressively reduce the spatial resolution while increasing the number of feature maps, until a small set of fully connected layers (green) maps the final embedding to the 1000 output classes.

The embedding is a central object in modern deep learning: because it distills the task-relevant content of the input into a fixed-size vector, the same embedding can often be reused for tasks other than the one it was trained on — an idea we return to when discussing pre-trained representations. A canonical example is the VGG network (Simonyan & Zisserman, 2015), sketched in Figure 2.3, a deep stack of small 3×3 convolutions and max-pooling layers that successively halve the spatial resolution while increasing the number of feature maps, followed by a few fully connected layers that map the final embedding to the output classes.

Beyond a certain depth, adding more layers makes the network harder to optimise. ResNets (He et al., 2016) resolved this with a simple architectural change. Rather than asking a stack of layers to fit a target mapping $\mathcal{H}(x)$ directly, a ResNet lets it fit only the *residual* $\mathcal{F}(x) = \mathcal{H}(x) - x$ and adds the input back through an identity *skip* connection, so that the block computes $\mathcal{F}(x) + x$ (Figure 2.4). The intuition is that if the optimal mapping is close to the identity — as it often is for the incremental refinements performed by successive deep layers — it is far easier to drive $\mathcal{F}(x)$ towards zero than to fit the identity with a stack of non-linear layers.

Visualisations in Li et al. (2018) show that skip connections dramatically smooth the $\mathcal{L}$ loss landscape: the rugged, highly non-convex surface of a deep plain network gives way to a far smoother, near-convex one that gradient descent can navigate much more easily (Figure 2.4).

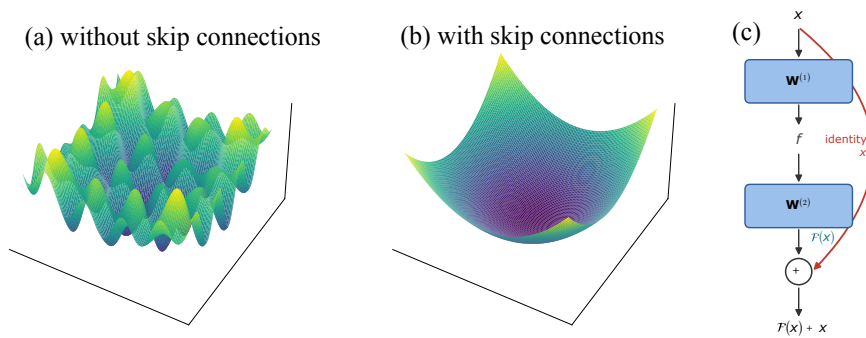


Figure 2.4: Residual networks. Effect of skip connections on the loss landscape (illustrative): (a) a deep plain network exhibits a rugged highly non-convex $\mathcal{L}$ surface with many poor local minima, whereas (b) the same network with skip connections has a near-convex landscape much easier to optimise, following the visualisation of Li et al. (2018). (c) A ResNet block: the two weight layers $\mathbf{W}^{(1)}$ and $\mathbf{W}^{(2)}$, with activation f , map the input through the feature maps $\mathbf{H}_1$ and $\mathbf{H}_2 = \mathcal{F}(x)$ (the residual), and an identity skip connection (red) adds the input back, so the block outputs $\mathcal{F}(x) + x$.

Combining the two preceding ideas yields one of the most widely used architectures in audio processing. Taking a VGG-like stack of convolutions and pooling as a *contracting* path that compresses the input into a low-resolution, high-channel representation, and mirroring it with a symmetric *expanding* path that upsamples back to the original resolution, gives an *encoder–decoder*. Adding skip connections in the spirit of ResNet — here linking each encoder layer to the decoder layer of matching resolution, so that fine-grained detail can bypass the bottleneck and reach the output directly — gives the U-Net (Ronneberger et al., 2015). Originally proposed for image segmentation, it has become the backbone of countless source-separation and speech-enhancement models.

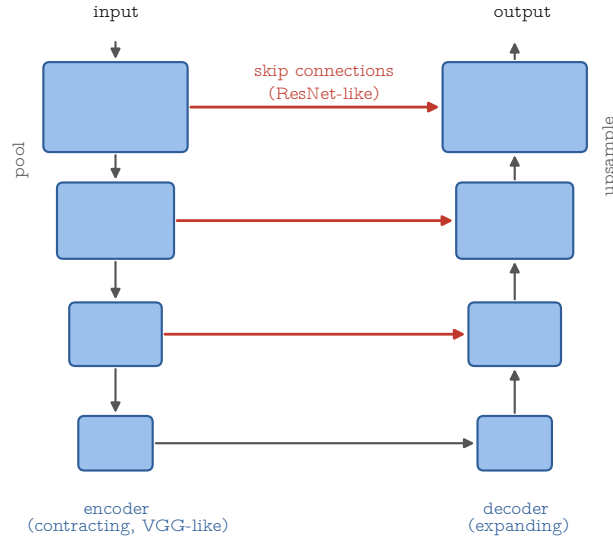


Figure 2.5: The U-Net as a combination of the two preceding models: a VGG-like encoder–decoder (blue) with ResNet-style skip connections (red). Details are omitted for clarity.

When an encoder–decoder is trained to reconstruct its own input rather than to predict a different target, it is called an *autoencoder*: the encoder compresses the input into the low-dimensional *embedding* of Section 2.1.1 — here acting as a *bottleneck* or *latent* — and the decoder rebuilds the input from it, so the network learns a compact representation with no labels at all. This unsupervised encoder–decoder is the basis of many of the generative models of Section 2.1.3.

2.1.2 Temporal networks (2016–2023)

The architectures of the previous section map a fixed-size input to its output in a single shot, with no notion of order. Audio, however, is inherently *sequential*: a waveform or a spectrogram is an ordered sequence of samples or frames whose interpretation depends on context. This section reviews three complementary ways of modelling that temporal structure — recurrence, autoregression, and dilated causal convolution.

Recurrent Neural Network (RNN) — processes a sequence one step at a time while carrying a hidden state h_t that summarises everything seen so far.

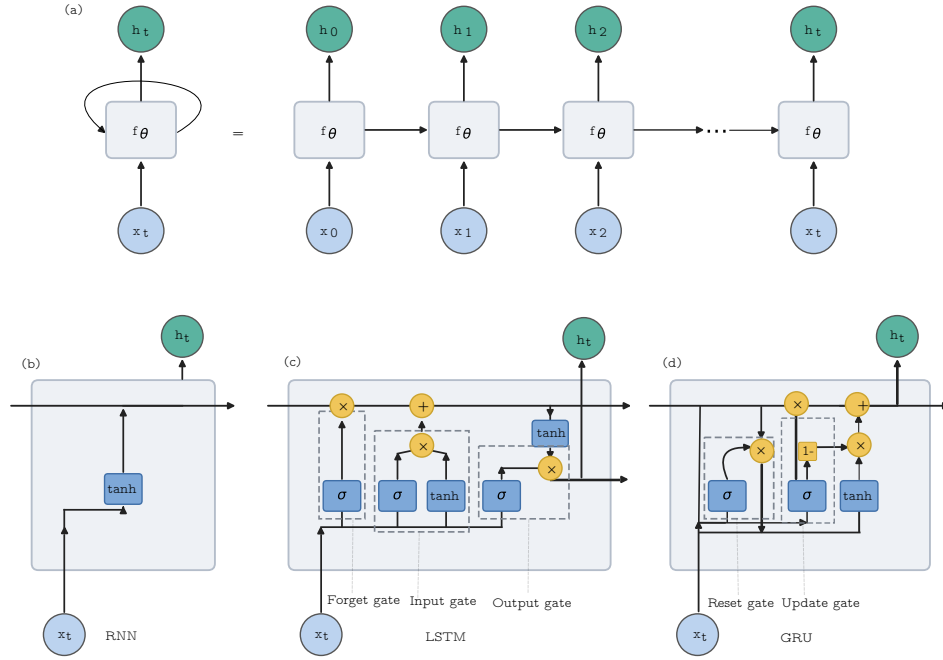


Figure 2.6: Recurrent networks. **(a)** A recurrent cell f_θ maps an input x_t to a hidden state h_t while feeding that state back into the next step; unrolling the recurrence over time (right) reveals a deep network that shares its weights across all steps. **(b)** Schematic of the plain RNN cell (a single $\tanh$), the LSTM (a cell-state highway regulated by forget, input and output gates), and the GRU (reset and update gates). σ denotes a sigmoid gate.

Writing x_t for the input at time t (e.g. a single spectrogram frame), the same cell is applied at every step,

$$h_t = f(\mathbf{W}_x x_t + \mathbf{W}_h h_{t-1} + b), \quad (2.3)$$

where f is the activation (typically $\tanh$), $\mathbf{W}_x$ and $\mathbf{W}_h$ are the input and recurrent weight matrices, and the feedback of h_{t-1} is the *recurrence*. Unrolling this loop over time (Figure 2.6a) shows that an RNN is effectively a very deep network that *shares* its parameters $\theta = \{\mathbf{W}_x, \mathbf{W}_h, b\}$ across all steps, trained by back-propagation through time. Plain RNNs struggle to retain information over long sequences, as gradients tend to vanish or explode; the gated Long Short-Term Memory (LSTM) (Hochreiter & Schmidhuber, 1997) and the lighter Gated Recurrent Unit (GRU) (Cho et al., 2014) mitigate this with multiplicative gates that learn what to keep, forget and output (Figure 2.6b). Their main drawback is practical: the recurrence in Equation 2.3 is inherently *sequential* — step t cannot be computed until h_{t-1} is available — which makes recurrent models hard to parallelise on GPUs and slow to train. They were nonetheless the dominant sequence models of their era and remain in use, underpinning music source-separation systems such as Open-Unmix (Stöter et al., 2019) and Spleeter (Hennequin et al., 2020) and text-to-speech models such as Tacotron 2 (Shen et al., 2018).

Autoregressive models — A second strategy models a sequence by *factorising* its joint distribution into a product of conditionals, predicting each element from the ones before it:

$$p(x) = \prod_t p(x_t \mid x_1, \dots, x_{t-1}). \quad (2.4)$$

WaveNet (Van Den Oord et al., 2016) was the landmark example, generating raw audio one sample at a time with a stack of dilated causal convolutions and reaching unprecedented synthesis quality (Figure 2.7). The same autoregressive principle underlies today’s GPT-style language models (Radford et al., 2019): applied to the discrete tokens produced by neural audio codecs (Section 1.1), it drives much of modern generative audio (Borsos et al., 2023). Autoregressive models are expressive density estimators, but inherently sequential at inference — each element requires all the previous ones — and therefore slow to sample.

Temporal convolutional networks — A third strategy keeps the convolutional machinery of Section 2.1.1 but adapts it to sequences. Temporal Convolutional Networks (TCNs) are one-dimensional CNNs built from *dilated causal* convolutions stacked in residual blocks: *causal*, because each output depends only on present and past inputs — suiting streaming and real-time use; and *dilated*, because inserting gaps between the kernel taps expands the receptive field *exponentially* with depth, so that long temporal context is captured

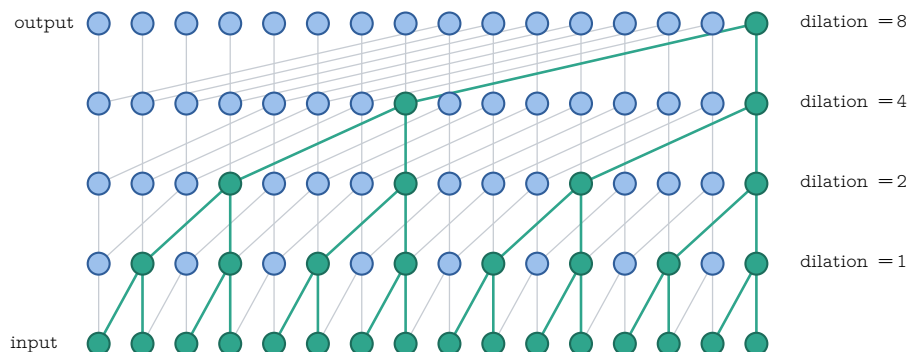


Figure 2.7: WaveNet model. Autoregression: each element x_t is predicted from all the previous ones; generation proceeds one element at a time (dashed). A stack of *dilated causal* convolutions: connections reach only into the past (causal) and skip an exponentially growing number of steps with depth (dilation 1, 2, 4, 8), so a single output (highlighted) sees a large receptive field with only a few layers.

with few layers while remaining fully parallelisable, unlike the recurrence of Equation 2.3. Introduced by WaveNet and characterised as a generic sequence model by Bai et al. (2018), dilated causal convolutions are the backbone of many audio-processing systems, including Conv-TasNet (Luo & Mesgarani, 2019) for source separation, the waveform-domain Wave-U-Net (Stoller et al., 2018), and general-purpose audio-effect removal (Rice et al., 2023). Their efficiency and causality made them particularly well suited to the real-time, on-device setting. Notably, many of these temporal models —WaveNet, Conv-TasNet and Wave-U-Net among them— operate directly on the raw waveform, and in doing so sidestep the magnitude–phase compromise of the early spectrogram CNNs of Section 2.1.1, which discarded the phase altogether.

Attention and Transformers — The Transformer (Vaswani et al., 2017) replaced recurrence with *attention*, yielding a better *parallelisable* architecture that processes all positions of a sequence at once rather than one step at a time — a decisive advantage for training on modern GPUs. Its structure is sketched in Figure 2.8. Both the input and the output are treated as sequences of discrete *tokens* — words or sub-words for text and, in the audio domain, short frames or the discrete codes produced by a neural codec. Each token is first mapped to an *embedding* — the same compact, learned vector representation introduced for CNNs in Section 2.1.1 — so that every token becomes a point in a continuous space the network can operate on.

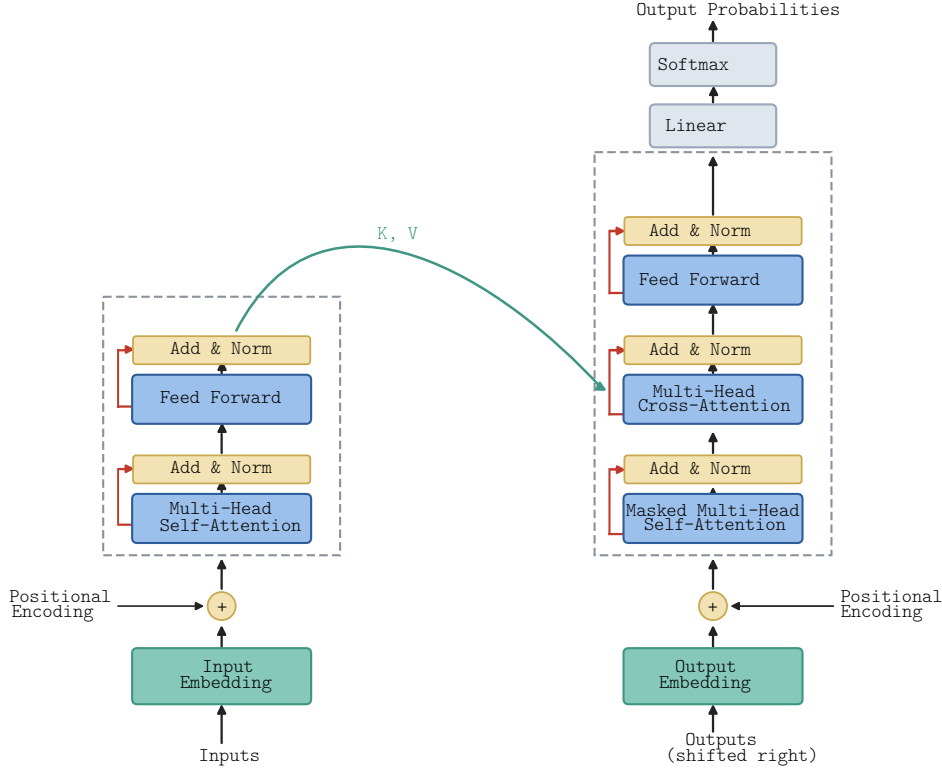


Figure 2.8: The Transformer (Vaswani et al., 2017). The encoder (left) refines the input embeddings through stacked *self-attention* and feed-forward sublayers, each wrapped in a residual “Add & Norm” connection (red). The decoder (right) generates the output autoregressively, combining self-attention with *cross-attention* (in green) that queries the encoder embeddings. This sketch is a simplified rendering of the original figure of Vaswani et al. (2017).

Because attention is order-agnostic, a *positional encoding* is then added to these embeddings to inject information about each token’s position in the sequence. The core operation is *self-attention*: within a sequence, every embedding is updated as a weighted combination of all the others, so that tokens exchange information directly regardless of how far apart they are. These attention sublayers are stacked into blocks, each wrapped in a *residual connection* (the “Add & Norm” of Figure 2.8) in exactly the spirit of the ResNets of Section 2.1.1, which is what keeps such deep stacks trainable. A second form, *cross-attention*, lets one sequence attend to another: the decoder queries the encoder’s embeddings, which is how a generator is *conditioned* on an input.

The architecture thus splits naturally into an *encoder*, which builds contextual embeddings of the input, and a *decoder*, which generates the output autoregressively; either half can be used on its own, as in the encoder-only BERT (Devlin et al., 2019) for representation learning or the decoder-only GPT-2 (Radford et al., 2019) for generation. Crucially, cross-attention does not require the two sequences to be of the same kind: a decoder can be conditioned on embeddings from a different *modality* altogether — text on audio, audio on text, images on text — which is precisely what has enabled the recent explosion of cross-modal tasks such as text-to-audio or audio captioning.

A representative audio example is Whisper (Radford et al., 2023), an encoder–decoder Transformer for speech recognition trained on hundreds of thousands of hours of weakly labelled audio: its encoder ingests a log-mel spectrogram and its decoder cross-attends to that representation to produce the transcript, with a single model also performing translation and language identification.

2.1.3 Generative models (2020–2026)

Every research objective of this thesis except the generation of audience musical noise (**RO5**) is addressed with *discriminative* models, which learn a direct mapping from an input to a target by estimating the conditional $p(y \mid x)$ — the classification and regression setting of the previous sections. Only **RO5** relies on *generative* models, which instead learn the data distribution $p(x)$ itself—or the conditional $p(x \mid c)$ if using a conditioning input c such as a text prompt or a reference voice—and can *sample* new, previously unseen data from it. Since the autoregressive generative models were already introduced in Section 2.1.2, this section briefly outlines the remaining generative families at the level of detail needed to follow the rest of the thesis.

GANs — A Generative Adversarial Network (GAN) from Goodfellow et al. (2014) casts generation as a two-player game: a *generator* maps random noise to samples, while a *discriminator* learns to tell real data from the generator’s output. The two are trained adversarially — the generator improving until its samples can no longer be distinguished from real ones — which produces sharp, realistic outputs but can be delicate to train. In audio, GANs are the standard choice for *vocoders* that invert a mel-spectrogram back into a waveform; a vocoder of this kind, HiFi-GAN (Kong et al., 2020), underlies the voice-conversion pipeline used in this thesis (RO5).

Beyond supervised learning — All the models discussed so far are trained with *supervised* learning, which requires paired examples (x, y) . *Self-supervised*

learning lifts this requirement by manufacturing the target from the unlabelled data itself, through a *pretext* task. The simplest is *reconstruction*: an autoencoder learns to rebuild its input, and a *denoising* autoencoder does so from a deliberately corrupted version, forcing the embedding to capture the signal’s structure rather than copying it verbatim. A second is *masking*: part of the signal, or of its latent representation, is hidden and the model must fill it in, much as masked-language modelling does for text; HuBERT (Hsu et al., 2021), for instance, masks spans of the audio latent and predicts pre-clustered targets at the masked positions. A third is *autoregression*: predicting the next frame or token from the past is itself a supervisory signal, which is why the autoregressive models of Section 2.1.2 (WaveNet, GPT-style decoders) are, in effect, trained in a self-supervised manner. A further, orthogonal ingredient is the *contrastive* objective, which neither reconstructs nor predicts a value but learns *by comparison*: the representation of a *positive* pair is pulled close to its context, while a set of *negative* distractors is pushed away. wav2vec 2.0 (Baevski et al., 2020) combines masking with exactly such an objective, learning to single out the correct quantised latent of a masked span among many distractors. Once trained, these label-free representations can feed a small *supervised* classifier — a *semi-supervised* pipeline — and, at scale, they are precisely what makes the pre-training of foundation models (Section 1.1) feasible.

Discrete latent codecs — A further family quantises the continuous latent of an autoencoder into a finite *codebook*, turning audio into a sequence of discrete tokens (Section 1.1). Building on the discrete-codebook idea of the VQ-VAE (Van Den Oord et al., 2017), residual vector quantisation (RVQ) stacks several codebooks, each quantising the residual left by the previous one, to encode the waveform at a very low bitrate; SoundStream (Zeghidour et al., 2021) and EnCodec (Défossez et al., 2023) are the standard neural codecs of this kind. Their training also illustrates how the pieces connect: EnCodec combines time- and frequency-domain *reconstruction* losses with an *adversarial* (GAN) loss from a set of multi-scale discriminators, bridging the discriminative models of the previous sections with the adversarial training above. The resulting discrete codes are exactly the tokens over which the autoregressive and Transformer models of Section 2.1.2 operate, and they underpin language-model-style audio generation (Borsos et al., 2023).

Diffusion — Diffusion models (Ho et al., 2020) define a fixed *forward* process that gradually corrupts data into pure noise, and learn the *reverse* process that removes it step by step; generation then starts from noise and iteratively denoises towards a clean sample. This yields high-quality, diverse samples, but at the cost of many sequential denoising steps. Running the process not on the raw signal but in the compressed latent space of an autoencoder, and parametrising the denoiser with a Transformer, gives *latent diffusion* models — the combination behind recent text-to-audio systems such as Evans et al. (2025); Nistal et al. (2024).

Flow matching — (Lipman et al., 2023) generalises diffusion. Instead of reversing a fixed noising process, it learns a continuous-time *velocity field* that transports a simple noise distribution to the data distribution along a chosen probability path, trained by directly regressing that velocity. The paths can be made much straighter than those implied by diffusion, so high-quality samples require far fewer integration steps, with diffusion itself recovered as a particular case. It has quickly become a competitive and often faster alternative for generative audio, used for instance in (Mehta et al., 2024).

This section has traced the DNN toolkit on which the rest of the thesis builds: from the perceptron and the supervised training loop, through convolutional networks, the residual connections and encoder–decoder patterns, the extension to sequences via recurrence, autoregression, dilated causal convolutions and attention. We presented *embeddings* — the compact learned representation that every such architecture produces and that later models reuse — and the distinction between *discriminative* models, which map an input to a target, and *generative* ones, which model the data distribution and sample from it.

However, a model is only half of the story: it is ultimately only as good as the *data* it is trained and evaluated on. This is doubly true in this thesis, where much of that data is not recorded but *synthesised* — sound fields spatialised and convolved with simulated room responses. Such synthetic data is only useful insofar as it is *ecologically valid*: faithful to the real signals the model will encounter at deployment. And faithfulness cannot be conjured from the generative pipeline alone; it presupposes a solid grasp of the physical phenomena being imitated — how sound propagates and is captured in space, and how rooms colour it. In short, **generating good synthetic data requires first understanding real data.** The remainder of this chapter therefore turns from the model to the data: first the capture and spatial representation of sound fields, and then the measurement and simulation of RIRs.

2.2 Spatial audio and Ambisonics

The next two chapters and the corresponding audio processing tasks — HAs (Chapter 3) and SE (Chapter 4), **RO1** to **RO4** — rely on spatial audio to build and reverberate their datasets. This is most evident in **RO1**, where the evaluation benchmark for HAs is created by reproducing complex acoustic scenes over Ambisonics. We additionally apply spatial audio concepts like binaural-rendering and directional room simulation across **RO2–RO4**. This section introduces the necessary background: first the general paradigms of spatial audio as a whole, and then a chronological account of Ambisonics development.

2.2.1 Spatial audio paradigms

Spatial audio is the body of techniques for capturing, representing and reproducing the directional properties of sound, so that a listener perceives sources at their intended positions in space. Human spatial hearing relies on a small set of cues: the Interaural Time Difference (ITD) and the Interaural Level Difference (ILD) —the differences in arrival time and in level between the two ears— together with the direction-dependent spectral colouring imposed by the head, torso and outer ears. Spatial-audio systems work by evoking or preserving these cues, and their methods are commonly grouped into three paradigms: channel-based, object-based and scene-based.

Channel-based audio, the oldest and most widespread, stores one signal per loudspeaker and assumes a fixed playback layout. It grew from two-channel stereo to multichannel *surround*, adding centre and surround channels around the listener in the now-ubiquitous 5.1 and 7.1 configurations. Dolby Laboratories was central to this evolution, popularising cinema surround with Dolby Stereo from the mid-1970s and discrete 5.1 with Dolby Digital in the early 1990s. In all of these formats, spatialisation is fixed at production time through *panning* — distributing a source’s energy across the available loudspeakers according to a panning law — so the spatial image is correct only for the intended layout and within a small *sweet spot*.

Object-based audio instead stores each source as a separate *audio object* together with metadata describing its position, and defers rendering to playback time, where it is adapted to whatever loudspeaker or headphone configuration is available. This decoupling of content from layout underlies modern immersive formats.

Scene-based audio sits between the two: it encodes the whole sound field around a point, independently of any loudspeaker layout, and decodes it to the available system at playback. Its prime example is *Ambisonics*: the sound field around a point is sampled into a series of directional components — the Spherical Harmonics (SH) — of increasing angular order. Ambisonics is a *complete* theory in the sense that it supports the entire chain: a sound field can be *recorded* directly with a compact microphone array; a synthetic source can be *encoded* into that same representation by panning it according to its direction; and the result can then be *decoded* to arbitrary loudspeaker layouts. The remainder of this section will trace its development and details these encoding and decoding steps.

Closely related to these paradigms, but concerned with reproduction over headphones, is *binaural* audio. Its starting point is that the head and body filter incoming sound in a direction-dependent way: most simply, the head casts an *acoustic shadow*, so that a source off to one side reaches the far ear later and weaker than the near ear, by an amount that depends on its direction. Beyond the head, the *pinna* (outer ear) and ear canal add their own direction-dependent spectral filtering — the peaks and notches that, in particular, disambiguate elevation — and, since these depend on each person’s anatomy, they vary substantially from one listener to another. This complete filter is precisely the Head-Related Transfer Function (HRTF), and is exactly what we exploit for building the datasets for **RO2**. Because measuring HRTFs individually is impractical, considerable effort goes into *personalising* them, for instance by predicting them from a 3D scan of the listener’s head and ears acquired with a depth sensor (Miccini & Spagnol, 2020).

Finally, with the rise of Extended Reality (XR), binaural rendering of Ambisonics has become especially attractive: because the sound field is represented in the SH domain, decoding it to the two ears reduces to multiplying the Ambisonics signals by a fixed matrix of HRTFs, and following the listener’s head rotations is itself just another matrix multiplication.

2.2.2 A chronological perspective of Ambisonics

The development of Ambisonics is best understood as a gradual generalisation of stereophony, followed by decades of refinement of its encoding and decoding. Figure 2.9 situates the milestones discussed below in time.



Figure 2.9: A personal, simplified selection of the Ambisonics, VBAP and binaural audio literature, restricted to the concepts most relevant to the work of Chapter 3; the review extends up to 2023.

Stereo and the Mid-Side (MS) technique. The earliest spatial-audio format, two-channel stereophony, was pioneered by Alan Blumlein around 1930 (Alexander, 2013), who introduced both coincident microphone techniques — *XY* (two coincident cardioids) and *MS* — and the panning laws (such as the sine law) still in use today. In the *MS* technique a forward-facing microphone, the *Mid* M , is combined with a sideways figure-of-eight, the *Side* S , and the stereo loudspeaker signals are obtained by a simple sum and difference,

$$L = M + S, \quad R = M - S, \quad (2.5)$$

so that the M/S ratio controls the stereo width. The Side microphone is the key element: being a figure-of-eight, it is positive towards the left and negative towards the right, encoding left–right direction as a single *signed* amplitude.

ITD and ILD. Wendt (1964) carried out careful listening experiments on two-channel stereophony, characterising how time and level differences between the loudspeakers map onto the perceived direction of a phantom source. These cues reflect the classical duplex theory of localisation: the ITD dominates at low frequencies, where diffraction keeps phase cues unambiguous, while the ILD dominates at high frequencies, where phase becomes ambiguous.

From MS to First Order Ambisonics (FOA). Gerzon (1973) took the step from two-channel stereo to a complete description of the sound field, in what he called *periphony* and we now know as First-Order Ambisonics. The idea generalises the *MS* pair: the Side figure-of-eight is just one first-order directional component, so capturing the *full* first-order field requires the omnidirectional pressure — the Mid, when it is an omni — together with *three* orthogonal figure-of-eight components aligned with the front–back, left–right and up–down axes. These four signals are the *B-format* channels $\mathbf{b} = [W, X, Y, Z]^\top$.

Mathematically, this is the expansion of the sound field around a point into Spherical Harmonics (SH), truncated at first order (Zotter & Frank, 2019). The SH are a set of functions on the sphere of increasing angular *order* ℓ (Figure 2.10): order $\ell = 0$ is a single omnidirectional function (the pressure W), order $\ell = 1$ adds the three dipoles X, Y, Z , and each higher order contributes $2\ell + 1$ functions of finer angular detail, used by Higher Order Ambisonics (HOA). First-Order Ambisonics keeps orders $\ell \leq 1$, the four B-format channels; in this work we use Ambisonics and SH interchangeably.

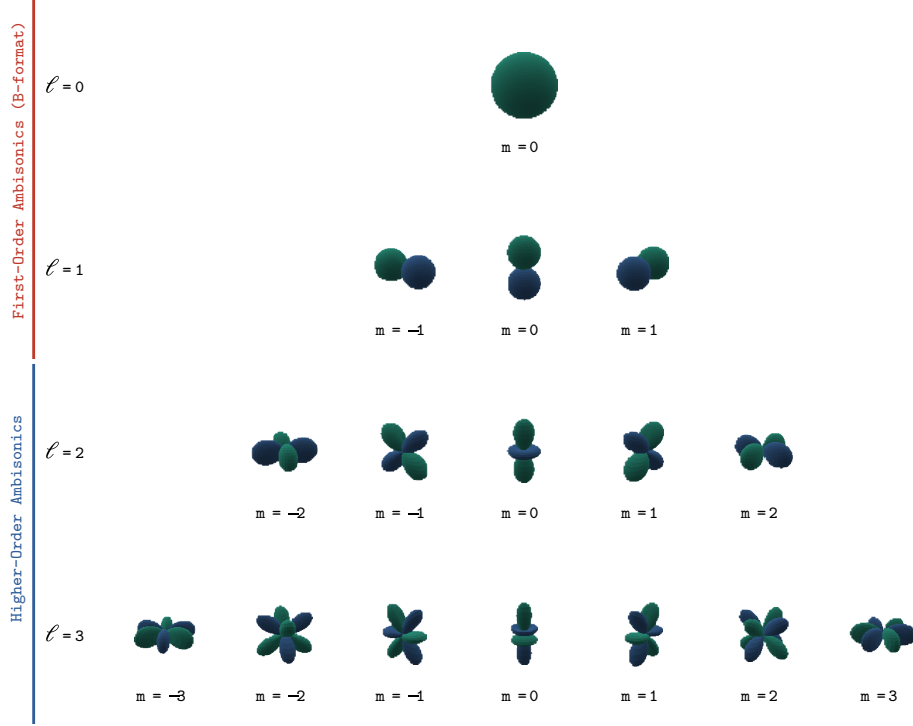


Figure 2.10: The real spherical harmonics up to order $\ell = 3$, the basis functions of the Ambisonics sound-field representation. Orders $\ell \leq 1$ form FOA, i.e. the B-format; each additional order adds $2\ell + 1$ components of finer angular resolution, used by HOA.

Encoding. A mono source s arriving from azimuth θ and elevation φ is panned into B-format by weighting it with the spherical harmonics evaluated at that direction. In the now-standard *ACN* channel ordering with *SN3D* normalisation,

$$\mathbf{b} = [W, Y, Z, X]^\top = s \mathbf{y}(\theta, \varphi), \quad \mathbf{y}(\theta, \varphi) = \begin{bmatrix} 1 \\ \sin \theta \cos \varphi \\ \sin \varphi \\ \cos \theta \cos \varphi \end{bmatrix}, \quad (2.6)$$

so that $W = s$ carries the pressure while X, Y, Z carry the projection of the source onto the three axes; a full sound field is the superposition of all such contributions. (The older FuMa B-format orders the channels $[W, X, Y, Z]$ and scales W by $1/\sqrt{2}$; the conventions differ only by a permutation and constant factors.)

A-format versus B-format. The same B-format can be captured acoustically. Ideally it would be recorded with one coincident omnidirectional microphone (for W) and three coincident figure-of-eight microphones aligned with

the axes (for X, Y, Z). In practice this is approximated by a compact tetrahedral array of four cardioid capsules, whose raw signals form the *A-format* $\mathbf{a} = [a_1, a_2, a_3, a_4]^\top$; B-format is then obtained through a fixed linear transform — sums yield the omni W , differences yield the dipoles — $\mathbf{b} = \mathbf{M}\mathbf{a}$, followed by a frequency correction for the finite capsule spacing. A-format is thus what the microphone records, whereas B-format is the layout-independent representation we work with. A-format is encountered only when physically recording with an Ambisonics microphone, and even then only as an internal intermediate; a large share of today’s Ambisonics content is instead *encoded* synthetically — panned, simulated, or rendered — straight into B-format. We introduce it here mainly to make the step from MS recording to Ambisonics concrete.

Decoding. To play a B-format signal over L loudspeakers at unit directions $\mathbf{n}_l = (\cos \theta_l \cos \varphi_l, \sin \theta_l \cos \varphi_l, \sin \varphi_l)$, each feed is obtained as a *virtual microphone* pointing towards that loudspeaker,

$$p_l = a_0 W + a_1 (\mathbf{n}_l \cdot [X, Y, Z]^\top), \quad (2.7)$$

where (a_0, a_1) set the directivity of the virtual first-order microphone (e.g. a cardioid), so that p_l predominantly picks up energy from the speaker’s direction; stacking the L feeds gives the matrix form $\mathbf{p} = \mathbf{D}\mathbf{b}$. This also makes the MS analogy precise: keeping only W and Y and decoding to two opposing cardioids ($\mathbf{n}_L = (0, 1, 0)$, $\mathbf{n}_R = (0, -1, 0)$) recovers exactly $L = W + Y$ and $R = W - Y$, i.e. Equation 2.5. B-format is therefore the MS pair completed: adding the front-back dipole X and the vertical dipole Z turns a fixed two-channel stereo image into a rotatable, layout-independent sound field.

The same Gerzon will later introduce in 1992 the energy and velocity localisation vectors r_E and r_V commonly used to summarise these cues (Gerzon, 1992), r_E in particular becoming a standard descriptor of directional concentration: $\|r_E\| = 0$ indicates no directionality and $\|r_E\| = 1$ a spatial Dirac.

A parallel line: amplitude panning. Developed independently of, and in parallel to, Ambisonics, Pulkki (1997) introduced Vector-Based Amplitude Panning (VBAP), which reproduces a source by decomposing space into loudspeaker pairs or triplets and panning within one. It is robust (only a few loudspeakers are active at a time) and easy to adapt to irregular layouts, but — unlike Ambisonics — it is not a complete theory (recording is not possible), and the apparent source size and coloration vary with position. Pulkki (1999) mitigates the latter by spreading each source across a pair of auxiliary dir-

ections at $r_E \pm \alpha$. Although a distinct paradigm, VBAP later re-enters the Ambisonics pipeline as the loudspeaker-panning stage of the AllRAD decoder (below).

Refining encoding and decoding. Within Ambisonics, subsequent work largely refines how the sound field is encoded and, especially, decoded — including to binaural. In Jot et al. (1999), techniques for combining binaural audio and Ambisonics are discussed: a basic binaural render is obtained by decoding Ambisonics to a virtual, ideally placed loudspeaker layout and convolving each virtual loudspeaker with its Head-Related Impulse Response (HRIR) pair — especially practical when real-time rotations must follow a head-tracking device.

Higher Order Ambisonics (HOA). In Daniel (2000), FOA is expanded to HOA for regular layouts, trading the constant source size and smooth panning of low orders for a larger sweet spot and sharper localisation, at the cost of as many loudspeakers as channels.

The Ambisonics signal becomes a vector $\mathbf{b} \in \mathbb{R}^{(N+1)^2}$ collecting all spherical harmonics up to order N , and encoding a source keeps the same form,

$$\mathbf{b} = s \mathbf{y}_N(\theta, \varphi), \quad \mathbf{y}_N(\theta, \varphi) = [Y_0^0, Y_1^{-1}, Y_1^0, Y_1^1, \dots, Y_N^N]^\top(\theta, \varphi), \quad (2.8)$$

with the $(N+1)^2$ real spherical harmonics Y_ℓ^m evaluated at the source direction (FOA is the case $N = 1$). Decoding to L loudspeakers is likewise a single linear map,

$$\mathbf{p} = \mathbf{D} \mathbf{b}, \quad \mathbf{D} \in \mathbb{R}^{L \times (N+1)^2}, \quad (2.9)$$

whose decoding matrix $\mathbf{D}$ is built from the loudspeaker directions (the basic decoder takes $\mathbf{D} = \mathbf{Y}^+$, the pseudo-inverse of the spherical-harmonic sampling matrix). Binaural rendering is simply the case $L = 2$, where the rows of $\mathbf{D}$ are the left- and right-ear HRTFs expressed in the SH domain. This basic decoding works well only at low frequencies; Daniel therefore robustifies the decoder by weighting the SHs so that the secondary lobes of the virtual microphones are attenuated ($\max r_E$) or suppressed (the in-phase decoder). Tapering the SH orders in this way is the angular counterpart of *windowing* a signal before a Fourier analysis: smoothing a truncated expansion lowers its side lobes (here, the off-axis leakage of the virtual microphones) at the price of a wider main lobe (a loss of angular resolution).

In Avni & Rafaely (2010), HRTFs are represented in the SH domain, enabling interpolation between their discrete measured directions, and in Zotter & Frank

(2012) AllRAD provides a physics-based HOA decoder for irregular layouts by first decoding to an ideal virtual layout and then rendering those signals with the available speaker triplets through VBAP. We use AllRAD in Chapter 3.

Ambisonics to Binaural. The remaining developments concern improving binaural rendering of limited-order Ambisonics. Bernschütz (2013) address low-frequency group delay in HRIR databases by replacing the low band with a synthetic spherical model called Low Frequency Extension (LFE), while Bernschütz et al. (2014) show that pairing high-order HRTFs with lower-order soundfields causes severe losses in the high frequencies in a frequency-dependent way; Ben-Hur et al. (2017) compensate the resulting low-pass effect of SH truncation with a global Spectral Equalisation (SpecEQ) filter. Zalusch et al. (2018) adapt the HRIRs for direct low-order rendering by removing the ITD linear-phase component and matching diffuse-field covariances (Time Alignment and diffuseness Constraint, TAC), and Schörkhuber et al. (2018) propose Magnitude Least Squares (MagLS), which in the high frequencies optimises the magnitudes alone, disregarding phase. In parallel, Armstrong et al. (2018) introduce a bi-radial render with two sets of virtual loudspeakers, one centred at each ear, gaining up to two Ambisonic orders of perceptual quality, and Hold et al. (2019) reduce angle-dependent artefacts by also tapering the SHs. Finally, Ben-Hur et al. (2021) time-align the HRIRs with respect to each ear for direct bilateral rendering, and these threads are combined in Engel et al. (2021), the current state of the art: Bilateral Magnitude Least Squares (BiMagLS), which ear-aligns the HRTFs, fits the low-frequency SH-HRTFs with an L2 approximation of the higher-order version, and disregards phase at high frequencies as in MagLS. We use BiMagLS in Chapters 3 and 4.

Beyond this scope. Several further directions, left out of the timeline above because they lie outside the scope of this thesis, are nonetheless worth signposting: HOA coding (Hold et al., 2023), DNN-based microphone-to-Ambisonics encoding (Heikkinen et al., 2025), continuous representations of the sound field, or of the room impulse responses themselves, so that the response at unmeasured positions can be interpolated from only a sparse set of probes (Luo et al., 2022; Masuyama et al., 2025; Erdem et al., 2026; Baek et al., 2025) or spatial audio generation from vision, producing Ambisonics soundtracks directly from 360° video (Morgado et al., 2018; Liu et al., 2025).

2.3 Room impulse responses

RIRs are recurrent in this thesis: in Chapter 3, to assemble the HRTFs of HAs; in Chapter 4 used to generate a SE dataset; and in Chapter 5 to measure a dataset of concert-hall responses for live MSS. We devote this section to defining and to reviewing RIRs and how to simulate them.

Sound propagation from a source to a receiver in a room is, to a good approximation, a linear time-invariant system. Its Room Impulse Response (RIR) $h(t)$ is the signal captured at the receiver when the source emits an ideal impulse — the Dirac delta $\delta(t)$ — and it characterises that source–receiver path completely: the signal $y(t)$ recorded for any emitted signal $s(t)$ is the convolution

$$y(t) = (s * h)(t) = \int_{-\infty}^{\infty} s(\tau) h(t - \tau) d\tau. \quad (2.10)$$

As illustrated in Figure 2.11, a RIR is conventionally split into three regions. The *direct sound* is the first arrival, travelling straight from source to receiver. It is followed by the *early reflections*, a sparse set of strong, temporally distinct reflections off the nearest boundaries, whose timing and direction shape the perceived location, width and timbre of the source. As time passes the reflections grow exponentially denser until they can no longer be resolved individually, forming the diffuse *late reverberation* (or *tail*), which decays approximately exponentially and is described statistically rather than reflection by reflection — most commonly through the reverberation time T_{60} , the time its energy takes to decay by 60 dB.

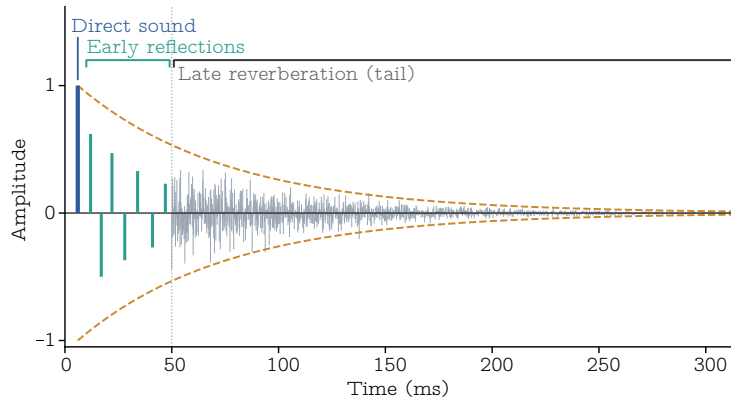


Figure 2.11: A typical room impulse response and its three regions: the direct sound, the early reflections, and the late reverberation tail.

2.3.1 Measuring RIRs with the exponential sweep

In principle, a RIR could be obtained by exciting the room acoustically with a physical impulse — like a pistol shot or a bursting balloon — and recording the result, but such sources are poorly repeatable, difficult to equalise and offer a limited Signal to Noise Ratio (SNR).

In practice, the emitted signal $s(t)$ of Equation 2.10 is an excitation reproduced by the *source* loudspeaker, and $y(t)$ is what a measurement microphone at the *receiver* position records; the goal is to recover $h(t)$ from this known input and recorded output. For the measured response to reflect the room rather than the measurement chain, both transducers are traditionally chosen to be as neutral and direction-independent as possible: the microphone is an omnidirectional capsule with a flat magnitude and linear-phase response, and the source approximates an ideal point emitter radiating uniformly in all directions — classically a dodecahedron loudspeaker, whose twelve drivers on the faces of the solid yield near-omnidirectional coverage.

A RIR can also be estimated from *continuous noise* excitation, as routinely done in live sound with dual-channel Fast Fourier Transform (FFT) analysers that recover the transfer function between the noise feeding the loudspeakers and a measurement microphone in the room. The now-standard alternative, however, is the Exponential Swept-Sine method of Farina et al. (2000), in which the room is excited with a sinusoid $x(t)$ whose instantaneous frequency rises exponentially from f_1 to f_2 over a duration T ,

$$x(t) = \sin \left[\frac{2\pi f_1 T}{\ln(f_2/f_1)} \left(e^{\frac{t}{T} \ln(f_2/f_1)} - 1 \right) \right], \quad 0 \leq t \leq T. \quad (2.11)$$

The RIR is recovered by *deconvolution* with an inverse filter $f(t)$ designed so that $x * f = \delta$. This filter is the time-reversed sweep, $x(T - t)$, weighted by an exponentially decaying envelope that whitens the sweep's pink spectrum,

$$f(t) \propto x(T - t) \left(\frac{f_1}{f_2} \right)^{t/T}, \quad (2.12)$$

up to a normalisation that sets $x * f = \delta$. Convolution of the recorded signal $y(t)$ with it,

$$h(t) = (y * f)(t), \quad (2.13)$$

collapses the sweep back to an impulse and returns the RIR. Applying a time window to $h(t)$ is itself a versatile analysis tool, since cropping the RIR in time selects which part of the source–receiver interaction is retained. Windowing

out everything before $t = 0$ discards the harmonic-distortion responses that the ESS maps to negative times. Keeping only a short segment after the direct sound, ending before the first early reflection, yields a *quasi-anechoic* response that captures the magnitude and phase of the measurement chain itself — loudspeaker, microphone and direct path — free of the room. Extending the window to include the early reflections brings in the strongly *position-dependent* part of the response, which varies markedly with source and receiver placement within the room. The late reverberation, by contrast, is diffuse and statistically similar throughout the room.

2.3.2 Room impulse response simulation

The most straightforward way to simulate a RIR is the Image Source Method (ISM) (Allen & Berkley, 1979). It rests on a simple geometric observation: a specular reflection off a flat wall is indistinguishable, at the receiver, from the direct sound of a *mirror image* of the source, placed behind the wall at the same perpendicular distance (Figure 2.12). Each boundary thus replaces a bounce by a virtual *image source*, and the reflected path is recovered as the straight line from that image to the receiver. Reflections off several walls are handled by mirroring recursively — an image reflected again across another wall is a second-order image, and so on — so that an order- N reflection corresponds to an image mirrored N times; for a rectangular (shoebox) room these images form a regular lattice of mirrored copies of the room.

Applying the method yields, for every reflection i , the coordinates of its image source and its order N_i , which is all that is needed to render the response. The image-to-receiver distance d_i fixes both the arrival time $\tau_i = d_i/c$ (with c the speed of sound) and the attenuation due to spherical spreading, which decays the pressure as $1/d_i$. The energy lost at each bounce is set by the walls' absorption coefficient α (the absorbed fraction of incident energy), or equivalently by the amplitude reflection coefficient $\beta = \sqrt{1 - \alpha}$; a path of order N_i has therefore been attenuated by the product of the β of the walls it visited, i.e. β^{N_i} for uniform boundaries. The RIR is then the sum of one delayed, attenuated impulse per image,

$$h(t) = \sum_i \frac{\beta^{N_i}}{d_i} \delta\left(t - \frac{d_i}{c}\right). \quad (2.14)$$

In practice the absorption is rarely known from material data; one usually targets a desired reverberation time instead. The two are linked by Sabine's

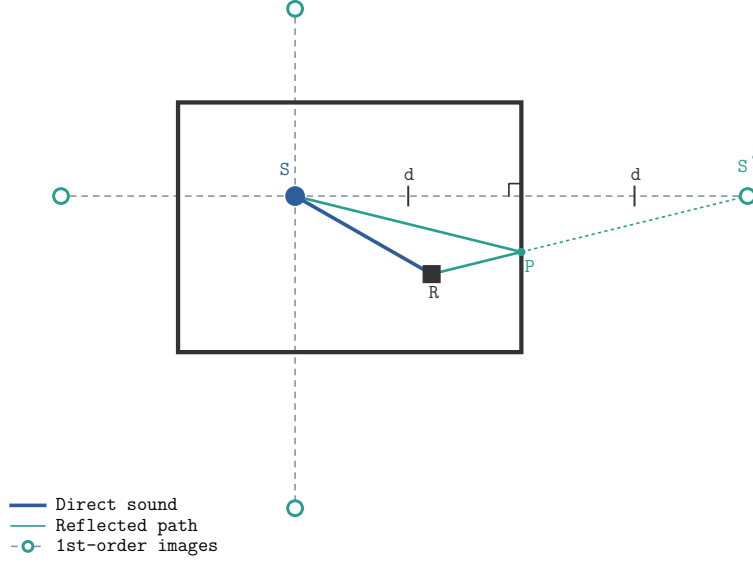


Figure 2.12: The ISM, illustrated to first order. Receiver R , source S , *image source* S' . The reflected path $S \rightarrow P \rightarrow R$ then coincides with the straight line from S' to R , so the reflection reduces to a delayed, attenuated copy of the direct sound.

equation (Sabine, 1927), $T_{60} = 0.161 V/A$, with V the room volume and $A = \sum_k S_k \alpha_k$ the total absorption (surface area times absorption, summed over boundaries). For uniform walls this inverts to

$$\alpha = \frac{0.161 V}{T_{60} S_{\text{tot}}}, \quad \beta = \sqrt{1 - \alpha}, \quad (2.15)$$

so a single *target* T_{60} determines the β used in Equation 2.14.

The ISM is exact for the specular reflections of a rigid-walled shoebox, which makes it an efficient and ubiquitous way to render the direct sound and early reflections. Some of its idealisations can be relaxed within the method itself: the real, frequency- and angle-dependent wall impedance can be modelled by replacing the scalar β with per-band reflection filters; source and receiver directivity can be applied per image; and air absorption can be added as a distance-dependent filter. A subtler artefact is that, with a real positive β and impulses placed at integer samples, every reflection is positive, so the simulated RIR is biased towards positive values, while measured responses are essentially zero-mean and symmetric about the time axis. This is sometimes corrected by rendering each arrival with a band-limited fractional-delay filter — a windowed sinc, whose side lobes restore the negative excursions (Peterson, 1986).

Finally, the regular-lattice construction is restricted to shoebox geometries, and the number of images grows cubically with order, so the method becomes impractical precisely where the late, diffuse reverberation begins. It is therefore sometimes truncated at low order and paired with a statistical model of the tail (Lehmann & Johansson, 2009).

Other assumptions require a different paradigm altogether. The ISM models only specular reflections, ignoring the diffuse *scattering* of rough surfaces and the *diffraction* around edges and obstacles; both call for ray- or beam-tracing with scattering coefficients, radiosity, or edge-diffraction models (Savioja & Svensson, 2015). Being a geometrical approximation, it also misses the modal behaviour and wave interference that dominate at low frequencies, which only wave-based solvers capture:

- **Ray tracing.** Rather than enumerating image sources, ray tracing shoots a large number of rays from the source and follows them as they bounce around the room, tallying those that reach a receiver volume. Its key advantage over the ISM is that reflections don't need to be ideally specular: at each surface a ray can be scattered into a random direction according to the boundary's scattering coefficient, so the diffuse reflections that the ISM ignores are captured through Monte-Carlo sampling. The cost is statistical: accuracy improves only as more rays are traced, and very late or weak paths are noisy.
- **Ray-radiosity.** Radiosity-based methods are instead *surface*-based. The boundaries are subdivided into patches, and the acoustic energy exchanged between every pair of patches is computed under the assumption of (ideally diffuse) reflection, yielding the time-dependent energy decay across the room without tracking individual paths. They model diffuse reverberation efficiently but, by construction, do not represent specular reflections or phase.
- **Wave-based methods.** At the opposite end, wave-based solvers discretise and solve the acoustic wave equation directly — through finite-difference time-domain, finite-element or boundary-element formulations — and therefore reproduce wave phenomena that geometrical methods omit, such as modal behaviour, interference and diffraction. They are more accurate at low frequencies compared to ISM, but their cost grows steeply with frequency, which restricts them in practice to the lower part of the spectrum.

- **Hybrid methods.** Since no single technique is accurate and affordable across the whole spectrum and the whole response, practical simulators are usually *hybrid*, assigning each regime to the method that suits it best: wave-based solvers for the low frequencies, where modes dominate; and a path-based method such as the ISM for the high-frequency early part, where specular reflections matter and timing is critical (Tang et al., 2022).
- **DNN-based methods.** More recently, DNNs generate RIRs directly, conditioned on a 3D scene mesh (Ratnarajah et al., 2022) or on perceptual descriptors via diffusion (Arellano et al., 2025). Classical solvers have also been accelerated on GPU (Diaz-Guerra et al., 2021).

2.3.3 Survey of publicly-available RIR datasets

A number of RIR datasets are publicly available; Table 2.1 summarises them. Two axes are worth tracking throughout, as they define our research gap: the sampling rate and the number of *distinct* rooms, since most datasets cover either very few rooms or only low sampling rates. We review the measured datasets first and then the simulated ones.

Real — RWCP-SSD (Nakamura et al., 2000) is one of the earliest resources, with 143 microphone-array responses at 48 kHz across 14 real offices and rooms for distant-speech research. The Aachen Impulse Response database (AIR) (Jeub et al., 2009) provides 344 binaural responses at 48 kHz measured with a dummy head across just 5 rooms, one of them a church, and was originally motivated by hearing-aid and dereverberation studies. C4DM (Stewart & Sandler, 2010) offers 468 mono and first-order Ambisonic (FOA) responses at a high 96 kHz, but only in 3 large London spaces, while OpenAIR^{2.2} is a community FOA library spanning some 46 real, often heritage or “extreme” environments (cathedrals, tunnels) at 48 kHz, without a fixed protocol or response count.

Several databases were conceived as speech-processing benchmarks, and are correspondingly small in rooms and sampling rate. Take, for example, the REVERB challenge database (Kinoshita et al., 2013), which offers microphone-array responses in only 3 room sizes at 16 kHz for dereverberation, ASR and SE; MIRD (Hadad et al., 2014) captures 234 eight-microphone-array responses at 48 kHz in a single room, at three different T_{60} values and several array spacings; and the ACE corpus (Eaton et al., 2015) records 7 rooms at 48 kHz

^{2.2}<https://www.openair.hosted.york.ac.uk/>

with a range of devices (laptop, phone, and an em32 Eigenmike, hence up to higher-order Ambisonics) for blind T_{60} and Direct-to-Reverberant Ratio (DRR) estimation. The MIT IR survey (Traer & McDermott, 2016) instead maximises room diversity, with 271 mono responses each measured in a distinct everyday space, but at 44.1 kHz and a single response per room, while BUT ReverbDB (Szöke et al., 2019) adds retransmitted speech and noise with about 1.3k responses across 9 real rooms, at 16 kHz.

A more recent trend targets *spatial* and densely sampled measurements, typically within a single room. TAU-SRIR (Politis et al., 2020) provides 114 FOA and tetrahedral-array responses at 24 kHz across 9 classrooms, gyms and halls, and underpins the DCASE sound-event localisation benchmarks. dEchorate (Carlo et al., 2021) annotates the timings of early reflections and the tail, together with the 3D positions of the corresponding image sources, with about 1.8k responses at 48 kHz in a single room; MeshRIR (Koyama et al., 2021) densely samples one room on a source-and-receiver grid (4.4k responses at 48 kHz) for sound-field reconstruction; Motus (Götz et al., 2021) measures 3.3k higher-order Ambisonics responses at 48 kHz, again in a single room, across 830 furniture configurations; and Arni^{2,3} records over 132k responses at 44.1 kHz within one variable-acoustics room, spanning 5342 configurations of movable absorptive panels.

The most recent measured datasets push count and automation further, but rarely the number of rooms. SoundCam (Wang et al., 2023a) records 5k responses at 48 kHz across 3 rooms with a moving human present, for tracking people from room acoustics; MIRACLE (Kujawski et al., 2024) contributes a large 856k responses at 48 kHz from a planar 64-microphone array on dense source grids, but in a single anechoic chamber (optionally with a reflecting panel); and GTU-RIR (Pekmezci & Genc, 2024) collects around 15k responses at 44.1 kHz over 11 rooms with a semi-automated rig. HOMULA-RIR (Miotello et al., 2024) models a teleconference in a single seminar room using higher-order microphones and a linear array (52 responses at 48 kHz), and MP-RIR (Friede et al., 2024) samples one complex-shaped room on a dense 3D grid with a robot-automated rig (68.7k responses at 48 kHz). FLAIR (Sundström et al., 2025) pairs 270 responses at 48 kHz in a single room with laser-measured 3D point clouds of the geometry; SRIRACHA (Pelling et al., 2025) provides 2.65M responses at 48 kHz as an interpolation benchmark, yet within a single shoebox with varying absorption; and UPV_RIR_DB (García-Gamborino et al., 2026)

^{2,3}<https://doi.org/10.5281/zenodo.6985104>

releases 166 multichannel responses at 48 kHz across 3 rooms with hierarchical metadata and a per-pair spatial grid. Finally, OpenSLR28 (Ko et al., 2017) is not a single campaign but the widely used augmentation *bundle* that aggregates real responses from RWCP, REVERB and AIR with simulated responses and point-source noises, all distributed at 16 kHz.

Synthetic — Obtaining room diversity at scale by measuring is impractical, which motivates simulating synthetic RIRs. BIRD (Grondin et al., 2020) provides 100k stereo responses at 16 kHz generated with the ISM across as many shoebox rooms; the spacing between the two microphones of each A/B pair is drawn uniformly as $\sim \mathcal{U}(0.1, 0.3)$ m, although the microphone directivity is left unspecified. SoundSpaces (Chen et al., 2022) renders about 16M responses at 16 kHz for the Matterport3D meshes and 44.1 kHz for the Replica subset, over 101 scanned indoor scenes with a bidirectional path-tracing simulator (continuous rendering in version 2.0); GWA (Tang et al., 2022) computes around 2M mono responses at 48 kHz over some 18.9k furnished rooms, using a hybrid simulator that couples a finite-difference time-domain wave solver (low frequencies) with ray tracing (high frequencies) and semantic acoustic-material assignment. Lastly, ReverbFX (Richter et al., 2025) departs from physical rooms altogether, deriving 1.85k stereo responses at 48 kHz from professional reverb effect plugins (convolution-based and algorithmic) to study singing-voice dereverberation under artificial reverberation.

Table 2.1 reveals a consistent trade-off along our two axes. Measured datasets are mostly single-room campaigns (dEchorate, MeshRIR, Motus, Arni, MIRACLE, HOMULA-RIR, MP-RIR, FLAIR and SRIRACHA), and even the broadest rarely exceed a dozen rooms (14 for RWCP-SSD, 11 for GTU-RIR, 9 for BUT ReverbDB and TAU-SRIR); only the MIT survey spans many rooms (271). Synthetic datasets, by contrast, can cover hundreds or thousands of rooms, but at the cost of lower sampling rates. BIRD covers 100k shoebox rooms and the Matterport3D subset of SoundSpaces 85 scanned houses, both at 16 kHz. The higher-rate options are in turn restricted in scene count — the Replica subset of SoundSpaces reaches 44.1 kHz but spans only 18 scenes — except for GWA which attains 48 kHz over some 18.9k rooms. The augmentation bundle most commonly used for speech enhancement, OpenSLR28 — the default choice in the DNS challenges — is likewise dated and limited to 16 kHz.

These limitations frame the gaps addressed in the following chapters: no dataset targets Hearing Aids (Chapter 3); no purpose-built, higher-fidelity synthetic RIRs exist for SE beyond GWA and the dated 16 kHz bundles (Chapter 4);

and no *concert-hall* or live-music-venue dataset is available at the scale and sampling rate required for source separation of live recordings (Chapter 5).

Table 2.1: Overview of publicly available RIR datasets, grouped into measured and simulated. Sampling rate is in kHz; fs denotes sampling rate.

	year	[kHz] fs	# IRs	# rooms	format / mic type	other notes
<i>Real</i>						
RWCP-SSD	'00	48	143	14	mic array	offices/real rooms
AIR	'09	48	344	5	binaural	dummy head
C4DM	'10	96	468	3	FOA	large spaces
OpenAIR	'10	48	—	46	FOA	heritage spaces
REVERB	'13	16	24	3	mic array	ASR and SE tasks
MIRD	'14	48	234	1	mic array	3 different T_{60} values
ACE	'15	48	—	7	HOA	laptop, phone, em32
MIT IR Survey	'16	44.1	271	271	mono	everyday real spaces
OpenSLR28	'17	16	—	—	mono	RWCP/REVERB/AIR + sim.
BUT Reverb	'19	16	1.3k	9	mic array	offices/real rooms
TAU-SRIR	'20	24	114	9	FOA + mic array	classrooms, gyms, halls
dEchorate	'21	48	1.8k	1	mic array	early & tail annotated
MeshRIR	'21	48	4.4k	1	mic & src array	sound field reconstruction
Motus	'21	48	3.3k	1	HOA	830 furniture cfgs
Arni	'22	44.1	132k	1	mono	5342 panel cfgs
SoundCam	'23	48	5k	3	mono	moving human present
MIRACLE	'23	48	856k	1	mic array	panel in anechoic chamber
GTU-RIR	'24	44.1	15k	11	mono	semi-automated rig
HOMULA-RIR	'24	48	52	1	HOA + mic array	model of teleconference
MP-RIR	'24	48	68.7k	1	mic array	robot-automated rig
FLAIR	'25	48	270	1	mono	laser-measured point clouds
SRIRACHA	'25	48	2.65M	1	mono	shoebox, varying absorption
UPV RIR DB	'26	48	166	3	multichannel	hierarchical metadata
<i>Synthetic</i>						
BIRD	'20	16	100k	100k	stereo	ISM shoebox A/B spaced pair
SoundSpaces	'20	16	16M	101	multi-format	scanned scenes; ray tracing
GWA	'22	48	2M	18.9k	mono	drawn scenes; wave+ray
ReverbFX	'25	48	1.85k	—	stereo	singing voice; plugins



An Objective Evaluation of Hearing Aids and DNN-based Binaural Speech Enhancement

In broad terms, HAs are body-worn devices that capture the acoustic scene with one or more microphones, processes it — amplifying and compressing it according to the user’s audiogram, and increasingly applying noise reduction and directional processing — and deliver the result to the ear through a small loudspeaker, conventionally called the *receiver*. Devices are commonly grouped by form factor. In-The-Ear (ITE) HAs house the entire electronics in a custom shell that sits in the concha or ear canal; they are discreet but constrained in size, battery life and microphone spacing. Behind-The-Ear (BTE) HAs instead place the electronics in a unit worn behind the pinna, which routes the amplified sound to the ear through a tube ending in an earmould; the larger housing affords more processing power and a wider spacing between microphones, at the cost of visibility. The Receiver-In-Cannal (RIC) style, now the most widespread, is a variant of the BTE in which the receiver is removed from the housing and placed directly in the ear canal, connected to the behind-the-ear unit by a thin wire and sealed by a soft dome (Figure 3.1). This separation keeps the BTE unit small while moving the loudspeaker close to the eardrum, reducing the risk of acoustic feedback and improving sound quality — which, together with its discreetness, explains its popularity and its choice for the present study.

HAs are the primary clinical remedy for hearing loss. Beyond restoring audibility through frequency-dependent amplification and dynamic range compression, their practical value rests largely on a harder problem: keeping speech in-

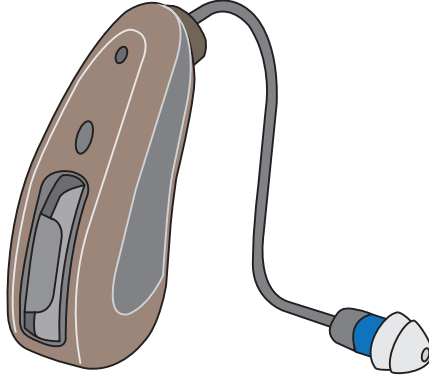


Figure 3.1: Schematic of the *audifon lewi R*, a Receiver-In-Cannal (RIC) HA, the form factor used in this study. The behind-the-ear unit houses the microphones, battery and processor; a thin wire carries the signal to the *receiver*, the miniature loudspeaker placed inside the ear canal, here fitted with a *power dome* — a closed silicone tip that seals the canal to maximise gain and minimise leakage for more severe hearing losses.

telligible amid competing sounds. To improve the SNR, HAs traditionally rely on directional microphones (Ricketts & Dhar, 1999) and adaptive beamformers (Luo et al., 2002; Gode & Doclo, 2022), complemented by single-channel noise reduction. This enhancement is typically validated with stationary noises under controlled laboratory conditions (IEC6011816, 2022). Everyday listening, however — a party, a restaurant, several competing talkers — is non-stationary and spatially complex, precisely the regime these methods handle worst. The resulting mismatch between the conditions under which a device is *evaluated* and those in which it is *used* is a question of ecological validity, and closing it for HAs is the aim of **RO1**.

In parallel, end-to-end DNNs (Section 2.1) have transformed SE, achieving strong denoising even in the monaural case (Reddy et al., 2021a). Their adoption in hearing devices has been constrained by the requirement of causality and by tight compute and latency budgets, but both are being addressed: causal architectures have been proposed (Tzinis et al., 2022) and scaled (Ali et al., 2023), binaural models preserve spatial cues (Han et al., 2020), algorithmic latencies now approach those of conventional HAs processing (Wang et al., 2023b; Stone et al., 2008), and network miniaturisation (Banbury et al., 2021), computation offloading (Sun et al., 2020) and the Clarity Challenge (Akeroyd et al., 2023) together make DNN-based HAs enhancement increasingly plausible. Indeed, some manufacturers already embed DNN post-filtering in commercial products (Andersen et al., 2021; Santurette et al., 2021; Brændgaard, 2020), yet without

disclosing their architectures or any ablation quantifying the gain they bring. Whether DNNs genuinely generalise better than the mature, hand-engineered enhancement of current devices, especially in the complex scenes where the latter struggle, is therefore an open question and the object of **RO2**.

This chapter, carried out as part of the GuestXR project in collaboration with Microson (Amplifon Group), addresses both objectives with a single evaluation setup. **We record five high-end commercial HAs exposed to complex speech-in-noise scenes, reproduced over an Ambisonics loudspeaker array and captured on a Neumann Dummy Head (KU100) dummy head, and we compare their built-in enhancement against the same recordings post-processed by two state-of-the-art DNNs.**

The main contribution is an evaluation which suggests that HAs traditional enhancement struggles under challenging acoustic conditions, and that DNN-based SE has the potential to improve current hearing devices. Along the way, and helping to fill the absence of any public HAs evaluation dataset noted in Chapter 2, we release an Ambisonics-to-binaural decoder tailored to HAs, a synthetic binaural training corpus, and a reverberant speech-in-noise test set with the corresponding recordings of each device.

3.1 Hearing aids measurement setup

We first describe the measurement and evaluation infrastructure, deferring the enhancement strategies it serves to compare to Section 3.2. The purpose here is only to explain how ecologically valid scenes are generated, reproduced, captured on the HAs, and turned into signals for the objective evaluation.

3.1.1 Overview

The setup, depicted in Figure 3.2, is designed to capture the signals a HA actually processes within complex acoustic scenes, and comprises three stages. In *scene generation*, realistic speech-in-noise scenes are built by combining existing databases with room-acoustic simulation, and rendered as sound fields in Ambisonics. In *recording*, these scenes are reproduced over an Ambisonics loudspeaker layout and captured with the microphones of a dummy head wearing each HA, with and without applying the device’s enhancement features. In *evaluation*, the HAs-enhanced recordings are compared against the same signals processed by the DNNs, using a set of objective metrics computed against a clean binaural reference. Next we detail each component of the pipeline.

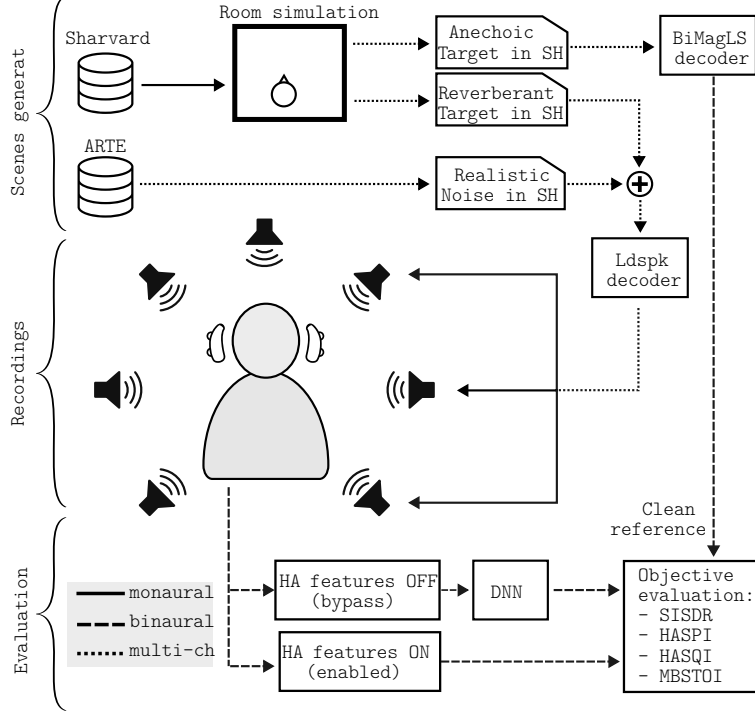


Figure 3.2: Hearing-aid measurement setup. *Scene generation:* complex acoustic scenes are created by combining existing databases (Sharvard, ARTE) with room-acoustic simulation. *Recording:* the material is played back over an Ambisonics reproduction system and the signals processed by the HAs are captured with the microphones of a dummy head, with and without its signal-enhancing features. *Evaluation:* the HAs-enhanced recordings are compared with the DNN-processed ones through a range of objective metrics.

3.1.2 Ambisonics complex scenes generation

To ensure that the sound reaching each device closely resembles real life, all scenes are represented and reproduced as Ambisonics sound fields (Section 2.2). We build three acoustic scenes of speech in noise, corresponding to three common environments: *party*, *restaurant* and *office*. The noise comes from the Ambisonic Recordings of Typical Environments (ARTE) database (Weisser et al., 2019), whose recordings were captured in real environments with a 62-channel microphone array and distributed as 31-channel *mixed-order* Ambisonics.

In a mixed-order representation, the horizontal plane is encoded at a higher spherical-harmonic order than the vertical one, exploiting the fact that both most sources and human localisation are concentrated near the horizontal: full-sphere resolution is kept only up to a low order, and the additional horizontal

detail is carried by the sectorial harmonics ($m = \pm\ell$, the outermost components of the harmonic pyramid in Figure 2.10) alone. ARTE keeps the full 3D field up to 4th order (25 channels) and adds horizontal-only components up to 7th order (6 further channels), for 31 in total. We zero-padded these into a full 10th-order set, placing each component in its ACN slot and filling the remaining, unmeasured harmonics with zeros. The target speech consisted of nine sentences spoken by a female speaker, drawn from the phonemically balanced Spanish Sharvard corpus (Aubanel et al., 2014).

Room acoustics are simulated with an adaptation of the Multichannel Acoustic Signal Processing (MASP) library^{3.1}, a shoebox ISM simulator (see Section 2.3.2) extended to output the field directly in Ambisonics domain. MASP is the translation from Politis (2016)’s MATLAB library into the Python programming language (Pérez López, 2020). Working at 10th order, which affords sufficient spatial resolution, we simulate for each scene two sound fields at the left-ear, right-ear and *head* positions: an anechoic one, propagating only the direct sound, and a reverberant one. The two ear positions were placed 17.5 cm apart to match the KU100 inter-ear distance. The three rooms measured $15 \times 10 \times 3.5$ m, $28 \times 17 \times 4.2$ m and $5 \times 2 \times 2.5$ m for the *party*, *restaurant* and *office* scenes respectively, and the target was placed 1 m from the head at either 0° or 30° to the right. The reverberation times were set by informal listening against the ARTE recordings, choosing T_{60} values around 60% of those reported for ARTE to account for the absorption of furniture and people. The result is a set of clean (anechoic) and reverberant speech and noise fields, all in Ambisonics.

3.1.3 Decoding to the loudspeaker array

Playback took place in Eurecat’s Spatial Audio recording and mixing studio, a three-dimensional irregular array of 25 Genelec 8040 loudspeakers surrounding the listening position. The studio is acoustically treated to keep its own reverberation short, but it is not anechoic (Figure 3.3). The signals captured at the dummy head therefore contain, on top of the reverberation deliberately simulated into each scene, the small residual response of the playback room itself; the acoustic treatment ensures this residual contribution stays well below the intended, simulated acoustics that dominate the reproduced field. To reproduce a scene, the speech and noise sound fields are first mixed at the *head* position, weighting the two so that the binaural signal-to-noise ratio is

^{3.1}<https://github.com/andresperezlopez/masp>

+5 dB once decoded to the ears (the binaural decoder of Section 3.1.5). The resulting Ambisonic mixture is then decoded to the 25 loudspeaker feeds with a fifth-order in-phase decoder (Section 2.2.2) optimised with IDHOA (Scaini & Arteaga, 2014) for this particular irregular layout; the in-phase design suppresses the side lobes of the virtual microphones, trading angular sharpness for robustness away from the centre. Finally, all sets of loudspeaker signals are normalised to equal energy to ease calibration.



Figure 3.3: Eurecat’s Spatial Audio studio: a three-dimensional irregular array of 25 Genelec 8040 loudspeakers arranged in floor, ear level and ceiling.

3.1.4 Dummy-head recordings

A Neumann KU100 dummy head is placed at the centre of the array. Each scene is calibrated to 70 dB Sound Pressure Level (SPL) at the ear position, next to the dummy head, using a Beyerdynamic MM1 omnidirectional measurement microphone (Figure 3.4). We record five high-end Receiver-In-Cannal (RIC) HAs available on the market at the time of writing — the GN ONE 961-DRWC and 561-DRWC, the Phonak Audéo P90-R and P70-R, and the Signia Pure C&G 3x — testing fifteen HAs–receiver combinations in total, with low, mid and high-power receivers for each device. For each recording, the HAs are fitted behind the ears of the KU100 and their receivers are inserted into its ear canals; to minimise the contribution of direct sound leaking past the

device, the ear-canal entrance is occluded with adhesive putty in addition to the HAs’ own power dome. Each scene is recorded twice with every HA. In the first pass the device is set to *bypass*: all signal-enhancing algorithms are deactivated, leaving only the feedback canceller and a linear amplification of approximately 20 dB. Since commercial devices offer no direct access to their microphone signals, these *bypass* recordings at the KU100 serve as an approximation of what the HAs’ microphones capture, and are therefore also the input fed to the offline DNN-based enhancement of Section 3.2. In the second pass the device is set to *enabled*, applying the signal-enhancement algorithms of its default factory settings, which in all tested devices include at least some form of adaptive beamforming and single-channel noise reduction. No hearing correction is applied in either mode.



Figure 3.4: The Neumann KU100 dummy head at the centre of the loudspeaker array, ready for calibration with the Beyerdynamic MM1 measurement microphone.

3.1.5 Ambisonics-to-binaural hearing aids decoder

Returning to the evaluation pipeline of Figure 3.2, the one element missing before being able to evaluate the performance is a way to obtain the clean references against which the metrics are computed. The intrusive metrics below require a clean binaural reference matching what the HAs’ microphones would capture in the absence of noise and reverberation. Since our scenes live in the Ambisonic domain — including the anechoic speech field simulated in MASP

(Section 3.1.2) — obtaining that reference amounts to decoding Ambisonics to binaural, which in turn calls for a set of HRTFs specific to the device-on-head configuration. Although Thiemann & van de Par (2019) published HRTFs for the microphones of a BTE HAs on a head-and-torso simulator, none were available for the KU100, and, evaluating commercial devices, we had no access to their microphone signals. We therefore measured our own HRTFs in a configuration as close as possible to the recording setup: in the same room, with a pair of *audifon lewi R* HAs coupled to the dummy-head ear canals, enhancement features disabled and only a linear gain applied.

The HRTFs were measured with the exponential-sweep method (Section 2.3.1) using a single Genelec 8020 loudspeaker positioned at each of the 50 directions of a *Lebedev grid* (Figure 3.5). Lebedev grids are sets of directions distributed as evenly as possible over the sphere, arranged so that each one has its mirror image across every axis. This symmetry covers all directions uniformly with comparatively few points: the 50 directions comprise the 6 vertices of an octahedron, the 8 vertices of a cube, the 12 edge midpoints between them, and a further symmetric orbit of 24 points. Grouped by elevation, they form horizontal rings whose population grows towards the horizontal plane — a single point at each pole, four at intermediate elevations and eight on the rings nearest and on the horizontal — since the rings near the equator are longer circles and fit more points at an even spacing. This concentrates angular resolution around the horizontal plane, where human localisation is most acute. Such efficiency is convenient for HRTF acquisition, where every added direction means another physical measurement.

The measured impulse responses were windowed before the arrival of the first wall reflection, yielding quasi-anechoic responses, and their low-frequency content was extended with the spherical-model Low Frequency Extension (LFE) of Bernschütz (2013). From these we build a 10th-order Ambisonics-to-binaural decoder following the BiMagLS method (Engel et al., 2021) (Section 2.2.2), with spherical-harmonic tapering above a cutoff of 6239 Hz — the theoretical limit for correct representation at 10th order. The clean binaural reference is then obtained by decoding the anechoic left and right-ear sound fields simulated in MASP. The measured HRTFs are released in the Spatially Oriented Format for Acoustics (SOFA) format (Audio Engineering Society, 2022). SOFA^{3,2}, standardised as AES69, is a self-describing container that stores acoustic impulse responses — here the HRIR pairs — together with the full measurement

^{3,2}we use the Python library from <https://github.com/Jencke/pySOFA>

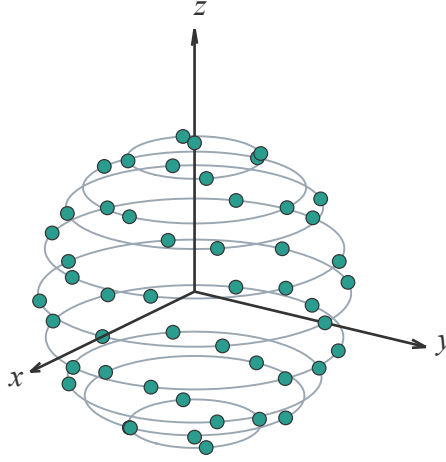


Figure 3.5: The 50-point Lebedev grid of HRTF measurement directions, shown on the unit sphere. The points group into horizontal rings of constant elevation: a single point at each pole, four at the intermediate elevations, and eight on the rings nearest and on the horizontal plane.

geometry (source and receiver positions and orientations) and metadata, so that a dataset can be exchanged and reused across tools without ambiguity.

3.1.6 Evaluation metrics

The objective evaluation compares four sets of recordings: *bypass*, *enabled*, *bypass* post-processed with the non-causal model (**DNN**) and *bypass* post-processed with its causal counterpart (**DNN-C**). The first represents the absence of signal enhancement and the remaining three the different enhancement strategies under comparison.

In a preliminary round we attempted to estimate the output SNR with the phase-inversion procedure of Hagerman & Olofsson (2004). The idea is to present the same scene twice: first the mixture of speech and noise, and then the same mixture with the *noise inverted in polarity*. If the device applies the same processing in both passes, adding the two recordings cancels the noise and leaves twice the processed speech, while subtracting them cancels the speech and leaves twice the processed noise. Having isolated the two components at the output, the output SNR — and hence the improvement over the input — follows directly. The procedure, however, assumes that the system behaves identically across the two presentations; this did not hold for several HAs, whose adaptive processing reacted differently to the polarity-inverted noise,

nor for any of the DNNs, whose non-linear mapping breaks the assumption outright, making the estimates unreliable. We therefore rely on intrusive, reference-based metrics computed against the clean binaural reference y of Section 3.1.5.

Given an enhanced estimate $\tilde{y}$ and the corresponding unprocessed recording $\hat{y}$, we time-align each pair to the reference by cross-correlation and compute four objective measures: the Hearing-Aid Speech Quality Index (HASQI) (Kates & Arehart, 2014), the Hearing-Aid Speech Perception Index (HASPI) (Kates & Arehart, 2021), the Modified Binaural Short-Time Objective Intelligibility (MBSTOI) (Andersen et al., 2018) and the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) (Le Roux et al., 2019). For the non-binaural measures (SI-SDR, HASPI, HASQI) we take the better ear and normalise the signals as in Akeroyd et al. (2023), using a flat normal-hearing audiogram as input to HASPI and HASQI. Because HASPI and HASQI rest on a level-dependent auditory model, they require a calibration relating digital amplitude to presentation level; both reference and estimate are scaled so that their root mean square match the 70 dB SPL at which the scenes are reproduced, rather than the 65 dB SPL assumed by default for a signal with Root Mean Square of one. The scale-invariant SI-SDR is defined as,

$$\text{SI-SDR}(\tilde{y}, y) = 10 \log_{10} \frac{\|\alpha y\|^2}{\|\alpha y - \tilde{y}\|^2}, \quad \alpha = \frac{\tilde{y}^\top y}{\|y\|^2}, \quad (3.1)$$

where the optimal scaling α makes the measure invariant to the overall gain of the estimate.

The remaining three measures are intrusive quality and intelligibility predictors specific to this domain. HASQI and HASPI were designed to predict, respectively, perceived *quality* and *intelligibility* for both normal-hearing and hearing-impaired listeners: each passes the reference and the estimate through an auditory model that incorporates the listener’s audiogram — here a flat normal-hearing one — and compares the two along cepstral and temporal-envelope features, returning a score in $[0, 1]$ where 1 denotes a signal indistinguishable from the reference. MBSTOI (Andersen et al., 2018) is the binaural extension of the short-time objective intelligibility measure: it correlates the short-time temporal envelopes of reference and estimate within frequency bands and across the two ears — thereby rewarding the preservation of interaural cues — and likewise yields a score in $[0, 1]$, monotonically related to predicted intelligibility. Unlike SI-SDR, which is a generic signal-fidelity ratio in decibels, these three are perceptually motivated and bounded, which is why

we report them alongside it.

Because absolute metric values conflate the enhancement with the difficulty of the scene, we report the *signal-enhancement benefit*: the improvement an enhancement brings over the unprocessed recording. For SI-SDR, for instance,

$$\text{SI-SDR}_i = \text{SI-SDR}(\tilde{y}, y) - \text{SI-SDR}(\hat{y}, y), \quad (3.2)$$

and ΔHASQI , ΔHASPI and ΔMBSTOI are defined analogously. A positive benefit means the enhancement improves on doing nothing; a negative one means it degrades the signal.

3.2 DNN-based binaural speech enhancement models

The recording procedure described so far already enables to evaluate the HAS' own SE. Our aim, however, is to compare it against a prototype DNN-based post-filter, so we now turn to training a binaural SE model. We approach it as a *supervised* learning problem (Section 2.1.1), which requires a large collection of noisy, reverberant mixtures paired with their corresponding clean-speech targets. Both inputs and targets are *binaural*, so that the model can exploit the spatial cues a hearing-aid user relies on (Section 2.2.1) instead of destroying them. Additionally, both must resemble the signals that the two frontal microphones of a hearing-aid device would capture, which is precisely what the HAS BiMagLS decoder of Section 3.1.5 provides.

3.2.1 Speech and noise material

Following common practice in the SE literature (Reddy et al., 2021a; Tzinis et al., 2022), the clean speech comes from audiobook recordings: specifically the Spanish subset of Multilingual LibriSpeech Studio (MLS) (Pratap et al., 2020), chosen to match the language of the Sharvard sentences used in the evaluation scenes. All material is resampled to 16 kHz and split into four-second chunks; for each utterance we retain the chunk with the highest energy, which avoids training on silence. This yields 221k utterances for training, 2408 for validation and 2385 for testing, amounting to some 251 hours of clean speech as shown in Table 3.1.

The noise comes from WHAM! (Wichern et al., 2019), a binaural corpus of real ambient recordings — babble, cafeteria noise and background music — captured in urban environments, and thus a good match for the everyday

Table 3.1: Binaural HA DNN-based SE training data splits and noise augmentations. # denotes the number of four-second utterances; Φ_{inv} denotes polarity inversion of the noise, $L \leftrightarrow R$ the permutation of its channels, and *stretch* factor.

set	MLS		WHAM!		noise augmentations		
	#	hours	#	hours	Φ_{inv}	$L \leftrightarrow R$	stretch
<i>train</i>	221k	245.6	52k	57.8			
			52k	57.8	✓		
			52k	57.8		✓	
			52k	57.8	✓	✓	
			6.5k	7.2			✓
			6.5k	7.2	✓	✓	✓
<i>valid</i>	2.4k	2.67	2.4k	2.67			
<i>test</i>	2.4k	2.67	2.4k	2.67			

situations the HAs are evaluated in. We keep the original splits of both corpora, which preserve gender balance and prevent speaker contamination across sets. Because WHAM! is considerably smaller than MLS, its training split is augmented to match the total duration of the speech material through three strategies: (i) inverting the polarity of the noise signal, (ii) swapping its left and right channels, a rough approximation of a 180° rotation of the noise field in the horizontal plane, and (iii) time-stretching it by a random factor between 90% and 110%. Combining them yields the six subsets of Table 3.1, whose 245.6 hours match the speech training set exactly. Validation and test noise is drawn, unaugmented, from the corresponding WHAM! splits.

3.2.2 Binaural room simulation

MLS consists of close-microphone studio recordings, which we take as a fair approximation of anechoic signals and therefore as suitable dry sources for simulation. Each training example is then built by placing a speech source and the listener inside a randomly drawn shoebox room, simulating the corresponding sound fields with MASP (Section 3.1.2) and decoding them to binaural with the HAs BiMagLS decoder of Section 3.1.5 — the very same pipeline used to generate the evaluation scenes, keeping training and test data consistent.

The random configurations are summarised in Table 3.2. Two of its choices deserve comment, as both encode acoustic common sense rather than independent sampling. The room depth r_y is drawn *relative* to its width r_x , which prevents the sampler from generating implausible, corridor-like geometries.

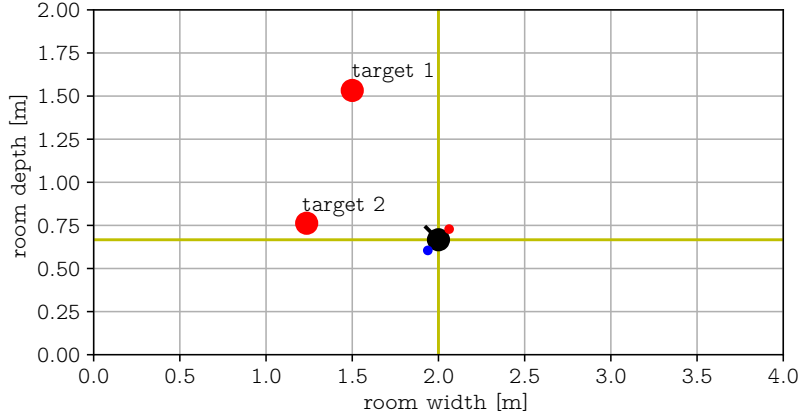


Figure 3.6: Top-down view of an example MASP room simulation. The head (black), with its left and right ears (blue and red) and its facing direction, is placed inside a randomly drawn shoebox room; the yellow lines mark its position along the room width and depth. Two speech sources (*target 0* and *target 1*, red) are positioned around it. The fields at the two ear positions are simulated in Ambisonics and decoded to binaural to build each training example.

Table 3.2: Random room configuration. Each parameter is drawn from a uniform distribution $\mathcal{U}$. Room dimensions r , *head* position and distances are in metres, SNR in dB, T_{60} in seconds, and the *head* azimuth θ , elevation ψ and angle to the target in degrees. The room depth r_y is drawn relative to the width r_x to avoid corridor-like rooms, and T_{60} decreases with SNR, since noisier rooms are also more crowded and hence more absorptive.

Variable	Description	Distribution
r_x	room width [meters]	$\mathcal{U}(3, 30)$
r_y	room depth [meters]	$r_x \cdot \mathcal{U}(0.5, 1)$
r_z	room height [meters]	$\mathcal{U}(2.5, 5)$
$head_x$	head position along x [meters]	$\mathcal{U}(0.35 r_x, 0.65 r_x)$
$head_y$	head position along y [meters]	$\mathcal{U}(0.35 r_y, 0.65 r_y)$
$head_z$	head position along z [meters]	$\mathcal{U}(1, 2)$
$head_\theta$	head orientation azimuth [degrees]	$\mathcal{U}(-45, 45)$
$head_\psi$	head elevation [degrees]	$\mathcal{U}(-10, 10)$
$\ head - target\ $	head to target distance [meters]	$\mathcal{U}(0.5, 3)$
$\angle head, target$	head to target angle [degrees]	$\mathcal{U}(-45, 45)$
SNR	signal-to-noise ratio [dB]	$\mathcal{U}(0, 6)$
T_{60}	reverberation time [seconds]	$\mathcal{U}(0.1, 0.5) \frac{SNR + 0.3}{5.3}$

More subtly, the reverberation time is made to depend on the SNR: noisier scenes are assigned shorter T_{60} values, on the grounds that a room crowded enough to be noisy is also a room whose occupants absorb a substantial part of its reverberation. Sampling the two independently would produce many physically implausible combinations — a loud, crowded room with a long, empty-hall decay — and training on them would waste capacity on situations the model will never meet.

One further correction is applied. Decoding through our own measured HRTFs imprints on every training example the particular colouration of the *audifon lewi R* device coupled to the KU100 ear canal — an artefact of our measurement rather than a property of HAs. We compensated the response of the RIC coupling by taking the direction-independent frequency response between the HRTFs of the bare KU100 and those of the KU100 wearing the HAs, approximating it with an Infinite Impulse Response (IIR) filter, and applying that filter to every utterance in the dataset.

3.2.3 Training setup

As the enhancement network we adopted *SuDo-RM-RF* (Tzinis et al., 2022), a well-tested architecture originally designed for source separation that has also performed well on HAs SE (Lee et al., 2022; Akeroyd et al., 2023). It follows the mask-based, time-domain paradigm of Conv-TasNet (Luo & Mesgarani, 2019) from Section 2.1.2: a learned one-dimensional convolutional encoder maps the waveform into a latent representation, a separator estimates a mask over it, and a decoder reconstructs the enhanced waveform, thus avoiding the magnitude–phase compromise of spectrogram-domain models. Its main building block is the U-ConvBlock, which — like the encoder-decoder structure of the U-Net of Section 2.1.1 — successively downsamples its input to extract features at multiple temporal resolutions and then resamples and sums them back, so that a wide temporal context is aggregated with far fewer parameters and operations than a deep stack of dilated convolutions would require. This efficiency is what makes the architecture attractive for a HAs setting.

We train two variants. *DNN* is the non-causal configuration named *SuDoRM-RF++GC* on their paper, which may look into future samples and therefore serves as an upper baseline on what the architecture can achieve. *DNN-C* is its causal counterpart, *C-SuDoRM-RF++*, in which convolutions and normalisation only access present and past samples; it is the variant that could conceivably run on a HAs device. Beyond configuring the encoder to accept

two-channel input, the topology was left unmodified.

Both models use 256 input and 512 output channels across 16 successive U-ConvBlocks with five downsampling/upsampling layers each, encoder and decoder kernels of size 21 producing embeddings of length 512, and four attention heads of depth 256 with a dropout of 0.1 applied during training only. All hyperparameters are shared between the two models except for the batch size, which had to be reduced from 12 for *DNN-C* to 2 for *DNN* to fit in GPU memory. We optimize with Adam at a learning rate of 10^{-3} , divided by three every eight epochs, for 25 epochs over the whole dataset. Fresh mixtures are generated at every epoch by permuting the reverberant speech and the noise within each batch, which effectively works as an additional augmentation, as no pairing is ever seen twice, and each mixture is normalised to zero mean and unit variance. We use the negative SI-SDR as the training loss and SI-SDR itself as the validation metric, reaching 11.7 dB on the test split.

3.3 Results

Figure 3.7 displays SI-SDR, HASPI, HASQI, and MBSTOI metrics obtained for each hearing device averaged across receiver types. In the *bypass* condition metrics range from -12.63dB to -8.01dB for SISDR, 0.57 to 0.73 for HASPI, 0.14 to 0.16 for HASQI, and 0.39 to 0.49 for MBSTOI. Enabling signal enhancement features in HAs can have a different effect on their performance depending on the specific device. For example, device 4 demonstrates improvement of 0.73dB on SISDR, 0.14 on HASPI, 0.02 on HASQI and 0.01 on MBSTOI while device 2 presents -0.54dB SISDR, -0.17 HASPI, -0.07 HASQI and -0.05 MBSTOI. Apart from these slight deviations, the overall impact of HA features is typically insignificant. In contrast, a much larger improvement can be observed in recordings processed with the DNNs. For the non-causal DNN, values reach -6.89dB to -1.21dB for SISDR, 0.60 to 0.77 HASPI, 0.14 to 0.21 for HASQI and 0.57 to 0.69 for MBSTOI.

Figure 3.8 depicts the signal enhancement benefit. Violin plots summarize the values obtained from the 15 hearing devices averaged along all scenes. For HAs' features the benefit is on average 0.014 dB for SI-SDR, -0.02 for HASQI, -0.01 for HASPI, and -0.01 for MBSTOI. A much larger improvement could be observed for DNN-based enhancement. For a causal model, the mean benefit was 4.96 dB for SI-SDR, -0.01 for HASQI, 0.02 for HASPI, and 0.15 for MBSTOI. The largest benefit was achieved by the non-causal DNN model, with

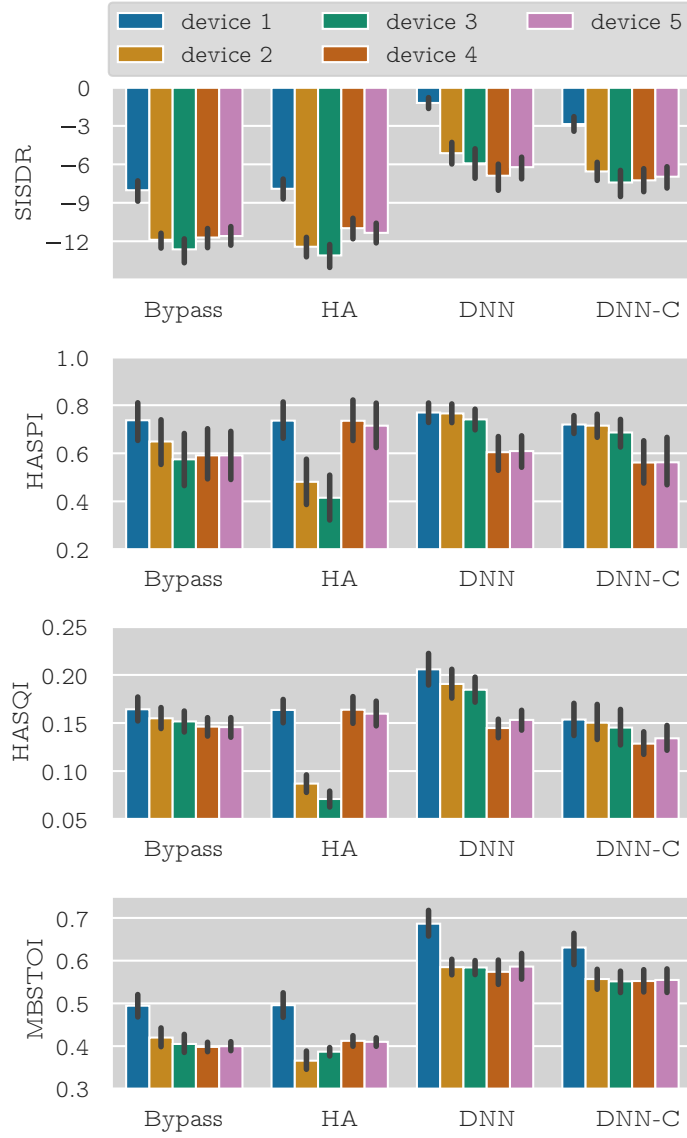


Figure 3.7: Objective metrics across devices, anonymized for avoiding any drawal of inappropriate interdevice conclusions.

mean values reaching 6.09 dB for SI-SDR, 0.02 for HASQI, 0.07 for HASPI, and 0.17 for MBSTOI.

3.4 Discussion

In this chapter we have presented a setup which allows to quantify the performance gap between the current HAs signal enhancement strategies and DNN-

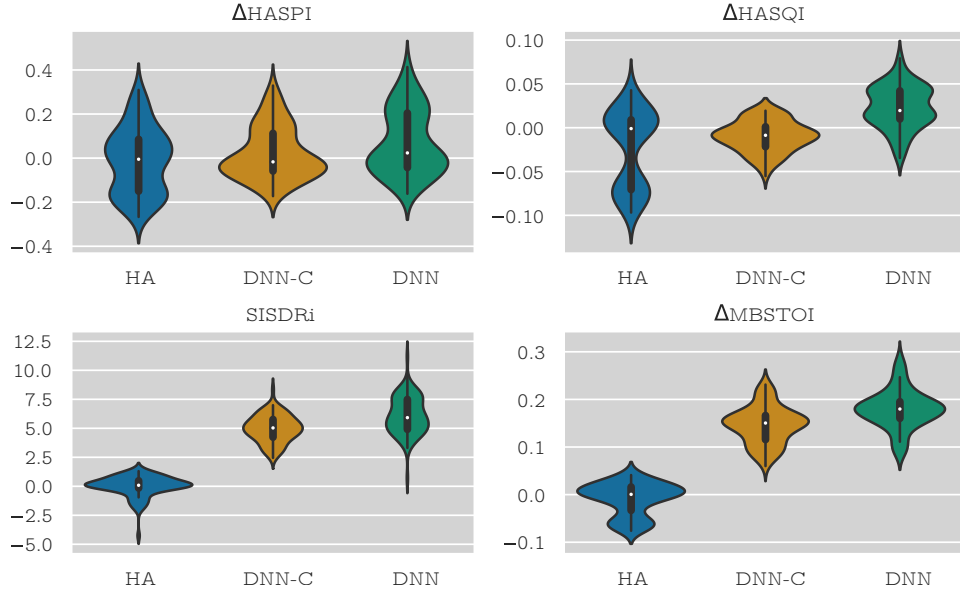


Figure 3.8: Signal enhancement benefit for HAs, non-causal DNN model (*DNN*) and causal DNN model (*DNN-C*). The corresponding mean values are collected in Table 3.3.

Table 3.3: Mean signal enhancement benefit across the 15 HAs-RIC combinations, averaged along all scenes, as depicted in Figure 3.8.

	<i>SISDRi</i> (dB)	ΔHASQI	ΔHASPI	ΔMBSTOI
HAs	0.014	−0.02	−0.01	−0.01
DNN-C	4.96	−0.01	0.02	0.15
DNN	6.09	0.02	0.07	0.17

based approaches. On the one hand, HA processing seems to struggle in these complex situations, even performing worse than in *bypass* for some devices, perhaps because beamformers cannot estimate the direction of the target speech due to the noise and competing talkers being non-stationary. On the other hand, this does not seem to affect DNNs, which improve *SISDR* and *MBSTOI* consistently. Interestingly, differences between devices in *bypass* are still observed after being processed by the DNNs, suggesting that DNN models are sensitive to the quality of the inputs. It would be interesting to study the concatenation of DNNs with traditional HAs strategies in future work.

We acknowledge that using the binaural decoding of the anechoic signal as reference can seem counter-intuitive because HAs are not designed to provide anechoic estimates yet. However, this reference was the one that matched better with our informal listening. Another limitation of the study is that we are

comparing real time HAs processing with DNNs applied as a post-processing, because we found it difficult to obtain signal insert points in consumer HA devices and also because no real time implementations of these models are publicly available yet. However, the slight performance difference between *DNN* and *DNN-C* and both being far better than the HA in terms of SI-SDR and MBSTOI should additionally encourage this research direction.

3.5 Follow-up work

The evaluation presented here has since been complemented by a perceptual study, published at Forum Acusticum Euronoise 2025 (Baig et al., 2025) and carried out by Martí Baig as part of his Master’s thesis. Being his contribution, it lies beyond the scope of this dissertation, and we summarise it here only for completeness. Using the same recordings and models, it runs a Multiple Stimuli with Hidden Reference and Anchor (MUSHRA)-like listening test with two groups of twenty participants — HAs users with moderate-to-severe hearing loss and normal-hearing listeners — rating the six processing conditions (*bypass*, *enabled*, and the causal and non-causal DNNs) across the six scenes, with the *bypass* recording as the hidden reference. Crucially, it adds a *mild* variant of both networks, trained on *pseudo-anechoic* targets that remove the noise while preserving some reverberation, in contrast to the *strong* variants studied in this chapter, whose fully anechoic targets suppress it entirely.

Both groups prefer the mild DNN models over every other condition — the non-causal mild model ranking first, and the causal one, the only variant that could run on a device, close behind — while the strong variants of this chapter are rated lowest of all, criticised for overly aggressive suppression that sacrifices naturalness and spatial perception. The HAs’ own *enabled* processing scores only moderately and with high variability. Statistically, the robust effect is precisely this contrast between mild and strong enhancement, rather than a clean separation between conventional and DNN processing; and a re-computation of the objective metrics with each listener’s own audiogram broadly tracks these preferences. Two conclusions carry back to the present work. First, the subjective results support the objective finding that DNN enhancement has the potential to improve on current HAs processing in complex scenes. Second, they indicate that the anechoic reference adopted here, flagged among the limitations in Section 3.4, is indeed more aggressive than HAs users would choose: a pseudo-anechoic target that leaves some reverberation intact appears to be the more ecologically sensible design, and a promising direction for the

speech-enhancement work of the following chapter.

3.6 Conclusions

This chapter has approached HAS SE as a problem of *ecological validity*: conventional devices are certified under stationary, laboratory conditions, yet users struggle precisely where everyday listening is non-stationary and spatially complex. By reproducing such scenes over an Ambisonic loudspeaker array and capturing them on real devices, we evaluated under conditions closer to real use, and could compare five HAS devices against DNN-based SE.

Regarding **RO1**, this ecologically valid evaluation exposed a clear performance gap: the signal-enhancement algorithms of the five devices struggle in complex, real-world speech-in-noise scenes, in several cases degrading the signal relative to applying no enhancement at all.

Regarding **RO2**, we have shown that DNN-based Speech Enhancement (SE) has the potential to outperform them substantially in denoising and objective intelligibility, and that the causal variant retains most of that advantage. Together, these findings encourage further work on the optimisation and pruning of such models for on-device deployment, and on their subjective validation with hearing-impaired listeners, like the follow-up study of Section 3.5.

Beyond the evaluation itself, the chapter contributes the infrastructure that made it possible, released as research materials^{3.3}: a set of HRTFs measured on a KU100 wearing a RIC HA, the corresponding Ambisonics-to-binaural BiMagLS decoder, a synthetic binaural training corpus for HA SE, and a reverberant speech-in-noise test set in Ambisonics and binaural formats, together with the recordings of each device.

^{3.3}https://github.com/enricguso/guso_waspaa23



Synthetic Room Impulse Responses for Monaural Speech Enhancement

The previous chapter highlights, for the specific case of HAs, the importance of the data a model is trained and evaluated on. This chapter carries the same idea to general-purpose SE, again within the context of the GuestXR project, and again keeping the focus on one particular ingredient of its training data: the Room Impulse Responses (RIRs) used to simulate reverberation. As we saw in Chapter 3, while HAs are beginning to adopt DNNs, they do so under tight latency and power constraints. For normal-hearing listeners in settings such as videoconferencing, where those constraints are far looser, DNN-based SE has instead thrived, to the point that neural noise suppression now runs in virtually every videoconferencing application. This success has been driven in no small part by ever larger and more diverse training sets, such as those of the Deep Noise Suppression (DNS) challenges (Dubey et al., 2024).

Scaling the amount and variety of data has produced models that generalise better and narrow the cross-dataset performance gaps discussed in Chapter 2 (Richter et al., 2023; Kadioğlu et al., 2020), and extending speech and noise material to a 48 kHz sampling rate has improved results further (Schröter et al., 2023; Zhang et al., 2023, 2024). Again, the standard recipe for injecting acoustic diversity is to convolve clean speech with a RIR, placing it in a variety of simulated rooms. The scaling and development of RIR datasets themselves, however, have not kept pace. As noted in the dataset review of Chapter 2, most SOTA systems still train on the RIRs of the early DNS challenges (Ko et al., 2017): shoebox rooms rendered with the Image Source Method (ISM)

(Section 2.3.2), at a 16 kHz sampling rate, and with a *single* absorption coefficient shared across the whole frequency spectrum. In fact, the RIRs of the DNS challenge were originally assembled for ASR rather than enhancement.

Increasing the realism of the RIRs has already been shown to help related tasks such as ASR for augmented-reality glasses (Arakawa et al., 2024) or keyword spotting (Bezzam et al., 2020), even when applied only as a post-processing augmentation (Ratnarajah et al., 2021), so the same should hold for monaural general-purpose SE. This raises the question of *which* properties of a synthetic RIR actually matter for SE generalisation to real acoustics (**RO3**).

Other recent efforts point in the same direction: the URGENT challenge (Zhang et al., 2024) pushes SE towards *universality* and *robustness*, bringing together denoising, dereverberation, declipping, bandwidth extension and other distortions in a single task, spanning a wide range of sampling rates, and, tightly related to our approach, evaluating on real recorded RIRs rather than only synthetic ones, which highlights the importance of ecological validity. However, the URGENT challenge still trains on the ten-year-old RIRs from DNS5. Generative augmentation of existing RIRs has been explored in Lin et al. (2025).

As we introduced in Chapter 2, some recent RIR datasets use path tracing over 3D meshes of realistic geometries, mainly for tasks beyond monaural SE such as audio-visual navigation (Kelley et al., 2024; Tang et al., 2022). Its best-known instance is the SoundSpaces dataset (Chen et al., 2022), which provides a dense grid of receiver positions within each room. However, it spans comparatively few distinct rooms, and how models trained on its RIRs perform on the SE task remains unclear. GWA (Tang et al., 2022) is likewise mesh-based and does report dereverberation and separation experiments, but only uses intrusive metrics on a very limited number of rooms. Whether such geometrically realistic yet scarce mesh-based RIRs are preferable to abundant but simpler shoebox ones is still not well understood (**RO4**).

To tackle both **RO3** and **RO4**, we evaluate six models trained on the same speech and noise datasets but on six different RIR datasets (two baselines, three of our own making and one set of SoundSpaces), with the intention of answering whether models benefit from using RIRs with frequency dependent (multiband) absorption coefficients, whether modeling the receiver directivity or the source directivity helps to generalize in monaural models, and whether mesh-based modeling might be helpful or more suitable than ISM (Bezzam et al., 2020). To this end, we propose an evaluation of the following RIRs:

- **DNS5**: the State-of-the-Art (SOTA) baseline from Ko et al. (2017).
- **SB**: recreation of the DNS5 at 48 kHz sampling rate using **single-band** absorption coefficients on the shoebox room material.
- **MB**: same as SB but using **multi-band** absorption coefficients instead.
- **REC+MB**: same as MB but adding **receiver** directivity by using HRTFs at the rendering stage.
- **SRC+REC+MB**: adding **source** directivity by modelling average human speech directivity in the ISM.
- **SSPA**: we use mesh-based RIRs from **SoundSpaces** instead.

Keeping the speech, the noise and the room geometries fixed, we train one SE model on each one of the RIR datasets (which are summarized in Table 4.1) and evaluate it on *real* RIRs, both objectively and through a listening test. We find that frequency-dependent absorption is always a beneficial feature, and we release the resulting multiband RIR dataset, *MB-RIRs*, as well as the multiband with source and receiver directivities, *SRC+REC+MB-RIRs*, as freely available research materials^{4.1}.

Table 4.1: The six synthetic RIR datasets under comparison and their sampling rate, absorption coefficients, receiver directivity, source directivity and rendering method. The first two are baselines, the next three are our own making, and the last one is a pre-computed mesh-based.

	f_s	Absorption coeff.	Receiver Dir.	Source Dir.	Renderer
DNS5	16 kHz	single	✗	✗	ISM
SB	48 kHz	single	✗	✗	ISM
MB	48 kHz	multi	✗	✗	ISM
REC+MB	48 kHz	multi	✓	✗	ISM
SRC+REC+MB	48 kHz	multi	✓	✓	ISM
SSPA	48 kHz	multi	✓	✗	mesh-based

^{4.1}<https://doi.org/10.5281/zenodo.15773093>

4.1 Training datasets

It is important to stress that the six models evaluated in this chapter differ *only* in the RIRs used to reverberate their training data; the clean speech and the noise are identical across all of them, so that any difference in performance can be attributed to the room acoustics alone. For both we use the entire DNS5 corpus (Dubey et al., 2024), a large and deliberately heterogeneous collection: emotional speech, English audiobook readings in a range of accents, singing voice from VocalSet, multilingual speech (French, Spanish, Italian, Russian and German from M-AILABS, German from Wikipedia, and the Spanish OpenSLR sets SLR73, SLR61, SLR39, SLR75, SLR74 and SLR71), together with noise from FreeSound and AudioSet. In total this amounts to 583k speech utterances (1.315 hours) and 63k noise samples (177 hours). We split both into 70% for training and the remaining 30% divided equally between validation and test, avoiding speaker contamination across splits wherever the metadata allowed it. The rest of this section describes the six RIR sets built on top of this shared material.

4.1.1 Baseline RIRs (DNS5)

Our baseline is the RIR set of the DNS5 challenge itself (Ko et al., 2017). These RIRs come from two companion collections released with that work: the roughly 60k *synthetic* shoebox responses of OpenSLR26, rendered with the Image Source Method (ISM) (Section 2.3.2) for small, medium and large rooms, and the 325 *real* measured responses of OpenSLR28, gathered from the RWCP, REVERB and Aachen (AIR) databases and spanning everything from small meeting rooms to large auditoria. Both were originally distributed at 16 kHz; we take 60k of them, distribute the small, medium and large rooms into the same 70%/15%/15% training, validation and test splits used throughout, and upsample everything to 48 kHz for compatibility with the other sets. This baseline carries the two limitations that motivate the rest of the chapter: it is effectively band-limited by its original 16 kHz rendering, and its shoebox responses use a *single* absorption coefficient across the whole spectrum.

4.1.2 Single-band absorption coefficients RIRs (SB)

The DNS5 baseline has two limitations at the same time: it is band-limited (rendered at 16 kHz) and uses single-band absorption coefficients. To isolate the first, and to obtain a controlled counterpart to our own datasets,

we re-create an approximation of it at 48 kHz while keeping the single absorption coefficient. We generate 60k shoebox RIRs with the MASP library (Pérez López, 2020; Politis, 2016), the same ISM simulator used for the hearing-aid scenes of Chapter 3. Comparing the resulting **SB** set with DNS5 therefore isolates the effect of sampling rate alone.

The random room configuration is summarised in Table 4.2. Crucially, it is drawn *once* and reused *identically* across all four of our ISM-based datasets (SB, MB, REC+MB and SRC+REC+MB): the same rooms, the same receiver and source placements, and the same reverberation times. The four sets thus differ only in the acoustic feature under study — the number of absorption bands and the receiver and source directivity — so that any change in performance can be attributed to that feature alone rather than to a different sampling of geometries. Only the pre-computed SoundSpaces set (SSPA) uses its own rooms, as it cannot share this configuration.

Within each room, the dimensions $\mathbf{r}$ are drawn from uniform distributions, with the depth r_y made relative to the width r_x to avoid implausible, corridor-like geometries. The receiver $\mathbf{rec}$ is placed at random, away from the walls, and the source $\mathbf{src}$ around it, between 0.5 and 3 m away and roughly in front. The single absorption coefficient of the six walls then follows from Sabine’s formula (Section 2.3.2) given $\mathbf{r}$ and a target reverberation time T_{60} . For consistency with the multiband case of Section 4.1.3 — where $\mathbf{T}_{60}$ is a vector, one value per frequency band — the scalar T_{60} used here is the mean of that vector, which turns out to be well approximated by a normal distribution.

Table 4.2: Random room configuration shared by the ISM-based datasets (SB, MB, REC+MB, SRC+REC+MB), sampled from the uniform distribution $\mathcal{U}$ (and, for the reverberation time, the Gamma and normal distributions). Dimensions, positions and distances are in metres and angles in degrees; bold symbols are vectors.

Variable	Description	Distribution
r_x	room width	$\mathcal{U}(3, 30)$
r_y	room depth	$r_x \cdot \mathcal{U}(0.5, 1)$
r_z	room height	$\mathcal{U}(2.5, 5)$
$\mathbf{rec}_x$	receiver position along x	$\mathcal{U}(0.35 r_x, 0.65 r_x)$
$\mathbf{rec}_y$	receiver position along y	$\mathcal{U}(0.35 r_y, 0.65 r_y)$
$\mathbf{rec}_z$	receiver height	$\mathcal{U}(1, 2)$
$\mathbf{rec}_\phi$	receiver azimuth	$\mathcal{U}(-45, 45)$
$\mathbf{rec}_\theta$	receiver elevation	$\mathcal{U}(-10, 10)$
$\ \mathbf{rec} - \mathbf{src}\ $	receiver–source distance	$\mathcal{U}(0.5, 3)$
$\angle \mathbf{rec}, \mathbf{src}$	receiver–source angle	$\mathcal{U}(-45, 45)$
$\mathbf{T}_{60}$	per-band reverberation time	$\text{Gamma}(\boldsymbol{\alpha}, \boldsymbol{\beta})$
T_{60}	broadband reverberation time	$\frac{1}{N} \sum_n \mathbf{T}_{60} \approx \mathcal{N}(0.4, 0.014)$

4.1.3 Multiband absorption coefficients RIRs (MB)

The first and central of our three ISM strategies is to replace the single broadband absorption coefficient of SB with a *frequency-dependent* one, so that each wall absorbs differently across the spectrum, as real materials do.

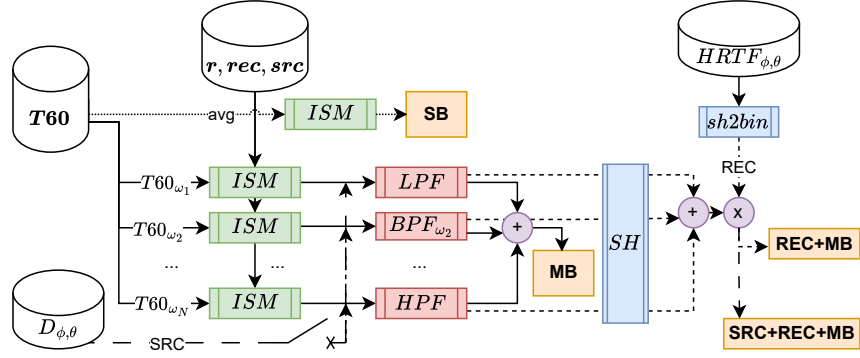


Figure 4.1: The RIR generation pipeline. All ISM-based datasets share the same geometric configurations. The dotted line is the single-band path (SB); the continuous line adds the per-band absorption and filterbank of MB; the short-dashed line adds receiver directivity (REC+MB) and the long-dashed line adds the source directivity (SRC+REC+MB).

Concretely, instead of a single scalar T_{60} we now sample a vector $\mathbf{T}_{60}$ of reverberation times, one per frequency band, over the six octave bands $\omega = \{125, 250, 500, 1k, 2k, 4k\}$ Hz, as illustrated in Figure 4.1. To keep these vectors realistic rather than arbitrary, we fit their distribution to measured data: we analysed 4.495 real $\mathbf{T}_{60}$ values from Sridhar et al. (2021) and modelled each band as an independent Gamma distribution, $T_{60} \sim \text{Gamma}(\alpha, \beta)$, fitting the shape α and scale β per band by minimising the negative log-likelihood. This yields

$$\alpha = \{1.72, 1.62, 1.93, 2.56, 4.17, 2.49\}, \quad (4.1)$$

$$\beta = \{0.39, 0.24, 0.14, 0.10, 0.09, 0.18\}, \quad (4.2)$$

so that drawing one sample from each band gives a physically plausible reverberation profile. As Figure 4.2 shows, samples drawn from these fitted distributions closely match the measured $\mathbf{T}_{60}$ values in every band. From it, the per-band absorption coefficients of the six walls follow from Sabine’s formula (Section 2.3.2) as in the single-band case.

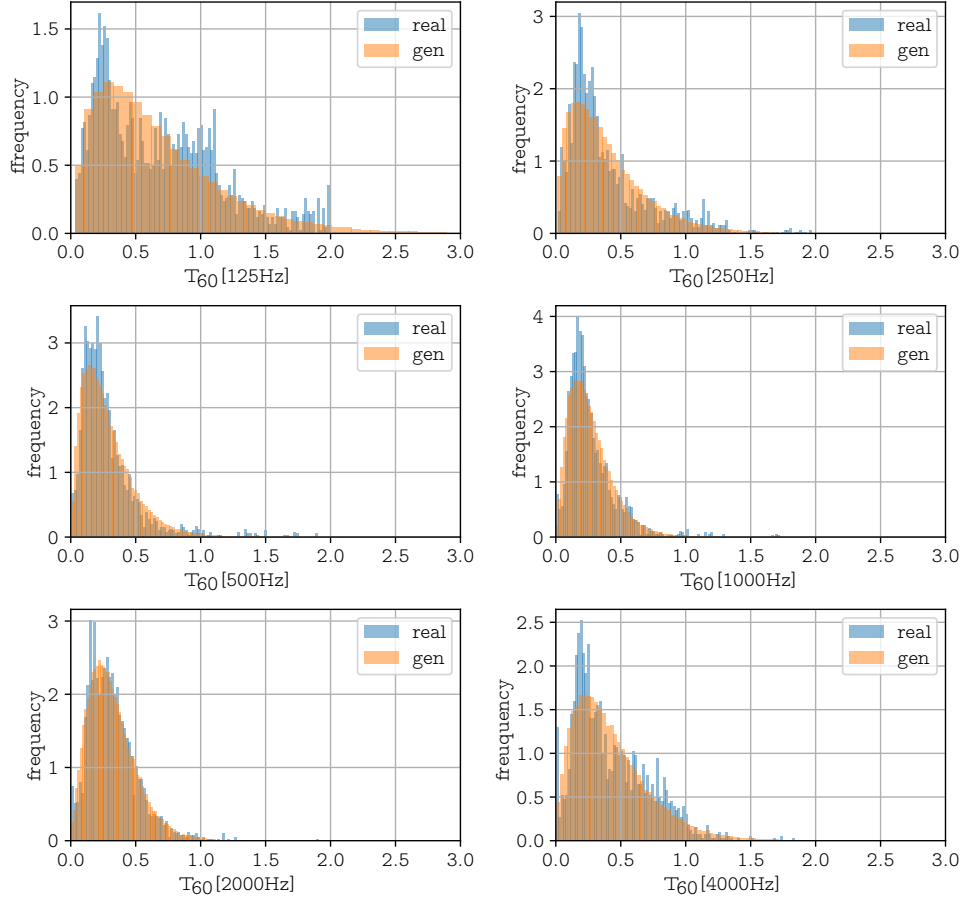


Figure 4.2: Per-band reverberation-time distributions: the 4.495 measured T_{60} values of Sridhar et al. (2021) (*real*) and those sampled from our fitted Gamma models (*gen*), for the six octave bands from 125 Hz to 4 kHz. The good overall agreement in every band confirms that the independent per-band Gamma fits reproduce the statistics of real rooms.

Because our ISM renderer assumes a single absorption per surface, we render one RIR per band and recombine them afterwards. For each band n we run an independent ISM simulation with that band’s absorption, and then sum the resulting sub-band RIRs h_n after passing each through the matching filter of an N -band filterbank —a low-pass for the lowest band, a band-pass for the intermediate ones and a high-pass for the highest:

$$h = \text{LPF}_{\omega_1} * h_1 + \sum_{n=2}^{N-1} \text{BPF}_{\omega_n} * h_n + \text{HPF}_{\omega_N} * h_N. \quad (4.3)$$

The result is a single broadband RIR whose decay differs across frequency, at the cost of N simulations per room instead of one.

4.1.4 Receiver directivity RIRs (REC+MB)

The second strategy models the directivity of the *receiver*. In the ISM above, both source and receiver are omnidirectional points, but the microphones of a real capture device are not: when the device is worn on or near the head, they inherit its acoustic shadow, so that a reflection arriving from one side is coloured and attenuated depending on its direction of arrival.

Our motivation for modelling the head shadow directivity is that beyond its synthetic training material, DNS5 provides two special test sets of *real* noisy recordings without ground truth — and thus evaluated only with non-intrusive metrics — grouped by capture device: a *headset* set, recorded with devices worn on the head such as in-ear and headset microphones, and a *speakerphone* set, recorded with hand-held or desk devices such as laptops and mobile phones. Since our measured HRTFs approximate the directivity of a head-worn microphone, an enhancement model trained on them should match the acoustics of the *headset* set more closely, and we would therefore expect it to improve on that set relative to a model trained without receiver directivity. Arakawa et al. (2024) indeed found this head-and-device directivity to be relevant for speech separation and recognition with augmented-reality glasses.

We model it by applying a HRTF to every reflection of the simulated RIR, reusing the pipeline of the hearing-aid decoder of Chapter 3. Since the RIRs produced by MASP are already expressed in the spherical-harmonic domain, decoding them to the ear with a Bilateral Magnitude Least Squares (BiMagLS) decoder (Engel et al., 2021) (Section 2.2.2) implicitly convolves each directional component with the corresponding HRTF. As the task is monaural, we restrict the decoding to the *left* ear, using a set of normal-hearing HRTFs of a Neumann

KU100 dummy head. Applied on top of the multiband set, this yields the **REC+MB** dataset.

4.1.5 Source directivity RIRs (SRC+REC+MB)

The third strategy models the directivity of the *source*. Unlike the general case, in SE the source is always speech, whose radiation is far from omnidirectional: the human head and torso make the voice progressively more *frontal* with frequency, so that a listener behind or to the side of the talker receives markedly less high-frequency energy. We take the average speech radiation pattern of Leishman et al. (2021), shown per octave band in Figure 4.3, and turn it into an azimuth–elevation lookup table $D_{\phi,\theta}$ giving the relative radiated level in each direction and band.

We apply this pattern *within* the ISM, before the final rendering, as shown in the pipeline of Figure 4.1. At that stage the simulation has produced an *echogram*: the list of reflections reaching the receiver, each with its arrival time, amplitude and direction. Modelling source directivity amounts to re-weighting the amplitude of every reflection in this list by $D_{\phi,\theta}$, according to the direction in which it *left the source*. The difficulty is that the echogram stores each reflection’s direction *as seen from the receiver*, not its emission angle at the source. We recover the latter through acoustic reciprocity (Samarasinghe et al., 2017): we run a second ISM simulation with the source and receiver positions swapped, whose echogram now lists the reflections’ directions at the (original) source location. Reading the emission angle of each reflection from this reciprocal echogram, we weight the corresponding reflection in the original one by the matching value of $D_{\phi,\theta}$, and only then proceed to the filterbank rendering of Section 4.1.3. Applied on top of the receiver-directivity set, this yields the **SRC+REC+MB** dataset.

4.1.6 Mesh-based RIRs (SoundSpaces, SSPA)

Rather than another strategy of our own, our last dataset is an external reference: the most recent and complex synthetic RIR dataset we could find at the time, against which to contrast our ISM-based sets. It is the one most directly comparable to our REC+MB set because it uses multi-band absorption coefficients and renders in Ambisonics and decodes to binaural, while additionally offering the geometric realism that a shoebox cannot. To keep the comparison fair, we take 60k of its RIRs and, as in REC+MB and SRC+REC+MB, use the left channel of the binaural downmix.

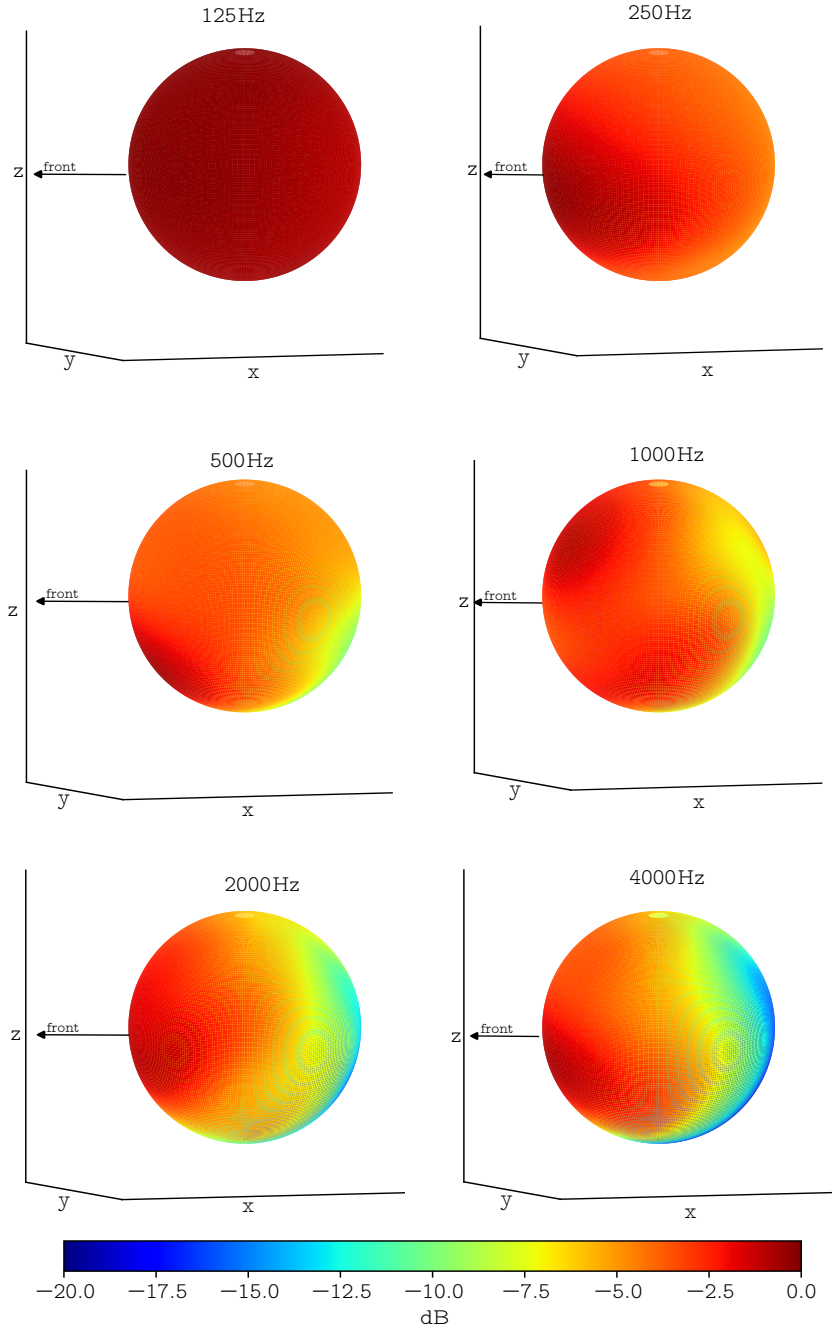


Figure 4.3: Average human-speech radiation pattern $D_{\phi,\theta}$, per octave band from 125 Hz to 4 kHz (Leishman et al., 2021). Each sphere shows the radiated level (in dB, relative to the frontal direction $+x$) as a function of direction. Radiation is essentially omnidirectional at low frequencies and becomes increasingly frontal with frequency, the rear and lateral directions being attenuated by up to about 20 dB at 4 kHz.

The crucial difference lies not in how each response is rendered, but in the number and acoustic diversity of the rooms available: SoundSpaces spans only 103 scanned scenes, whereas our ISM-based sets draw tens of thousands of independent room configurations, sampling a far wider range of sizes, geometries and per-band reverberation times. Whether the geometric realism of a mesh outweighs this loss of acoustic coverage is precisely **RO4**.

4.2 Experimental setup

Given the shared speech and noise and the six RIR datasets described above, we train six Speech Enhancement (SE) models — one per dataset — and evaluate them in three complementary ways. First, we apply each model to monaural noisy utterances convolved with *real* RIRs and compute a battery of intrusive and non-intrusive metrics (Section 4.2.1). Second, we validate a subset of these perceptually through a Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) listening test (Section 4.3.1). Third, we apply the models to the *headset* and *speakerphone* test sets of DNS5 to probe the effect of receiver directivity (Section 4.3.3); as these contain real mixtures without a separate clean-speech ground truth, only non-intrusive metrics can be computed there.

As the enhancement network we use DeepFilterNet3 (Schröter et al., 2023), chosen because it combines state-of-the-art performance with an open-source, real-time implementation — the same properties that make it a plausible candidate for deployment. All six models share its default hyperparameters, except for a smaller maximum batch size of 38 to fit our GPUs and a reverberation probability $p_{\text{reverb}} = 1$, so that a RIR is applied to every utterance. Each model is trained on its own dataset for 117 to 120 epochs, depending on early stopping.

4.2.1 Objective evaluation

The objective evaluation measures the SE performance of each model with a large set of metrics, over two kinds of signals, both of which involve *real* room acoustics. The first are *synthetic* mixtures, in which clean speech and noise are combined and convolved with real RIRs; here a clean reference is available, so both intrusive and non-intrusive metrics can be computed. The second are *real recordings* (the *headset* and *speakerphone* sets of DNS5 introduced in Section 4.1.4) already captured as noisy mixtures in real rooms, and therefore carrying real acoustics by construction; as they provide no separate clean

reference, only non-intrusive metrics apply. In both cases the room acoustics are real, so the evaluation measures how well each model, trained on *synthetic* RIRs, generalises to *real* ones.

For the synthetic mixtures, we take from the DNS5 test split 10k high-quality read-speech samples — 7k from VCTK (Veaux et al., 2017) and 3k from LibriVox (Panayotov et al., 2015) — and 397 real RIRs gathered from the ACE (Eaton et al., 2015), MIT IR Survey (Traer & McDermott, 2016), OpenAIR (Murphy & Shelley, 2010) and BUT ReverbDB (Szöke et al., 2019) datasets, together with the OpenSLR28 (Ko et al., 2017) responses not used in training. Each clean four-second chunk y is convolved with a real RIR h and mixed with a noise sample n at $\text{SNR} = \mathcal{U}(0, 30)$ dB, giving the noisy, reverberant input x .

Two methodological points deserve mention. First, preserving the time alignment between x and its target y proved less trivial than expected: the onset of a *real* RIR is harder to detect than that of a clean synthetic one, so rather than the widely used correlation-based synchronisation — which we found less robust to the noise in real responses — we search the vicinity of the peak amplitude for any earlier sample exceeding a -6 dB threshold, and align to that. Second, the conventional signal model $x = (y * h) + n$ leaves the noise dry while the speech is reverberant; we therefore also ran the whole evaluation with $x = (y + n) * h$, in which speech and noise share a spectrally coherent reverberation at the cost of placing them at the exact same point in space (a physically infeasible simplification). The results were very similar, so for brevity we report only the conventional model.

The metrics we report fall into four families. *Signal-fidelity* measures compare the estimated waveform with the reference on a sample level: the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) (Section 3.1.6) and the plain Signal-to-Distortion Ratio (SDR), in decibels. *Spectral* measures compare their time–frequency representations: the Log-Spectral Distance (LSD) and the Mel-Frequency Cepstral Distance (MSD), both distances for which lower is better. *Perceptual* measures predict human judgements of quality and intelligibility from an auditory model: wideband Perceptual Evaluation of Speech Quality (PESQ) (roughly $[1, 4.5]$) and Short-Time Objective Intelligibility (STOI) ($[0, 1]$). Finally, *semantic* measures, less common in SE, gauge whether linguistic and speaker content survive enhancement: a phonetic similarity, a speaker-embedding cosine similarity, and a Bidirectional Encoder Representations from Transformers (BERT)-based textual similarity between transcripts.

Because some of these rely on a clean reference and others do not, we group them into *intrusive* metrics — computed against the reference y — and *non-intrusive* ones, computed from the estimate alone. The latter are increasingly estimated by neural networks trained to mimic human ratings: we use Deep Noise Suppression Mean Opinion Score (DNSMOS, Reddy et al., 2021b) and Non-Intrusive Speech Quality Assessment (NISQA) (Mittag et al., 2021), which predict Mean Opinion Scores (MOS), and the Speech Quality and Intelligibility Measures (SQUIM) estimators (Kumar et al., 2023), which predict reference-free versions of SI-SDR, PESQ, STOI and MOS directly from the noisy signal (denoted with the subscript _{squim}). Most of these follow the metric set of the URGENT challenge (Zhang et al., 2024).

As in Chapter 3, and because reverberation itself can bias several of these measures, we report each metric as an *increment* over the unprocessed noisy, reverberant signal rather than as an absolute value — denoting it SI-SDR_i for the SI-SDR and Δm for the remaining metrics m . For a model f with estimate $\hat{y} = f(x)$, non-intrusive metrics are computed as $m(\hat{y}) - m(x)$ and intrusive ones, which need the reference y , as $m(\hat{y}, y) - m(x, y)$. On the *headset* and *speakerphone* real recordings, where no clean reference exists, only the non-intrusive metrics apply and we report their absolute values.

For context, our evaluation deliberately reports *increments* on unseen real RIRs rather than absolute scores on a standard benchmark. On the widely used Voicebank+Demand benchmark, DeepFilterNet reaches strong figures: a PESQ of 3.17 and STOI of 0.944 for the DeepFilterNet3 version we use (Schröter et al., 2023, Table 1)— good results for a real-time model, so we start from a strong baseline. Our concern here is not to improve on those benchmarks but to test how well such a model generalises to real room acoustics depending on the RIRs it was trained on.

4.3 Results and discussion

The results confirm three findings, which we take in turn: extending the sampling rate to 48 kHz already helps; frequency-dependent absorption (MB) brings the clearest benefit; and neither source/receiver directivity nor mesh-based rendering brings a clear further gain, the latter being outweighed by the smaller variety and quantity of rooms.

4.3.1 Subjective results

Figure 4.4 shows the MUSHRA scores; the means are collected, alongside the objective metrics, in Table 4.3. We compare conditions with pairwise t -tests at a significance threshold of $p < 0.05$. Two baselines frame the picture. First, SB significantly outperforms the 16 kHz DNS5 baseline ($p = 3 \times 10^{-3}$), confirming that extending the sampling rate to 48 kHz already improves perceived quality. Second, DNS5 and SSPA are statistically indistinguishable ($p = 0.55$): despite modelling complex geometries and furniture, and despite having the same number of RIRs as SB, the mesh-based set spans far fewer distinct rooms, indicating that the variety and quantity of rooms matter on top of the complexity of each response.

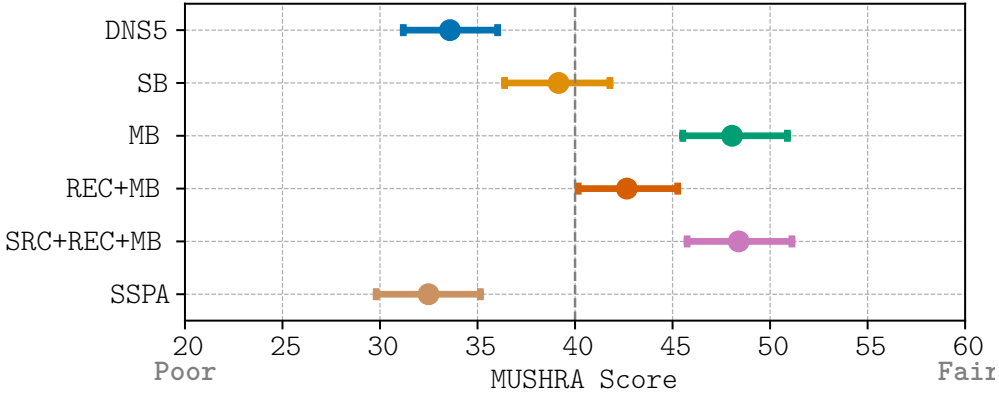


Figure 4.4: MUSHRA scores (mean and 95% confidence interval) for the six models. Higher is better.

Every dataset except SSPA scores significantly above the DNS5 baseline (SB: $p = 3 \times 10^{-3}$; MB: $p = 7 \times 10^{-14}$; REC+MB: $p = 8 \times 10^{-7}$; SRC+REC+MB: $p = 2 \times 10^{-14}$), so that both the higher sampling rate and the added acoustic features improve enhancement. The best scores go to MB (48.05) and SRC+REC+MB (48.39), which are themselves indistinguishable ($p = 0.86$); this last equality is our first piece of evidence that modelling directivity, while it does not help here, at least does not *degrade* monaural SE on real RIRs.

4.3.2 Objective results on real RIRs

The objective metrics on the same real-RIR test set, in Table 4.3, broadly echo the listening test: MB RIRs tend to outperform the rest. There is, however, no full consensus. Most intrusive metrics favour MB and SRC+REC+MB, in line with the subjective scores, but SI-SDR, MCD and phonetic similarity

Table 4.3: Evaluation on real RIRs with DNS5 read speech (mean $\pm$ standard deviation). Higher is better ($\uparrow$) except for LSD and MCD ($\downarrow$). SQUIM, NISQA and DNSMOS are non-intrusive neural estimators. Best value per row in bold.

		DNS5	SB	MB	REC+ MB	SRC+REC+ MB	SSPA
<i>Intrusive</i>							
SI-SDRi	$\uparrow$	1.105 $\pm$ 3.50	1.335 $\pm$ 3.49	1.361 $\pm$ 3.45	1.965 $\pm$ 4.52	1.900 $\pm$ 4.41	0.888 $\pm$ 3.27
Δ SDR	$\uparrow$	2.817 $\pm$ 3.17	3.298 $\pm$ 3.02	3.328 $\pm$ 3.00	2.674 $\pm$ 3.19	2.706 $\pm$ 3.17	2.025 $\pm$ 3.11
Δ LSD	$\downarrow$	1.213 $\pm$ 1.41	1.340 $\pm$ 1.42	1.260 $\pm$ 1.40	0.841 $\pm$ 1.43	0.778 $\pm$ 1.43	1.105 $\pm$ 1.43
Δ MCD	$\downarrow$	-1.434 $\pm$ 2.00	-1.479 $\pm$ 1.98	-1.504 $\pm$ 1.99	-1.542 $\pm$ 1.97	-1.533 $\pm$ 1.98	-1.454 $\pm$ 2.00
Δ PESQ	$\uparrow$	0.520 $\pm$ 0.47	0.583 $\pm$ 0.47	0.593 $\pm$ 0.47	0.595 $\pm$ 0.46	0.609 $\pm$ 0.46	0.510 $\pm$ 0.45
Δ STOI	$\uparrow$	0.088 $\pm$ 0.07	0.101 $\pm$ 0.07	0.102 $\pm$ 0.07	0.101 $\pm$ 0.06	0.101 $\pm$ 0.06	0.082 $\pm$ 0.06
Δ PhonSim	$\uparrow$	0.051 $\pm$ 0.15	0.058 $\pm$ 0.15	0.058 $\pm$ 0.15	0.063 $\pm$ 0.15	0.063 $\pm$ 0.15	0.061 $\pm$ 0.15
Δ SpkSim	$\uparrow$	-0.055 $\pm$ 0.13	-0.058 $\pm$ 0.14	-0.057 $\pm$ 0.13	-0.034 $\pm$ 0.12	-0.030 $\pm$ 0.11	-0.053 $\pm$ 0.13
Δ BertSim	$\uparrow$	0.081 $\pm$ 0.07	0.086 $\pm$ 0.07	0.087 $\pm$ 0.07	0.089 $\pm$ 0.07	0.090 $\pm$ 0.07	0.082 $\pm$ 0.07
<i>Non-intrusive</i>							
SI-SDRi _{squim}	$\uparrow$	3.793 $\pm$ 5.87	4.456 $\pm$ 5.71	4.544 $\pm$ 5.67	3.814 $\pm$ 5.65	4.038 $\pm$ 5.63	2.291 $\pm$ 5.61
Δ MOS _{squim}	$\uparrow$	0.547 $\pm$ 0.71	0.537 $\pm$ 0.71	0.541 $\pm$ 0.71	0.523 $\pm$ 0.70	0.506 $\pm$ 0.71	0.541 $\pm$ 0.70
Δ MOS _{dnsmos}	$\uparrow$	0.884 $\pm$ 0.53	0.935 $\pm$ 0.54	0.937 $\pm$ 0.53	0.895 $\pm$ 0.53	0.906 $\pm$ 0.53	0.811 $\pm$ 0.52
Δ MOS _{nisqa}	$\uparrow$	1.158 $\pm$ 0.73	1.198 $\pm$ 0.73	1.224 $\pm$ 0.72	1.109 $\pm$ 0.72	1.099 $\pm$ 0.72	0.977 $\pm$ 0.73
Δ PESQ _{squim}	$\uparrow$	0.636 $\pm$ 0.62	0.667 $\pm$ 0.59	0.678 $\pm$ 0.59	0.613 $\pm$ 0.58	0.637 $\pm$ 0.58	0.525 $\pm$ 0.57
Δ STOI _{squim}	$\uparrow$	0.060 $\pm$ 0.08	0.070 $\pm$ 0.08	0.071 $\pm$ 0.08	0.063 $\pm$ 0.08	0.066 $\pm$ 0.08	0.048 $\pm$ 0.08
<i>Subjective</i>							
MUSHRA	$\uparrow$	33.59 $\pm$ 20.3	39.15 $\pm$ 22.6	48.05 $\pm$ 22.8	42.65 $\pm$ 22.5	48.39 $\pm$ 23.0	32.49 $\pm$ 21.7

instead point to REC+MB. The non-intrusive metrics agree more closely, with MB slightly ahead ($p < 0.05$) — possibly a bias of the NISQA and SQUIM predictors, though, interestingly, one that favours MB rather than the simpler SB. Overall, the multiband sets behave very similarly to one another and clearly above DNS5 and SSPA, so the objective and subjective evidence agree on the main point: multiband absorption is the change that brings the clearest benefit, with MB and SRC+REC+MB at the top and indistinguishable from one another.

4.3.3 Directivity and the device-specific sets

The *headset* and *speakerphone* sets were meant to isolate the effect of directivity, under the hypothesis that receiver-directivity models (REC+MB, SRC+REC+MB) should match the head-worn *headset* recordings more closely.

Table 4.4 shows this hypothesis is not borne out. On both sets, most non-intrusive metrics again put the MB models marginally ahead, as on the real-RIR test, but the differences are not statistically significant (e.g. $p = 0.9$ for

Table 4.4: Non-intrusive evaluation on the real DNS5 *headset* (HDS) and *speakerphone* (SPK) recordings. Higher is better; best per row in bold.

		DNS5	SB	MB	REC+MB	SRC+REC+MB	SSPA
SI-SDR _{squim}	HDS	15.65	15.75	15.80	15.35	15.42	14.21
	SPK	16.81	16.86	17.05	16.65	16.82	15.51
MOS _{squim}	HDS	3.92	3.89	3.90	3.91	3.94	3.91
	SPK	4.15	4.15	4.15	4.14	4.14	4.14
MOS _{dnsmos}	HDS	3.01	3.04	3.05	3.04	3.02	3.01
	SPK	3.03	3.05	3.06	3.03	3.04	2.99
MOS _{nisqa}	HDS	3.54	3.64	3.68	3.61	3.56	3.43
	SPK	3.77	3.82	3.82	3.77	3.76	3.62
PESQ _{squim}	HDS	2.58	2.59	2.62	2.55	2.54	2.45
	SPK	2.64	2.63	2.64	2.56	2.59	2.49
STOI _{squim}	HDS	0.92	0.92	0.92	0.92	0.92	0.91
	SPK	0.94	0.94	0.94	0.94	0.94	0.93

SI-SDR_{squim} between MB and SB), and REC+MB and SRC+REC+MB show no advantage on the *headset* set. We therefore find no evidence of a benefit from directivity modelling.

Although Arakawa et al. (2024) showed that directivity can be exploited even by monaural models, we could not reproduce that effect — whether because of the limitations of the NISQA and SQUIM predictors or of the DNS5 device sets themselves. This leaves open a question we cannot settle here: do MB RIRs help because they are more *domain-consistent* with real rooms, or simply because they are more *diverse* than SB (Jeon et al., 2024)? Directivity does not degrade performance and may yet help, but on this evidence we cannot claim a clear improvement.

4.3.4 Summary

The results recommend training SE models on varied, high-sampling-rate multiband RIRs, optionally adding source and receiver directivity depending on the use case. This answers **RO3** in the affirmative for frequency-dependent absorption — which benefits modern SE as it had benefited keyword spotting — and more cautiously for directivity, which does no harm but shows no clear gain. It answers **RO4** in the negative: geometric realism does not, on its own, compensate for a loss of room diversity, as the amount and variety of RIRs prove as important as the complexity of each one. We release the two most useful sets, MB and SRC+REC+MB, as the *MB-RIRs* dataset.

4.4 Conclusions

This chapter has asked which properties of a synthetic RIR improve the *ecological validity* of the training data — and thereby the generalisation of monaural SE models to real room acoustics — and has answered it by training a state-of-the-art, real-time model (DeepFilterNet3) on six RIR datasets that differ in a single, controlled feature at a time.

Regarding **RO3**, extending the RIRs with frequency-dependent absorption coefficients — an idea previously shown to help keyword spotting — also benefits modern SE, and is the change that brings the clearest and most consistent improvement across objective and subjective evaluation. Extending the sampling rate from 16 to 48 kHz already helps on its own. Source and receiver directivity, by contrast, neither degrade performance nor yield a clear gain: they may help, but on our evidence we cannot claim a definite improvement, and the hypothesis that receiver directivity would benefit the head-worn *headset* recordings was not borne out.

Regarding **RO4**, geometric realism does not, on its own, compensate for a smaller variety and quantity of rooms: the mesh-based SoundSpaces set, despite modelling complex geometries, did not outperform the far more numerous ISM shoebox rooms, and was statistically indistinguishable from the baseline. The amount and variability of the RIRs thus prove as important as the complexity of each individual response: for the ecological validity of a RIR dataset, *coverage* of the space of real rooms weighs as heavily as the physical fidelity of any single response — a theme that runs through this thesis.

One limitation of this answer to **RO4** is that the mesh-based condition confounds two factors: SoundSpaces is at once geometrically realistic and limited to few rooms, so its poor performance cannot be attributed to the mesh-based rendering alone. Disentangling them would require a mesh-based dataset that also offers a large acoustic variety, such as GWA (Tang et al., 2022), which renders millions of responses over tens of thousands of rooms with a hybrid wave-based and ray-tracing simulator (Chapter 2). Repeating our evaluation with GWA in place of SoundSpaces would test whether mesh-based rendering helps once the loss of room variety is removed; we leave this comparison for future work.

In conclusion, improving the ecological validity of the training RIRs measurably improves SE generalisation to real rooms. We recommend training on varied, 48 kHz multiband RIRs, optionally adding source and receiver directiv-

ity depending on the use case, and we release the two most useful datasets, MB and SRC+REC+MB, as *MB-RIRs*, freely available as research material.^{4.2}

^{4.2}<https://doi.org/10.5281/zenodo.15773093>



Extending Music Source Separation to Live Recordings with Audience Noise and Concert-Venue Impulse Responses

The two previous chapters improve the ecological validity of training data for speech; this final chapter brings the same idea to music, in line with the research interests of the Music Technology Group (MTG), by proposing a new task: adapting existing Music Source Separation (MSS) datasets to a new use case, live recordings made from the audience.

MSS — decomposing a musical mixture into its constituent sources, conventionally *vocals*, *drums*, *bass* and *other* — has advanced rapidly since the standardisation of the task (Rafii et al., 2019; Stöter et al., 2019), with successive architectures (Rouard et al., 2023; Luo & Yu, 2023; Tong et al., 2024) and community challenges (Fabbro et al., 2024b; Mitsufuji et al., 2022) steadily improving separation quality. Yet almost all of this progress rests on a single assumption: that the sources mix *linearly*, as they do in the studio. That assumption breaks down for *live* music recorded from the audience — the kind of phone recording anyone attending a concert might make — where three factors reshape the signal before it reaches the microphone: the venue acoustics, the response of the Public Address (PA) loudspeaker system, and the noise of the crowd itself. Applied to such recordings, studio-trained models

degrade sharply, and the few prior works in live settings are confined either to a specific tradition (Plaja-Roglans et al., 2023) or to music enhancement without separation (Kandpal et al., 2022).

The situation mirrors one already met in speech separation, where the WHAM! (Wichern et al., 2019) and WHAMR! (Maciejewski et al., 2020) datasets bridged an analogous gap by augmenting the clean WSJ0-2mix corpus with ambient noise and reverberation. We follow the same idea for music: we *formulate* a new Music Information Retrieval (MIR) task, which we call *Live* MSS — the separation of music recorded from within a live audience, where venue acoustics, the PA system and crowd noise break the linear-mixing assumption of studio MSS, as illustrated in Figure 5.1. Rather than relying on the scarce and hard-to-annotate real live recordings, we *synthesise* live conditions on top of existing studio datasets — exactly the ecological-validity-through-synthesis strategy of the rest of this thesis.

Importantly, our use of *live* refers to the *acoustic conditions* of music captured from within a live audience — venue reverberation, PA colouration and crowd noise — and *not* to latency. It is therefore unrelated to the parallel line of work on *real-time*, low-latency MSS for streaming and on-stage use (Venkatesh et al., 2024), where *live* instead denotes low-latency processing: our task concerns *where* the recording was made, not *how fast* it is separated, and applies whether separation is performed online or offline.

Concretely, and targeting *audience-side* recordings degraded by crowd noise and venue acoustics, this chapter contributes:

- **CrowdioSet**, an *audience* noise dataset: a curated selection of real ambience and event recordings from Freesound (Anastasopoulou et al., 2025) (**RO6**), together with a generative pipeline that produces synthetic *sing-alongs* by zero-shot singing-voice conversion of the vocals in existing datasets (**RO5**);
- **PaRIRset**, a stereo RIR dataset measured across 40 professional concert venues with a microphone array, the first captured in real venues through their PA systems (**RO7**); and
- a *Live* MSS baseline, obtained by re-training the open-source SCNet model (Tong et al., 2024) on studio data augmented with both datasets;
- the formulation of *Live* MSS as a new MIR task, together with a reproducible recipe for synthesising its training and evaluation data from

studio corpora;

- We release the datasets, code, model weights and audio examples publicly.

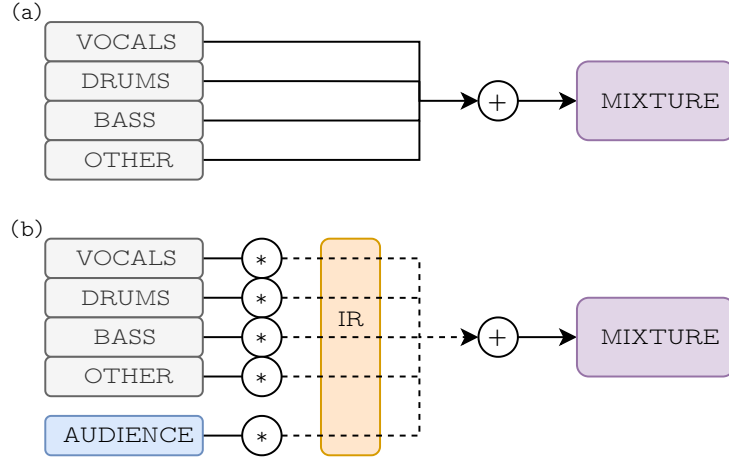


Figure 5.1: We propose to extend (a) traditional four-source Music Source Separation (MSS), in which *vocals*, *drums*, *bass* and *other* are assumed to mix linearly, to (b) *Live* MSS for recordings from the audience. Two ingredients are added to approximate an audience-side recording: an extra *audience* source, capturing crowd noise and sing-alongs, and a convolution of every source with a concert-venue RIR, which introduces the room reverberation and PA colouration absent from studio mixtures. Both are applied before the sources are summed into the mixture that the model receives.

The rest of the chapter is organised as follows. Section 5.1 reviews the background specific to MSS. Sections 5.2 and 5.3 present CrowdioSet and PaRIRset, Section 5.4 the experimental setup, and Section 5.5 the results; Sections 5.6 and 5.7 discuss limitations and conclude.

5.1 Music source separation

Having motivated the task, we now review the models and data that define it. Modern separators fall into three broad families (Section 2.1): waveform-domain models such as the Demucs line (Rouard et al., 2023), spectrogram-domain models such as the Band-Split RNN (Luo & Yu, 2023), and hybrid models that operate in both, such as SCNet (Tong et al., 2024) — the backbone we adopt in this chapter for its strong performance and open-source availability.

Around this core, recent work has broadened the task in several directions: generative approaches to separation (Plaja-Roglans et al., 2025), lightweight

models for real-time or on-device use (Hung et al., 2025; Venkatesh et al., 2024), revisited evaluation metrics that better track perception (Jaffe & Burgoyne, 2025; Sheridan & Benetos, 2025), and a move beyond the fixed *vocals/drums/bass/other* taxonomy towards universal (Postolache et al., 2023) and text-prompted separation (Shi et al., 2025), the latter of which we include as a baseline.

For all this progress, the models are almost invariably trained and evaluated on *studio* material, where the stems mix linearly and cleanly. Public corpora reflect this: MUSDB18-HQ (Raffi et al., 2019) and the larger MOISESDB (Pereira et al., 2023) both provide clean, close-miked studio stems, with no venue acoustics, no loudspeaker colouration and no audience. A parallel diagnosis motivates the recent Music Source Restoration task (Zang et al., 2025), which observes that MSS treats a mixture as a plain sum of sources, ignoring the equalisation, compression and reverberation applied during music *production*, and instead models it as a degraded sum of degraded sources to be recovered. Live music, and in particular the audience-side recordings that motivate this chapter, violates the linear-mixing assumption in a different way, through the acoustics and noise of the venue rather than the studio production chain. The few works that venture there remain narrow: Plaja-Roglans et al. (2023) addresses separation for Carnatic music, and Kandpal et al. (2022) enhances degraded music without separating it, using general-purpose RIRs. As noted in Chapter 2, no public dataset captures the acoustics of concert venues, nor the noise of a crowd — the two gaps this chapter sets out to fill. Related efforts underline the value of doing so: the Cadenza challenges (Roa-Dabike et al., 2025) have shown that MSS can assist HAs users, a use case that extends naturally to the live setting.

5.2 CrowdioSet: an audience noise dataset

The first ingredient of a live recording that a studio dataset lacks is the *audience*. Through informal listening to concert recordings made from the crowd, we identified three categories of audience sound: *ambiences* — the room tone and background chatter that are almost always present; *events* — punctual reactions such as applause, cheering and shouts of approval or disapproval; and *sing-alongs* — passages where the crowd sings the lyrics together with the performer. CrowdioSet accordingly has two parts: a curated selection of real Freesound recordings covering *ambiences* and *events*, which is the core of our contribution and directly addresses **RO6**, and a preliminary generative

pipeline for *sing-alongs* (**RO5**), released as a first step towards the hardest of the three categories (Section 5.6).

5.2.1 Ambiences and events

We retrieved 11,375 files from Freesound (Anastasopoulou et al., 2025) with the queries *crowd*, *audience*, *cheering*, *applause*, *chatter* and *protest*, restricted to clips shorter than eight minutes and to permissive licences (CC0, CC-BY 3.0/4.0, CC-BY-NC 3.0/4.0 and Sampling+ 1.0), resampling each to 44.1 kHz 16-bit stereo. After manually discarding recordings that could not plausibly occur at a concert (traffic, nature or machinery), 4,819 files remained, which we labelled as either *ambiences* or *events* as detailed in Table 5.1, amounting to 105 hours of audience noise.

Table 5.1: CrowdioSet query distribution across the *ambiences* and *events* categories (105.12 hours in total).

	<i>ambiences</i>		<i>events</i>	
	#files	hours	#files	hours
crowd	1,997	73.26	1,398	10.40
audience	262	8.91	690	3.65
chatter	105	4.01	10	0.04
cheering	15	0.72	119	1.10
applause	0	0.00	182	1.70
protest	30	1.21	11	0.13
Total	2,409	88.11	2,410	17.02

Per-file attribution and licence metadata are released with the dataset, and because most files (92%, of which 59% are CC0) permit commercial use, the 8% under non-commercial licences can be excluded without retraining from scratch. To align with the source-separation corpora, we reserve 50 files per category for testing and 14 for validation.

5.2.2 Synthetic sing-alongs

Sing-alongs cannot be curated in the same way: isolated recordings of a crowd singing a known song, cleanly separated from the performance, essentially do not exist. We therefore *synthesise* them from the *vocals* stems already present in the source datasets, turning a single lead vocal into a plausible crowd singing along, through the pipeline of Figure 5.2. For each of the $N = 384$ *vocals* stems

in MUSDB18-HQ and MOISESDB we combine two techniques. The first is the Antares AVOX Choir plugin, a choir effect (vibrato, detuning and delay) in its 32-voice configuration. The second, and the core of **RO5**, uses a pool of $I = 200$ vocal samples retrieved from Freesound (queries *a cappella* and *vocals*) to convert each vocals stem with the pre-trained zero-shot voice-conversion model HQ-SVC (Bai et al., 2026): half of the conversions preserve the original f_0 contour (in tune, but timbrally similar to the lead) and half adapt it to the range of the zero-shot sample (timbrally diverse, but less in tune).

A realistic crowd should be *timbrally diverse* yet only *loosely coupled* to the lead melody, so we rank the conversions by a score balancing these two criteria. For each candidate v_i and the original stem v_o we compute a chroma mean absolute error $d_i = \text{MAE}(\Phi(v_i), \Phi(v_o))$, where $\Phi(\cdot)$ is the 12-bin chromagram (a 2048-sample Hann window, hop 512), and a singer-identity similarity $s_i = \cos(\Psi(v_i), \Psi(v_o))$, where $\Psi(\cdot)$ is the singer embedding of Torres et al. (2023). A small d_i means the candidate agrees with the lead on pitch content, while a small s_i means a *distinct* vocal identity; standardising both across the $N \cdot I$ pool and sorting by $\arg \min(\tilde{d}_i + \tilde{s}_i)$ favours voices that are in tune yet timbrally different from the lead.

The best 64 candidates are then humanised independently to break their unnatural synchrony. Using onset detection (McFee et al., 2025) on v_o to locate syllables, we perturb each syllable’s gain by $\mathcal{U}(0.3, 1)$ and its timing by delays of $\mathcal{U}(0, 0.5)$ s, delay each whole voice by a further $\mathcal{U}(0, 0.3)$ s, and pan it with independent left/right gains $\mathcal{U}(0, 1)$ before summing all voices into a stereo mix. This HQ-SVC mix is finally balanced by ear with the AVOX Choir mix to form the *sing-along*.

Sing-alongs are provided for every MOISESDB song and for the MUSDB18-HQ training set. At training time the *audience* stem is assembled *on the fly* inside the data loader: starting from silence, each of the three components (sing-along, ambience, event) is added independently with probability $p_v = p_a = p_e = 0.5$ and a random gain $\mathcal{U}(0, 1)$. This stochastic assembly exposes the model to an effectively unlimited variety of audience conditions without storing pre-mixed material, while for the MUSDB18-HQ test and validation sets we instead provide fixed, pre-generated audience mixes to keep evaluation reproducible^{5.1}.

^{5.1}<https://doi.org/10.57967/hf/9773>

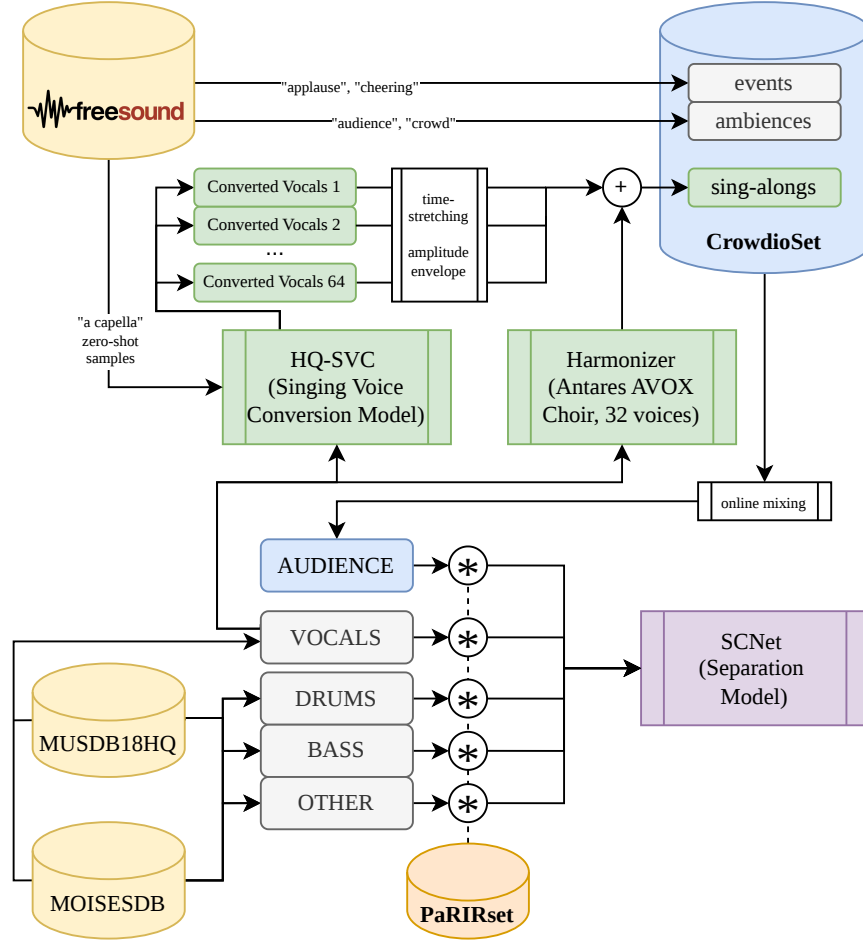


Figure 5.2: Data generation pipeline for the *audience* source. The offline stage (green) synthesises a *sing-along* for every *vocals* stem, by combining zero-shot singing-voice conversions from the HQ-SVC model with the 32-voice AVOX Choir effect, and is run once before training. The online stage (blue), executed inside the data loader at every iteration, assembles the *audience* stem by drawing and gain-scaling the pre-computed *sing-alongs* together with the curated *ambiences* and *events* of CrowdioSet, so that each training example sees a fresh audience.

5.3 PaRIRset: a concert-venue impulse response dataset

The other component absent from studio data is the venue acoustics. Following Kandpal et al. (2022), one could reuse the RIR collections common in Speech Enhancement (SE) — the ACE Challenge (Eaton et al., 2015), the MIT IR Survey (Traer & McDermott, 2016) and the OpenSLR28 corpus (Ko et al., 2017), the same real RIRs we evaluated on in Chapter 4 — but these are ill-suited to live music in two ways. First, the rooms they capture (offices, classrooms, lecture halls) bear little resemblance to concert venues. Second, and more fundamentally, modern live sound is delivered through a Public Address (PA) system, typically formed by left and right main hangs of a line array, complemented by flown or ground-stacked subwoofers and point-source frontfill loudspeakers, as depicted in Figure 5.3.

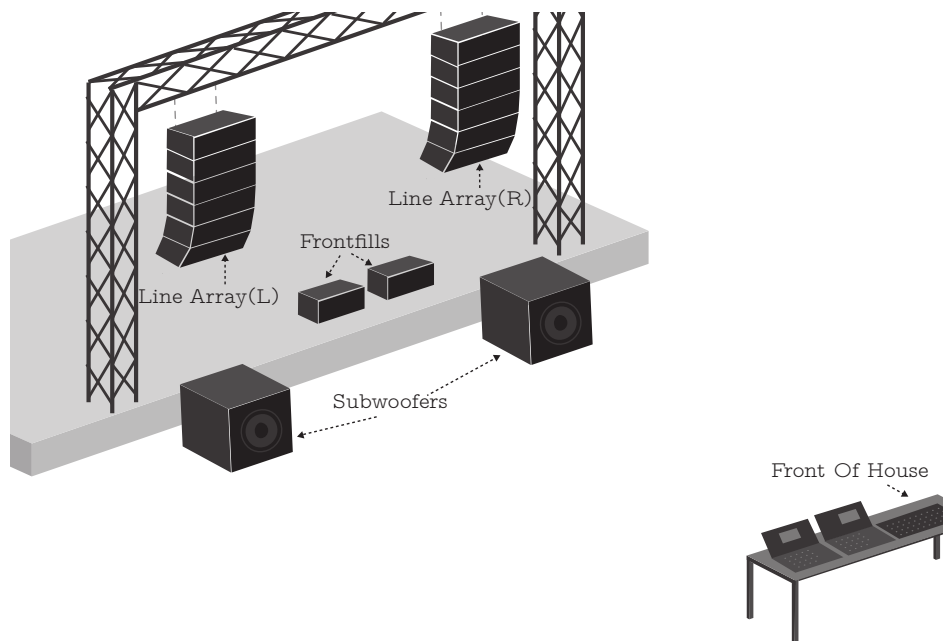


Figure 5.3: Schematic of a typical concert Public Address (PA) system. Two *line arrays*, flown left and right, cover the audience where the *front of house* position normally is; ground-stacked *subwoofers* reinforce the low end; and *frontfills* on the stage lip cover the first rows.

Because no SE dataset contains real RIRs from concert venues, we therefore measured our own set of responses, the *Public Address Room Impulse Response Set* (PaRIRset) — to our knowledge, the first dataset of real RIRs captured

in professional concert venues through their PA systems, and the contribution behind **RO7**.

PaRIRset covers 40 professional concert venues across a wide geographical area (Figure 5.4) and, more importantly, a wide range of PA systems. Table 5.2 summarises the collection: it spans 16 manufacturers (d&b audiotechnik, L-Acoustics, Nexo, TW Audio and Electro-Voice, among others) and 35 distinct main-array models, from compact two-element-per-side systems in small clubs to 24-element arena arrays, with 0 to 12 subwoofers per venue and frontfills present in 24 of them; 9 of the 40 venues were measured outdoors. The full per-venue detail — location, indoor or outdoor, and the specific model and element count of every main, sub and frontfill — is listed in Appendix D (Tables D.1 and D.2). We release the raw mono and multichannel recordings at 48 kHz^{5.2}, together with the processed, augmented stereo RIRs used for training, downsampled to 44.1 kHz to match the source-separation corpora^{5.3}.

Table 5.2: Composition of PaRIRset (40 venues, measured between 2023 and 2025).

Venues (indoor / outdoor)	40 (32 / 8)
Distinct cities	31
PA manufacturers	16
Distinct main / sub models	35 / 33
Main array size (elements per side)	2–24 (median 6)
Subwoofers per venue	0–12 (median 4)
Venues with frontfills	24 of 40

5.3.1 Measurement methodology

Access to professional venues and their PA systems is tightly constrained — typically a short window during a soundcheck — so we limited each measurement to a single exponential sine sweep per PA side (left and right). The microphone was placed at the Front Of House (FOH) position, next to the mixing console, which is the reference listening spot in the room. As shown in Figure 5.5, we captured each sweep with a Zylia ZM-1 microphone array alongside a beyerdynamic MM1 measurement microphone through a Zoom AMS-24 interface.

The RIRs were obtained with the exponential-sine-sweep method (Farina et al., 2000) (Section 2.3.1), with $f_1 = 20$ Hz, $f_2 = 20$ kHz and a sweep duration of 6 s. After deconvolution, two post-processing steps were applied. First, we

^{5.2}https://huggingface.co/datasets/enricguso/parirset_dev

^{5.3}<https://doi.org/10.57967/hf/9774>

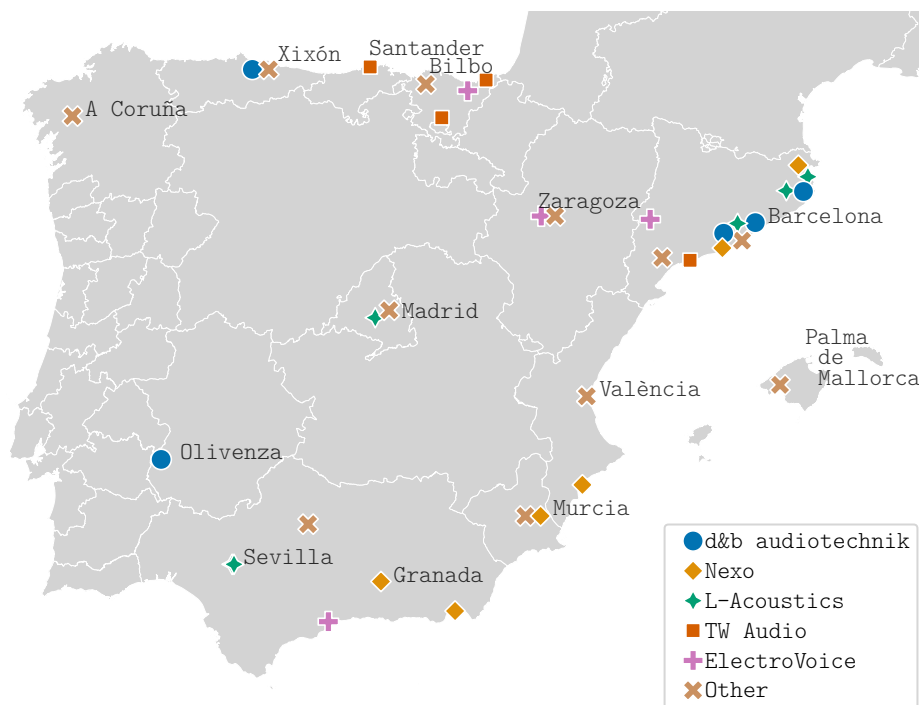


Figure 5.4: PaRIRset venue locations, coloured by the manufacturer of each venue’s PA system.

reduced noise by windowing each RIR where its envelope fell below the noise floor, keeping 100 ms before the direct sound and applying 50 ms fades at the start and before the noise-floor onset. Second, because the FOH position is not always equidistant from the two PA sides, the left and right RIRs were time-aligned by cross-correlation. Finally, taking the flat-response beyerdynamic MM1 as a reference, we corrected the colouration of the Zylia-derived responses with a second-order high-shelf filter ($f_0 = 3.5$ kHz, gain -4 dB, $Q = 0.5$).

A subtlety of this per-side procedure is worth clarifying. In a typical PA, only the main hangs are strictly left and right; the subwoofers and frontfills are usually fed from a matrix of the two channels (most often a mono L + R sum). Measuring the left side therefore does not isolate the left main hang alone: the captured response also contains the contribution of any system driven by that channel through the matrix. At the FOH position this is acceptable, and in fact desirable. The frontfills, aimed at the first rows and far from FOH, contribute negligibly there, so they do not meaningfully colour the measurement; the subwoofers, on the other hand, do reach FOH and their low-frequency energy is genuinely part of what a listener — and an audience recording — receives at

that position. Capturing the mains together with the subs, as our sweeps do, thus yields a stereo pair of responses that reflects the full low-to-high-frequency field actually present at FOH, which is exactly what we want to convolve our sources with.

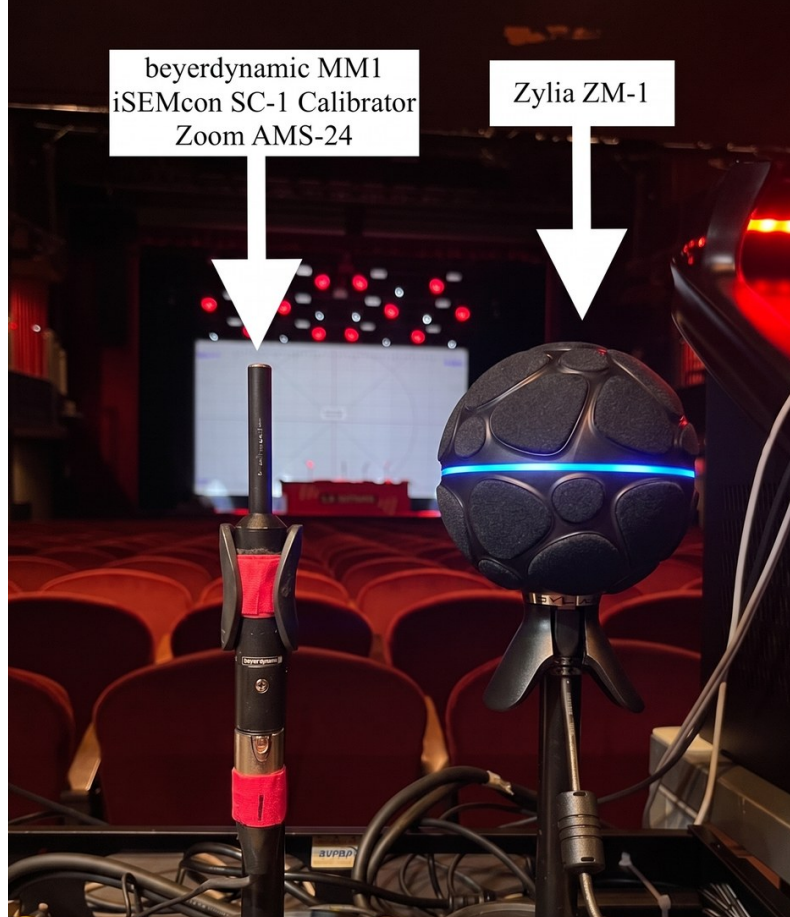


Figure 5.5: PaRIRset recording setup at the Front Of House (FOH) position, next to the sound engineer’s console.

5.3.2 RIR Augmentation

Forty measurements are too few to train on directly, so we explored four augmentation strategies of increasing scope. D_1 operates on the beyerdynamic RIRs alone: splitting each response h at $\arg \max(|h|) + 2 \text{ ms}$ into a direct part d_h and a tail t_h , we generate variants by flipping polarity and rescaling the tail by gains $g = \{-6, -1.5, 0, 1.5, 3.5\} \text{ dB}$, augmenting each measurement 20-fold. D_2 adds the raw signals of the 19 Zylia capsules ($\text{zylia}_{\text{caps}}$), each with a

slightly different position; D_3 adds 19 virtual microphones synthesised by the array’s proprietary beamformer with random orientations and polar patterns ($zylia_{\text{beams}}$); and D_4 permutes the direct part d_h of one venue with the tail t_h of another (in what we call PA_{permute}).

To assess them, we retrained an SCNet (Tong et al., 2024) model from scratch on MUSDB18-HQ with default hyperparameters, convolving all sources and targets with the RIRs and adding one augmentation set at a time; we evaluated on 8 held-out venues (augmented with D_1) forming the PaRIRset test set, and report the Signal-to-Distortion Ratio (SDR) across all sources. The results in Figure 5.6 yield three observations.

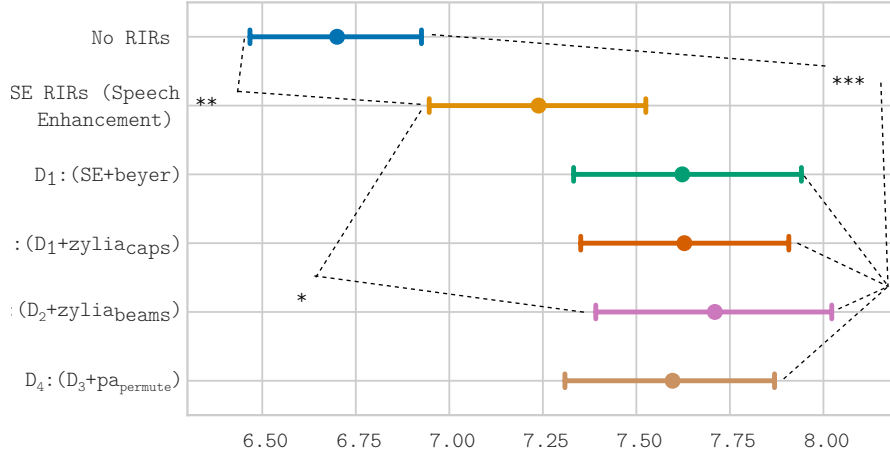


Figure 5.6: Ablation of the PaRIRset augmentation strategies, adding one set at a time. Significance from paired t -tests: * $p \leq 0.05$, ** $p \leq 0.01$; error bars are 95% confidence intervals.

First, every configuration that includes RIRs significantly outperforms the no-reverberation baseline, confirming that exposing the model to acoustics is what matters most. Second, the four strategies differ little from one another, which suggests that *including* real venue responses matters more than the particular way they are augmented. Third, D_3 (the beamformed virtual microphones) improves significantly over the already strong SE-only baseline (paired t -test, $p = 0.027$) supporting the premise behind **RO7**. A plausible reason for this is that its randomly oriented polar patterns inject directional diversity that the raw capsule signals of D_2 do not, though our evidence does not let us claim this conclusively. Conversely, D_4 (PA_{permute}) brought no gain and was discarded; recombining the direct sound of one venue with the tail of another appears not to create useful new variety. Henceforth, *PaRIRset* denotes the combination of the SE RIRs, D_1 , D_2 and D_3 , totalling approximately 2,200 stereo RIRs.

A final subtlety concerns the reverberation level. Convolver every source with a full-length response produced mixtures that were audibly too reverberant, because PaRIRset was measured in *empty* venues during soundcheck, whereas a full audience shortens reverberation substantially (Beranek, 1960). To approximate the occupied condition, we attenuate the tail of each RIR — that is, we scale every sample except the direct-sound peak at $\arg \max(|h|)$, leaving the direct sound untouched while reducing the reflections — by a random $\mathcal{U}(0, 1)$ factor during training and by a fixed -6 dB for evaluation, which we found to produce a more realistic reverberation level by ear.

5.4 Experimental setup

Our clean sources come from two studio corpora. MUSDB18-HQ (Raffi et al., 2019) is the standard MSS benchmark: 150 full-length stereo tracks at 44.1 kHz (100 for training, 50 for testing), each provided as four isolated stems — *vocals*, *drums*, *bass* and *other*. From the 100 training tracks we hold out the conventional 14-track validation split.^{5.4} MOISESDB (Pereira et al., 2023) is a larger and more recent corpus of 240 tracks, but with a finer, hierarchical stem taxonomy (vocals, bass, drums, guitar, piano, and further categories). To co-train it with MUSDB18-HQ we *parse* it into the same four-stem taxonomy, mapping vocals, drums and bass directly and collapsing all remaining instruments into *other*. Together with CrowdioSet for audience noise and PaRIRset for venue acoustics, these constitute the training material for a first *Live* MSS baseline. As in the PaRIRset ablation, we adopt SCNet (Tong et al., 2024) as the backbone (Section 5.1): it is open-source and among the most competitive models on MUSDB18-HQ, and thus a realistic starting point.

5.4.1 Training details

All models are trained from scratch with SCNet’s default hyperparameters, save for a few adjustments. When co-training on both MUSDB18-HQ and MOISESDB we lower the initial learning rate from 5×10^{-4} to 4×10^{-4} , as prescribed by Tong et al. (2024), and we remove the exponential moving average of the weights, which simplifies the pipeline at negligible cost: our co-

^{5.4}Actions – One Minute Smile; Alexander Ross – Goodbye Bolero; ANiMAL – Rockshow; Clara Berry And Wooldog – Waltz For My Victims; Fergessen – Nos Palpitants; James May – On The Line; Johnny Lokke – Promises & Lies; Leaf – Summerghost; Meaxic – Take A Step; Patrick Talbot – A Reason To Leave; Skelpolu – Human Mistakes; Traffic Experiment – Sirens; Triviul – Angelsaint; Young Griffo – Pennies.

trained baseline reaches 9.0 dB SDR on the MUSDB18-HQ test set, matching the 9.0 dB reported for SCNet trained on MUSDB18-HQ alone and close to the 9.2 dB of its co-training setting. Each model is trained for 200 epochs on a single NVIDIA L40S GPU (about seven days per model), with SCNet’s default random time-shift augmentation of up to two seconds.

To expose the model to as many source combinations as possible, we use batch sizes of 10 or 8 (with or without audience), split into two groups per batch so that sources can be remixed across songs within a batch (Jeon et al., 2024). This cross-song remixing is standard practice in MSS, but its benefit is not universal: Jeon et al. (2024) show that it stems from the increased diversity of otherwise *inconsistent* mixtures, and that for highly consistent sources it can help far less — in vocal-harmony separation, where sources are especially correlated, they note that random mixing was even found “detrimental on a small dataset”. This is a concern in our setting, because our synthetic *sing-alongs* are, by construction, strongly correlated with one another and with the lead vocal (some converted voices sing in near-unison, differing only in timbre), so remixing the *audience* source across songs might not help, or might even hurt it. We therefore trained the model both with and without remixing and compared them (Table 5.3). Remixing left the *audience* source essentially unchanged (+0.07 dB SDR, paired t -test $p = 0.07$) while improving the musical sources overall, so we retained it for all subsequent experiments.

Table 5.3: Effects of cross-song source remixing, trained and evaluated on the MUSDB18-HQ+MOISESDB set with audience: mean SDR (dB) per source over the 50 test songs, with and without remixing. Remixing does not significantly affect the *audience* source (paired t -test, $p = 0.07$), while it improves the musical sources ($p \approx 10^{-3}$ overall).

Source	Remixing	No remixing	Δ
vocals	7.50	7.49	+0.02
drums	9.98	9.76	+0.22
bass	8.41	7.90	+0.51
other	5.95	5.74	+0.22
audience	2.85	2.79	+0.07
overall	6.94	6.74	+0.20

The core of the experiment is a comparison of four training configurations, which differ only in the degradations applied to the same clean sources and which we name after them:

- **clean**: MUSDB18-HQ + MOISESDB;
- **rev**: (MUSDB18-HQ + MOISESDB) $\otimes$ PaRIRset;
- **noisy**: MUSDB18-HQ + MOISESDB + CrowdioSet;
- **noisyrev**: (MUSDB18-HQ + MOISESDB + CrowdioSet) $\otimes$ PaRIRset;

where $\otimes$ denotes convolution with a PaRIRset RIR. Both the audience mixing and the reverberation are applied *inside* the training loop, as described in Sections 5.2 and 5.3, so that every epoch presents novel combinations of sources, audiences and rooms. Because the added degradations make optimisation progressively harder, we lower the initial learning rate accordingly across conditions, to $\{4, 3.5, 2, 2\} \times 10^{-4}$ for **clean**, **rev**, **noisy** and **noisyrev**, whose best validation checkpoints are reached at epochs 198, 199, 197 and 144. In the reverberant conditions our goal is to *separate* the sources while *preserving* reverberation: both the targets and the estimates remain reverberant, so the model is not asked to dereverberate.

5.4.2 Evaluation

We evaluate on the MUSDB18-HQ test set — keeping our results comparable with prior work — augmented with the held-out *audience* test stems and the PaRIRset test set. Reverberation is applied here *deterministically*, so that every model sees the same conditions: 25% of the samples with no reverberation, 25% with the full RIR, and the remaining 50% split equally between tails attenuated by -6 and -12 dB. Each of the four models is evaluated under the same four conditions (**clean**, **rev**, **noisy**, **noisyrev**), giving a full cross of training and test degradations.

To place the audience-separation results in context, we add SAM Audio (Shi et al., 2025) as a baseline capable of isolating arbitrary sources from a text prompt. We use its large pre-trained variant in text-prompting mode, with a single re-ranking candidate and non-overlapping 20s windows to fit memory, querying *vocals*, *drums*, *bass*, *other* and *crowd* (the last outperforming *audience*).

As in SCNet, our primary objective metric is the Signal-to-Distortion Ratio (SDR) (Le Roux et al., 2019), computed against non-reverberant or reverberant references according to the evaluation condition. However, SDR does not always track perceived quality (Gusó et al., 2022; Jaffe & Burgoyne, 2025) — and it is especially unforgiving of diffusion-based models such as SAM Audio,

whose outputs are not sample-aligned and are therefore penalised regardless of how they sound — so we complement it with a subjective AB listening test. To retain statistical power with a modest sample, we restrict the test to two tasks, *vocals isolation* and *audience isolation*, on 10 real audience recordings sourced from social media and split into two groups of five. For vocals isolation, participants compared the vocals from our **noisy** model against those of the **clean** baseline; for audience isolation, the audience stem from the **noisy** model against SAM Audio; in both cases they chose which separation they preferred, if any. Twenty-six participants took part — media researchers and professional musicians or sound engineers — using the web-based AB interface of Barry et al. (2021), with the presentation order of the two systems randomised.

5.5 Results and discussion

What follows are the per-source results on **noisyrev** and **clean** conditions, audience isolation for both **noisy** and **noisyrev** conditions, average SDR across all sources and on vocal isolation alone considering all evaluation conditions, and the listening test results.

5.5.1 Objective results

Table 5.4 reports the per-source SDR in the most demanding evaluation condition (**noisyrev**) and, in parentheses, in the **clean** one. The first observation concerns the SAM Audio baseline: it performs substantially worse than all our models under both conditions — only 3.72 dB on clean *vocals*, against the 10.05 dB of our **clean** model. We attribute this mainly to the mismatch between its open-vocabulary formulation and the fixed four-stem taxonomy of MUSDB18-HQ: SAM Audio tends either to extract a source or to fail outright, and its successful extractions still yield positive SDR, which suggests the gap is not merely a sample-alignment artefact of its diffusion outputs. We nonetheless rely primarily on the subjective test below.

Among our four models, no single variant dominates the **noisyrev** condition: **noisy** gives the best *vocals* (SDR 3.26 dB), **noisyrev** the best *bass* (5.66 dB), and **rev** the best *drums* (8.23 dB) and *other* (3.63 dB). Two patterns stand out. First, reverberant training (**rev**, **noisyrev**) is especially helpful for the instrumental sources. Second, and more useful in practice, the **noisy** model retains near-baseline performance on clean inputs (9.97 vs. 10.05 dB on vocals), so

Table 5.4: Per-source objective results: mean SDR $\pm$ standard deviation (dB) in the **noisyrev** evaluation condition and, in parentheses, in the **clean** one. Best per column in bold. [†]SDR for the diffusion-based SAM Audio should be read with caution, as its outputs may be sample-misaligned.

	vocals	drums	bass	other
clean	1.46 ± 3.4 (10.05)	7.04 ± 3.6 (10.62)	4.15 ± 3.5 (8.97)	1.43 ± 3.5 (6.69)
rev	1.10 ± 3.6 (9.01)	8.23 ± 3.1 (9.86)	5.36 ± 3.7 (7.77)	3.63 ± 2.4 (5.98)
noisy	3.26 ± 2.9 (9.97)	7.00 ± 3.3 (10.68)	3.91 ± 3.6 (8.85)	1.79 ± 3.2 (6.69)
noisyrev	3.15 ± 3.1 (9.07)	8.19 ± 3.3 (9.98)	5.66 ± 3.6 (7.86)	3.55 ± 2.7 (5.88)
SAM Audio [†]	-0.45 ± 1.9 (3.72)	3.30 ± 2.6 (5.63)	1.30 ± 3.0 (5.14)	-1.82 ± 2.0 (-1.74)

Table 5.5: Audience isolation results (the three audience-capable models): mean SDR $\pm$ standard deviation (dB). [†]SDR for the diffusion-based SAM Audio should be read with caution, as its outputs may be sample-misaligned.

	audience isolation	
	non-reverberant	reverberant
noisy	2.85 ± 1.3	1.25 ± 1.4
noisyrev	1.55 ± 1.0	1.56 ± 0.9
SAM Audio [†]	-1.94 ± 3.0	-1.60 ± 2.7

audience-noise augmentation alone does not meaningfully harm studio-quality separation. We therefore recommend the **noisy** model for vocal extraction when the recording conditions are unknown.

Table 5.5 reports audience isolation, of which only the audience-trained models are capable. Under reverberant evaluation **noisy** and **noisyrev** are nearly tied (1.25 vs. 1.56 dB), but on non-reverberant mixtures **noisy** is clearly superior (2.85 vs. 1.55 dB); both beat SAM Audio by more than 3 dB, which despite being an unreliable metric for diffusion-based models is in line with the subjective evaluation results below. Together with its strong vocals, this supports selecting **noisy** for the subjective test.

Figure 5.7 extends the comparison to the full cross of the four models and four evaluation conditions. The upper heatmap (all sources combined) confirms the expected pattern: in each condition the best model is the one whose training degradations match it (with **noisyrev** marginally ahead in the **rev** column).

Crucially, adding the audience source is nearly free on matched-clean data: **noisy** trails **clean** by only 0.03 dB (9.05 vs. 9.08) in clean evaluation, and **noisyrev** matches **rev** (7.83 vs. 7.82) in reverberant evaluation. The lower

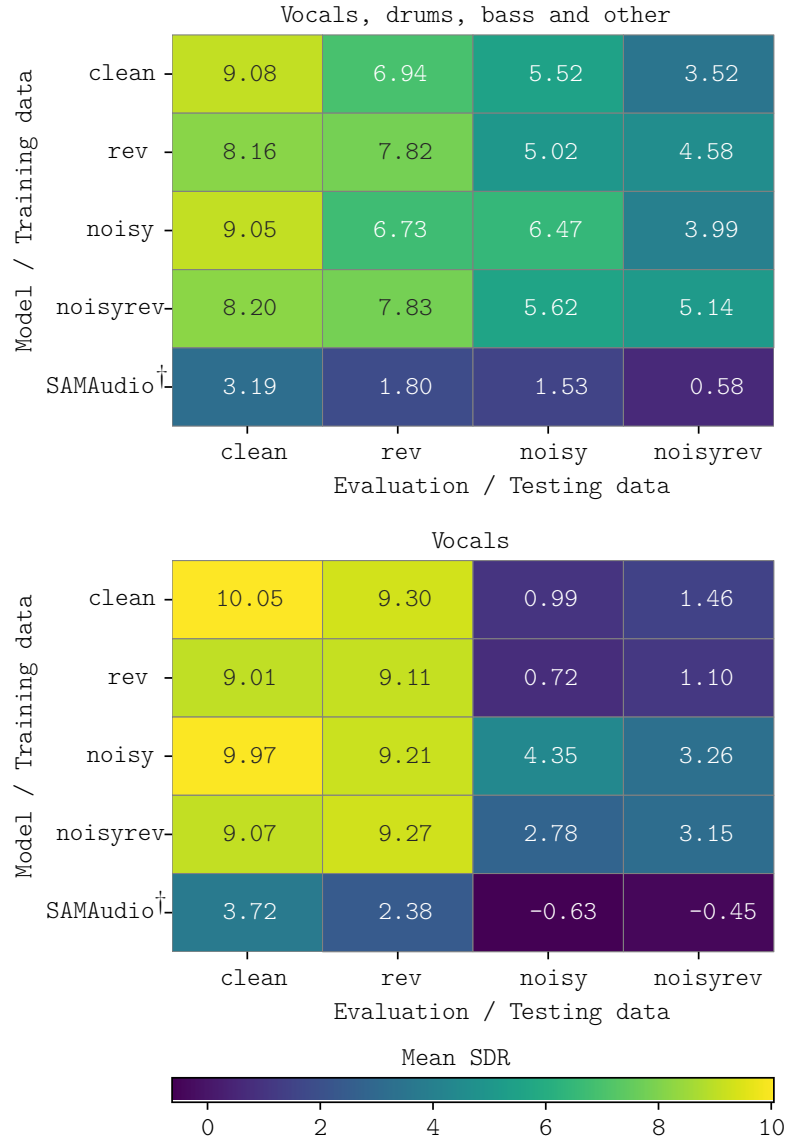


Figure 5.7: Mean SDR (dB) for all sources combined (top) and *vocals* isolation (bottom). Rows correspond to MSS models, columns correspond to evaluation conditions. [†]SDR for the diffusion-based SAM Audio should be read with caution, as its outputs may be sample-misaligned.

heatmap (*vocals* only) shows **noisy** to be the most robust across conditions: in the **noisyrev** column **noisyrev** beats **noisy** by only 0.39 dB, whereas **noisy** beats **noisyrev** by 1.57 dB in the **noisy** column — reinforcing our choice of **noisy** for the listening test.

5.5.2 Subjective results

Figure 5.8 summarises the AB listening test. Counting ties as votes for the baseline and applying a one-sided binomial test ($H : P > 0.5$), listeners significantly preferred our **noisy** model in both tasks: for *vocals* isolation, 105 of 130 votes ($p < 10^{-12}$); for *audience* isolation, 119 of 130 votes ($p < 10^{-23}$). The subjective preference is thus far more decisive than the objective margins, consistent with the known limitations of SDR on this kind of material, and confirms that the improvements are perceptually real.

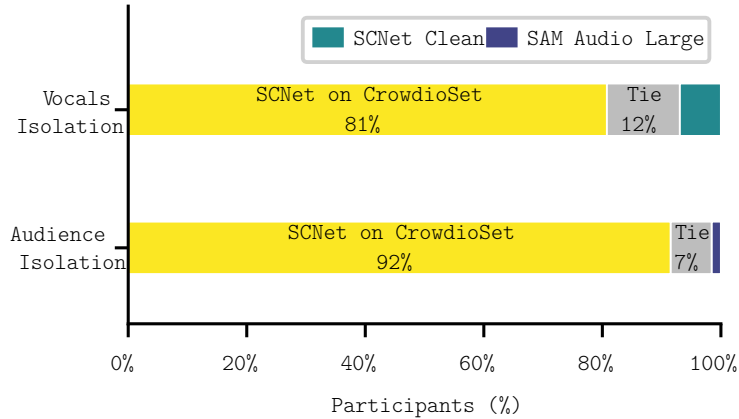


Figure 5.8: Subjective AB results comparing our **noisy** model against the two baselines: **clean** for *vocals* separation and **SAM Audio** for *audience* separation.

5.6 Limitations and future work

Several limitations temper these results. We did not systematically search the online mixing parameters (the probabilities p_v , p_a , p_e and the gains): the values used were chosen by informal listening and may be suboptimal. More fundamentally, listening to the model on real concert recordings shows that, while it separates *ambiences* and *events* effectively — acting as a denoiser — it rarely isolates the *sing-along* reliably. We attribute this to the synthesis itself: the voice-conversion sing-alongs remain too similar to the original *vocals* stem (some converted voices sing in exact unison, differing only in timbre),

and the audio effects already present on those stems (reverb, delay) further blur the boundary between the lead vocal and its synthetic crowd. A future CrowdioSet could start from *dry* vocals (Zang et al., 2025), use more expressive synthesis, and model the performers’ reactions to the crowd besides the crowd’s reaction to the performance. Finally, the early convergence of the **noisyrev** model (epoch 144 against ~ 200 for the others) hints that the combined noise-and-reverberation setting may need architectural changes or a fuller hyperparameter search to be exploited fully.

5.7 Conclusions

This chapter introduced the *Live* MSS task — the separation of music recorded from within a live audience — and, in line with the rest of the thesis, tackled it as a problem of *ecological validity*: studio-trained separators fail on audience recordings because their training data omits the venue acoustics, the PA colouration and the crowd noise that such recordings inevitably contain. Rather than only collecting scarce and unlabelled real live material, we closed this gap by *synthesising* those conditions on top of existing studio datasets.

Concretely, we addressed the three research objectives set out for this chapter:

- **RO5** — generating synthetic audiences that sing along — was preliminarily addressed through the voice-conversion pipeline of CrowdioSet, which turns the vocals of existing datasets into humanised, timbrally diverse crowd sing-alongs;
- **RO6** — curating audience noise from freely available recordings — was addressed by assembling the ambiances and events of CrowdioSet from Freesound, with documented licences and reproducible splits; and
- **RO7** — capturing impulse responses in real concert venues — was addressed with PaRIRset, the first set of stereo RIRs measured across 40 concert venues through their PA systems.

Experiments with SCNet show that training with CrowdioSet improves vocal and audience separation on live recordings while barely affecting clean studio performance, and that PaRIRset yields statistically significant gains over the general-purpose RIRs used in SE (Chapter 4). A listening test confirms the objective findings. As with the previous chapters, the improvement comes not from a new architecture but from training data that better reflects the

conditions of real use. We release the datasets^{5.5,5.6}, the code and model weights^{5.7} and audio examples publicly.^{5.8}

^{5.5}<https://doi.org/10.57967/hf/9773>

^{5.6}<https://doi.org/10.57967/hf/9774>

^{5.7}https://github.com/enricguso/guso_ismir26

^{5.8}https://enricguso.github.io/crowdioset_parirset



Conclusion and future perspectives

This dissertation has pursued a single idea across three application fields: that the *ecological validity* of the data used to train and evaluate DNNs (i.e. how faithfully it reflects the conditions of real use) is as decisive as the models themselves, and sometimes a more accessible lever for improving them. Chapter 3 applied this idea to the *evaluation* of HAs, Chapter 4 to the *training* data of monaural SE, and Chapter 5 to both evaluation and training, in the newly framed task of Live Music Source Separation.

This chapter summarizes the contributions across the three research fields and the seven Research Objectives (RO) they addressed (Section 1.4). Because in the MIR field a lot can change during the course of four years, we then update the changes in the SOTA for each published work to discuss how the field has moved since: which of our intuitions proved well-grounded, where industry has converged on the same questions, and which limitations have become apparent only in hindsight (Section 6.2). Widening the scope further, we reflect on the broader relevance of ecological validity beyond the three tasks studied here (Section 6.3) and, finally, we outline the directions we find most promising going forward, towards deployable, interactive tools built on ecologically valid synthetic data (Section 6.4), and close with some final thoughts.

6.1 Summary of contributions

This thesis set out to improve the *ecological validity* of the synthetic datasets used to train and evaluate DNNs for audio processing, across three application fields. A common constraint shaped how we pursued it: as a small, compute-constrained academic group, we could not compete on model scale or compute, so we competed on *data* instead — building material that better reflects real use — and released every dataset, decoder, model and recording, so that the resulting improvements are available to any laboratory. All seven research objectives were addressed; as the summaries below make clear, this does not mean all were answered affirmatively — a negative or inconclusive result resolves an objective as much as a positive one.

Hearing Aids (HAs) (Chapter 3). Our main contribution here is a *methodology* for ecologically valid evaluation of HA Speech Enhancement (SE): we reproduce complex speech-in-noise scenes over a loudspeaker array using Ambisonics and VBAP, capture them on real devices, and compare the devices’ own processing against DNN-based enhancement under conditions closer to real use than the stationary laboratory tests devices are usually certified with.

RO1 *Expose the performance gap between conventional HA enhancement and DNN-based approaches in ecologically valid, complex acoustic scenes.* → *Addressed* → the evaluation revealed a clear gap, with conventional enhancement occasionally performing worse than no processing at all.

RO2 *Investigate the potential of DNN-based SE to outperform conventional HA processing.* → *Addressed affirmatively* → DNNs substantially improved denoising and objective intelligibility, and the causal variant retained most of that advantage.

Beyond these findings, the lasting contribution is the evaluation methodology itself, and the case it makes for pursuing DNN-based HA enhancement: to support further work in this direction we released a HA HRTF set, 10th-order Ambisonics-to-binaural BiMagLS decoders, a synthetic binaural training corpus and a reverberant test set with the recordings of each device.

Monaural Speech Enhancement (SE) (Chapter 4). Within the GuestXR project, we asked which properties of a synthetic RIR drive generalisation to real rooms, training one model per dataset on six controlled RIR sets.

- RO3** *Determine whether frequency-dependent absorption and source or receiver directivity improve generalisation to real acoustics.* → Addressed with a mixed result → frequency-dependent absorption brings the clearest benefit, whereas the evidence on source and receiver directivity was inconclusive.
- RO4** *Determine whether fewer but geometrically more realistic mesh-based RIRs are preferable to abundant Image Source Method (ISM) ones.* → Addressed with a negative result → mesh-based realism did not compensate for a smaller variety and quantity of rooms, though a mesh-based set that is also varied would be needed to fully disentangle the two (Chapter 4).

We released the resulting MB-RIRs dataset, among the most downloaded resources of the GuestXR project repository.

Live Music Source Separation (MSS) (Chapter 5). More in line with the research of the MTG, we formulated *Live* MSS (i.e. separating music recorded from within a live audience) and made it trainable by synthesising live conditions on top of studio data.

- RO5** *Generate synthetic audiences that sing along, through singing-voice conversion.* → Addressed as a first step → the voice-conversion pipeline produces usable sing-alongs, though reliably isolating them remains difficult and is left as a limitation.
- RO6** *Curate audience noise from freely available recordings.* → Addressed affirmatively → the curated ambiances and events of CrowdioSet improve separation on live recordings while barely affecting clean studio performance.
- RO7** *Determine whether impulse responses measured in real concert venues improve over general-purpose ones.* → Addressed affirmatively → PaRI-Rset yielded statistically significant gains over the SE RIRs of Chapter 4.

For convenience, all the datasets, models, decoders and recordings released throughout this thesis are collected, with their repositories and licences, in Appendix C.

6.2 Recent developments and closing discussion

We now revisit each of the three fields we have addressed, discussing how its SOTA has evolved since our work was published.

6.2.1 Hearing aids

Since the evaluation of Chapter 3 was published, the Clarity Challenge series has established itself as the main forum for DNN-based HA enhancement, and its findings echo ours. Evaluated, like our study, with HASPI and HASQI (Cox et al., 2023), its editions reached the same two conclusions we drew: conventional HA processing struggles in complex, noisy scenes, and DNN-based enhancement has the potential to improve on it. Two details are worth noting. First, HASQI was added alongside HASPI precisely because some systems raised intelligibility at the expense of naturalness (Cox et al., 2023) — the same tension our follow-up test surfaced between *strong* and *mild* enhancement (Section 3.5). Second, and most relevant to this thesis, that challenge paired a simulated evaluation set with one based on real-room measurements, and found that systems improving on the simulated data degraded sharply on the measured recordings (Cox et al., 2023) — an independent, large-scale instance of the ecological-validity gap that motivated our loudspeaker-based evaluation.

Our own contribution — a dataset centred on binaural RIRs for HAs SE — was mirrored, the same year, by BinauRec (Delebecque & Serizel, 2023), which likewise studies the influence of RIRs on binaural speech enhancement, and, the following year, by an extension to the multichannel case (Westhausen et al., 2024). That the same idea was pursued independently and elsewhere suggests the topic was a timely one.

A second, clear trend is the tightening focus on the real-time and low-latency constraints of hearing devices, which we identified as the main obstacle to deployment. Multi-stage systems now meet the sub-5 ms latency budget while improving HASPI and HASQI over the challenge baseline (Ouyang et al., 2023), and this push has continued towards *sub-millisecond* latency on hearables (Dementyev et al., 2024). In the same spirit as our own remark that model size remains a bottleneck for hearing aids, some works have gone as far as transmitting only a low-bit-rate binaural link between the two devices to afford the enhancement (Westhausen & Meyer, 2023).

Overall, the questions our study raised — how to evaluate HA enhancement under ecologically valid conditions, and how to make DNN-based processing de-

ployable on-device — remain central to the field. Our work has since been taken up in directions ranging from optimised filter design for low-power devices (Rawandale & Kolte, 2024) to broader surveys of the area (Jespersen, 2023; Kanev & Shapovalova, 2024; Cao et al., 2025), which suggests our study has helped inform the work that followed.

6.2.2 Speech enhancement

The premise of Chapter 4 — that the realism of synthetic RIRs, and frequency-dependent absorption in particular, improves generalisation to real rooms — has since been taken up, and largely confirmed, by industry. The mesh-based responses we used come from SoundSpaces (Chen et al., 2022), a simulator developed at Meta; and Treble Technologies has released a commercial wave-based simulation Software Development Kit (SDK) together with large, physically accurate RIR datasets built on exactly the same rationale (Mullins et al., 2025). Most tellingly, a Treble study reproduced our experiment at a larger scale and in the multichannel far-field setting: training the same enhancement model on RIRs of increasing simulation fidelity, they report up to a 38% relative reduction in word error rate on *measured* data for the high-fidelity set (Götz et al., 2026). Both works cite MB-RIRs, placing it in the lineage of increasing RIR realism alongside GWA (Tang et al., 2022) and other efforts.

In that context, our contribution was less a whole novel simulator and more of an ecologically valid RIR set made openly available, so that compute-constrained groups could benefit from the same idea that industry was pursuing behind commercial SDKs. Made freely downloadable, MB-RIRs has since become the most downloaded resource in the GuestXR project’s Zenodo repository, which we take as a sign that open data of this kind meets a real need. The comparison with Treble is also instructive about our own limitations: their focus on far-field, multichannel conditions is not incidental — it is there, rather than in our near-field monaural setup, that the gap between simulation methods is widest, which may be why the source- and receiver-directivity benefits we looked for turned out inconclusive (**RO3**). The effect we sought was likely clearest precisely where we did not look.

A related point concerns **RO4**. We found that abundant ISM rooms outperformed fewer mesh-based ones, but the mesh-based set available to us (SoundSpaces) was also the least varied, confounding realism with diversity. The natural way to disentangle the two is a mesh-based set that is itself large and varied, such as GWA (Tang et al., 2022), which we did not evaluate, as at

the time it sat outside our usual audio venues, having been presented in the computer-graphics community. Rather than a mere oversight, the growing interest in GWA reinforces our reading that such datasets are valuable: the Treble challenge mixes Treble and GWA rooms (Lin et al., 2025), and the multichannel study lists GWA beside MB-RIRs (Götz et al., 2026). We see the most promising direction not as choosing between paradigms but as *combining* them — for instance, the roughly 20,000 mesh-based rooms of GWA together with the 60,000 ISM rooms of MB-RIRs — pairing geometric realism with sheer variety, and we encourage the community to explore it.

More broadly, room-acoustic data has become a remarkably active research area since our work. On the simulation and measurement side, new resources keep appearing: Meta released a 1,000-room corpus for scene-wide acoustic parameter estimation (Falcon-Perez et al., 2025), open ray-tracing toolkits such as GSound-SIR (Zang & Kong, 2025) lower the barrier to spatial RIR synthesis, and measured databases now capture more challenging conditions, including moving microphones along a controlled trajectory (Damiano et al., 2026). In parallel, a complementary line attacks the problem from the estimation side, inferring RIRs or their parameters blindly from reverberant speech (Götz et al., 2025; Wang & Li, 2026). Together with the industrial efforts above, this activity confirms that the realism and coverage of room-acoustic data — the very concern of Chapter 4 — is widely recognised as a problem worth addressing.

6.2.3 Music source separation

Being the most recent of the three studies, our work on live MSS (Chapter 5) has the least state of the art to catch up with. What has been research during this thesis is largely orthogonal to our contribution, which concerns *data* rather than architecture. On the modelling side, band-split Rotary Positional Embedding (RoPE) transformers have consolidated their lead on studio benchmarks (Lu et al., 2024), and lightweight variants now bring comparable quality to real-time and embedded settings — the low-latency line we distinguished from our own use of *live* in Section 5.1. Because our proposal is a training and evaluation recipe rather than a network, these advances are complementary: any of these separators could be trained on CrowdioSet and PaRIRset in place of the SCNet backbone we used.

More closely related in spirit is the emergence of Music Source Restoration (Zang et al., 2025), run as an ICASSP 2026 challenge, which likewise rejects the

linear-mixing assumption — there, to undo studio production artefacts rather than to add live-venue degradations. The two settings are complementary halves of the same idea: that the gap between clean stems and the signals encountered in the wild must be modelled explicitly, whether that gap comes from a mastering chain or from a concert PA and its audience. Beyond this conceptual parallel, their dataset would also be of direct practical use to us: as noted in Chapter 5, the *dry* vocal stems it provides would be an ideal starting point for recreating our *sing-alongs*, which our current pipeline synthesises from already-processed vocals. Finally, generative approaches to separation have begun to appear, synthesising rather than masking the target source (Plaja-Roglans et al., 2025); as discussed in Section 6.4, we see this as a promising route to the more realistic sing-alongs that our own pipeline still struggles to isolate.

6.3 The broader relevance of ecological validity

The concern that runs through this thesis — the importance of ecological validity — is not specific to the three tasks we studied. It resurfaces, often unremarked, wherever a field quietly substitutes convenient data for representative data. Two examples from music information retrieval make the point.

Automatic music mixing is the first. The field has seen remarkable progress: recent systems repurpose out-of-domain data to overcome the scarcity of dry/wet pairs (Martínez-Ramírez et al., 2022), model the one-to-many nature of mixing with generative approaches (Moliner et al., 2026), and learn transferable mixing styles directly from separated stems (Cheng et al., 2026), with listening tests that increasingly approach human-level quality. These are genuine breakthroughs. Yet across this work it is rarely made explicit *which* human, mixing *which* music. In practice, the reference for a “professional mix” is very often MUSDB18-HQ (Raffi et al., 2019): a corpus assembled for source *separation*, whose mixtures were never selected for being exemplary. Training or evaluating automatic mixing on it thereby equates “professional” with a body of mostly amateur or semi-professional rock and pop recordings by a handful of engineers — a bias acknowledged even by the community that popularised the dataset, which notes its focus on Western pop and its limited pool of mixing engineers (Fabbro et al., 2024a). A model that matches such references has learned to reproduce competent demos, not the mixing decisions behind commercially successful records.

The second example is music identification, where the stakes are directly economic: audio fingerprinting underpins the monitoring of public performances for royalty distribution. Classical fingerprints are designed to recognise the exact studio master and do not generalise to live renditions, where venue reverberation, a PA system and crowd noise reshape the signal. Araz et al. (2025) tackle this head-on, improving neural fingerprint robustness by training with realistic room acoustics and evaluating on real microphone recordings made across bars, nightclubs and concert halls — and, in doing so, uncovering evaluation flaws in a prior system that had flattered its reported performance. Yet even this careful work inherits the limitation at the heart of our Chapter 5: its room acoustics come from general-purpose RIR datasets such as OpenAIR (Murphy & Shelley, 2010), measured in halls and churches *without* a live PA system. Such responses capture a room’s reverberation but not the colouration of the line arrays and subwoofers through which concert audiences actually hear music, nor the crowd that accompanies it — precisely the gap PaRIRset and CrowdioSet were built to fill. Tellingly, the authors themselves note that at festival distances it is the low frequencies from the speakers that survive the crowd noise, yet the acoustics they train on contain no such system.

The parallel with our own domains is exact. Just as training speech enhancement on oversimplified RIRs teaches a model to undo a reverberation that does not occur in real rooms (Chapter 4), training an automatic mixer on incidental mixtures teaches it to imitate a target that no listener would call a great mix, and training a fingerprinter without a PA teaches it to recognise a concert it will never actually hear. The failure is not in the models but in the implicit claim that the available data stands in for the real objective. It is the cooking-school version of learning to cook by averaging the recipes of whichever restaurants happen to be nearby: one may become fluent, and still never learn to cook well.

None of this is a criticism of these methods, several of which are genuinely sophisticated; it is an encouragement to spend as much care on the ecological validity of training and —more importantly— evaluation data as on the models themselves. It is, we believe, the most transferable recommendation of this thesis.

6.4 Future directions and final thoughts

In light of the discussion above, several next steps follow naturally.

Growing PaRIRset, CrowdioSet and its sing-alongs. CrowdioSet has proven surprisingly effective for the modest effort invested in it, which makes it the natural place to build. The obvious first step is to enlarge it and, above all, to improve the synthetic sing-alongs, whose isolation remains its weakest point (**RO5**). Starting from clean *dry* vocal stems, such as those released for Music Source Restoration (Zang et al., 2025), would remove the processing our current pipeline inherits from already-mixed vocals. So far we have been unable to generate convincing sing-alongs with general-purpose generative audio models such as Stable Audio (Evans et al., 2024); should these mature, they would offer a route to far more varied and realistic crowds. Combined with tiny, deployable architectures, richer versions of these datasets could support on-device denoising for the hearing impaired, connecting the live-music work of Chapter 5 back to the hearing aids of Chapter 3. Likewise, should PaRIRset gain traction and find use in other tasks, we intend to keep measuring further venues and to release a larger second version in the future.

Synthetic pre-training with PaRIRset as real-data reward. A recent and, to us, compelling recipe comes from MuScriptor (Rouard et al., 2026), a multi-instrument transcription model trained in three stages: pre-training on abundant *synthetic* data, fine-tuning on *real* recordings, and a final reinforcement-learning stage that aligns the model using only a small, manually verified set of high-quality examples. Its guiding principle — “data, not architecture” — is exactly ours. We see a direct application to RIR-based training: a speech-enhancement or separation model could be pre-trained on abundant synthetic responses — combining the mesh-based rooms of GWA (Tang et al., 2022) with the ISM rooms of MB-RIRs (Chapter 4) — and then refined with a reinforcement-learning stage rewarded on the smaller set of *real* concert measurements in PaRIRset. This would let a large synthetic corpus supply coverage while a handful of real measurements supply the final, ecologically valid calibration — turning the synthetic-versus-real trade-off of this thesis into a pipeline rather than a choice.

A discarded chapter: ecologically valid audio doubling. The initial plan for this thesis included a fourth study, on *automatic audio doubling*, which we ultimately set aside but which this recipe could revive. When recording an album, musicians commonly play the same part — a rhythm guitar, say — two or more times and pan the takes left and right, to obtain a wider, larger-than-life sound. The traditional Digital Signal Processing (DSP) methods that imitate this effect simply add short, flipped-polarity delays; because the copy is identical to the original, their high correlation produces the hollow colouration of comb filtering. When a human plays the part twice, the small imperfections between takes lower that correlation and yield a far more natural result. During the planning stage we saw that sampling audio-conditioned diffusion models was a promising way to model this phenomenon in an ecologically valid manner — learning the distribution of a musician’s imperfections rather than a fixed delay. The obstacle was data: we could assemble only about two hours of paired doubled takes, far too little to train a model that generalises. The MuScriptor recipe above resolves exactly this: synthesise abundant doublings for pre-training, then align the model with our small real set through reinforcement learning. Unlike most generative tools now reaching the market, such a system would *assist* musicians rather than replace them, helping them widen a performance they themselves played.

We began this thesis by arguing that, when scaling is out of reach, the fidelity of the data becomes a productive vector for improvement. We have shown this in three application fields — hearing aids, speech enhancement and music source separation — and it may prove useful in others we did not set out to study. The work presented here, however, has been largely incremental, and concerned with offline, non-real-time algorithms; what we are ultimately drawn to is turning these datasets and findings into deployable, interactive tools — systems a practitioner can use, rather than benchmarks alone.



Artificial intelligence usage statement

In the interest of transparency, this appendix declares the use of artificial intelligence tools in the preparation of this dissertation.

For the peer-reviewed articles on which the results chapters are based, no generative AI was used in their writing, with the exception of the last one, where it was used solely to polish the text. In assembling the present manuscript, however, generative AI (Claude Opus 4.8) was used to complement and expand those articles into the extended prose of the corresponding chapters.

No AI tool wrote any of the code released as research materials with this thesis (Appendix C). Generative AI was also used as an aid in producing Figures 1.1, 1.2 and 3.5, whose underlying data we verified manually.

In all cases, the tools were used under our supervision and as an aid rather than a source: the authors take full responsibility for the content of this dissertation, including any part assisted by these tools.



Publications

This section lists all peer-reviewed publications resulting from the doctoral research undertaken during this thesis. The works are divided into two categories: main contributions directly within the scope of the thesis, followed by secondary collaborative works outside its core scope (ordered by degree of contribution).

B.1 Directly related to this thesis

WASPAA 2023. Enric Gusó, Joanna Luberadзка, Martí Baig, Umut Sayın, & Xavier Serra. *An Objective Evaluation of Hearing Aids and DNN-Based Binaural Speech Enhancement in Complex Acoustic Scenes*. In: 2023 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA), New Paltz, USA.

Contributions: Training and evaluation of the models, generation of the binaural decoders, adaptation of the MASP library, benchmark of the HAS recordings, writing of the manuscript, publication of the code and SOFA files, and dissemination.

WASPAA 2025. Enric Gusó, Joanna Luberadзка, Umut Sayın, & Xavier Serra. *MB-RIRs: a Synthetic Room Impulse Response Dataset with Frequency-Dependent Absorption Coefficients*. In: 2025 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA), Lake Tahoe, USA.

Contributions: Adaptation of MASP to multi-CPU processing. Implementation of source and receiver directivity in the ISM algorithm. Generation of

the dataset, deployment in an AWS environment for generation, training and evaluation of the models, writing of the manuscript, publication of the dataset and dissemination.

ISMIR 2026. Enric Gusó & Xavier Serra. *CrowdioSet and PaRIRset: Two Datasets Towards Live Music Source Separation*. In: Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR), Abu Dhabi, UAE.

Contributions: Selection of the topic, data scraping and manually labeling the CrowdioSet dataset, recording and processing of the PaRIRset, training and evaluation of the RIR ablations and the SCNet models, writing and revision of the manuscript, publication of the datasets, project webpage, dissemination.

B.2 Additional publications

ICASSP 2022. Enric Gusó, Jordi Pons, Santiago Pascual, Joan Serrà. *On Loss Functions and Evaluation Metrics for Music Source Separation*. In 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 306-310). IEEE.

Contributions: Dissemination during the first year of the PhD. Previously: selection of the topic, model training and evaluation, first draft of the manuscript.

WASPAA 2025. Joanna Luberadzka, Enric Gusó, Umut Sayın. *Conditioned Wave-U-Net for Acoustic Matching of Speech in Shared XR Environments*. In: 2025 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA), Lake Tahoe, USA.

Contributions: Part of the evaluation methodology, code reviews, manuscript revision and dissemination.

IEEE VR 2025. Joanna Luberadzka, Enric Gusó, Umut Sayın & Adan Garriga. *Using Virtual Reality to Raise Awareness of Communication Challenges Faced by Individuals with Hearing Loss*. In: 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Saint Malo, France.

Contributions: The SE models.

Forum Acusticum 2025. Martí Baig, Enric Gusó, Joanna Luberadzka & Umut Sayin. *Perceptual Evaluation of Deep Learning-Based Speech Enhancement for Hearing Aids*. In: 11th Convention of the European Acoustics Association (Forum Acusticum Euronoise), Málaga, Spain.

Contributions: Extension of our WASPAA 2023 work to include a perceptual evaluation of the SE models, and manuscript revision.



Materials generated during this thesis

All datasets, models and code produced during this thesis are publicly available, as detailed below. Unless noted otherwise, resources are released under the CC0 1.0 public-domain dedication.

From Chapter 3 Released at https://github.com/enricguso/guso_waspaa23:

- SOFA file of the KU100 dummy head;
- SOFA file of the KU100 wearing the RIC HAs;
- BiMagLS and MagLS Ambisonics-to-binaural decoders;
- code (MIT License) and model weights.

From Chapter 4

- MB-RIRs dataset (CC0 1.0): <https://doi.org/10.5281/zenodo.15773093>;
- code (MIT License) and model weights: https://github.com/enricguso/guso_waspaa25.

From Chapter 5

- Original RIR recordings and processing scripts: https://huggingface.co/datasets/enricguso/parirset_dev;

- PaRIRset dataset (CC0 1.0): <https://doi.org/10.57967/hf/9774>;
- CrowdioSet dataset: <https://doi.org/10.57967/hf/9773>. Its ambience and event recordings are sourced from Freesound and *retain their original per-file licences* (58.8% CC0, 33.2% CC-BY, 8.0% CC-BY-NC, and a small Sampling+ 1.0 subset), documented individually in the accompanying `crowdioset_metadata.csv`; the synthetic sing-alongs inherit the terms of their source datasets and generation tools;
- code (MIT License) and model weights: https://github.com/enricguso/guso_ismir26;
- samples and project website: https://enricguso.github.io/crowdioset_parirset/.



PaRIRset venues and their PA systems

Table D.1: PaRIRset venues (part 1): location and PA manufacturer. I/O denotes indoor or outdoor.

#	Venue	City	I/O	Manufacturer
1	Recinto Ferial	Olivenza	O	d&b audiotechnik
2	Plaça de la palmera	Figueres	O	Nexo
3	Cafè del Teatre	Lleida	I	ElectroVoice
4	Las Armas	Saragossa	I	ElectroVoice
5	Paral · lel 62	Barcelona	I	Meyer Sound
6	Moon	València	I	Lynx Pro-Audio
7	Magma	Alicante	I	Nexo
8	La Riviera	Madrid	I	L-Acoustics
9	Es Gremi	Palma de Mal- lorca	I	Kling&Freitag
10	Zero	Tarragona	I	TW Audio
11	El Tren	Granada	I	Nexo
12	Apolo	Barcelona	I	d&b audiotechnik
13	La Mirona	Girona	I	L-Acoustics
14	Acapulco	Gijón	I	HK Audio
15	REM	Murcia	I	Outline Audio
16	Escenario Sant- ander	Santander	I	TW Audio
17	Trinchera	Málaga	I	ElectroVoice

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Table D.1 (continued)

#	Venue	City	I/O	Manufacturer
18	Antzokia	Bilbo	I	Adamson
19	Dabadaba	Donosti	I	TW Audio
20	Hangar	Córdoba	I	EAW
21	Berlin Club	Almeria	I	Nexo
22	Sanagustin Kultur- gunea	Azpeitia	I	ElectroVoice
23	Sala X	Sevilla	I	L-Acoustics
24	Auditori Municipal Maria Rosa Jun- cosa	Vilafranca del Penedès	I	d&b audiotechnik
25	Sala Ramon Roma- gosa	Cornellà de Llobregat	I	QSC
26	Plaça del Castell	Sant Pere de Ribes	O	Nexo
27	Apolo	Barcelona	I	d&b audiotechnik
28	Moon	València	I	Lynx Pro-Audio
29	Mamba!	Murcia	I	Nexo
30	Custom	Sevilla	I	L-Acoustics
31	Teatro Eslava	Madrid	I	Kv2
32	Capitol	A Coruña	I	EAW
33	Teatro Albeniz	Gijon	I	d&b audiotechnik
34	Jimmy Jazz	Gasteiz	I	TW Audio
35	Sala Oasis	Saragossa	I	Stage Accompany
36	Auditori Municipal Maria Rosa Jun- cosa	Vilafranca del Penedès	O	d&b audiotechnik
37	Museu de la Terra- cota	La Bisbal	O	d&b audiotechnik
38	Bodega Cooper- ativa de Falset	Falset	O	DAS
39	Fòrum Romà D'Empúries	L'Escala	O	L-Acoustics
40	Fòrum de Bar- celona	Barcelona	O	L-Acoustics

Table D.2: PaRIRset venues (part 2): PA system composition. Quantities are elements per side (main, frontfill) or total (subwoofers).

#	Venue	Main $\times$ qty	Sub $\times$ qty	FF $\times$ qty
1	Recinto Ferial	J12 $\times$ 24	J-SUB $\times$ 12	J12 $\times$ 2
2	Plaça de la palmera	M12 $\times$ 12	STM S118 $\times$ 9	–
3	Cafè del Teatre	Xi1183-764 $\times$ 2	RAMPA 2x18 $\times$ 1	–
4	Las Armas	X-Array XN $\times$ 2	XDS $\times$ 1	–
5	Paral·lel 62	M'Elodie $\times$ 18	HP600 $\times$ 4	UPA- 1P $\times$ 2
6	Moon	LX-V8 $\times$ 12	LX-218S $\times$ 4	GR12 $\times$ 2
7	Magma	Geo S1210 $\times$ 4	RS15 $\times$ 2	PS10 $\times$ 2
8	La Riviera	V-DOSC $\times$ 12	–	–
9	Es Gremi	LINE 212-6 $\times$ 6	SW215-D $\times$ 4	–
10	Zero	T20i $\times$ 2	BS30 $\times$ 2	M8 $\times$ 2
11	El Tren	Alpha M4 $\times$ 4	Alpha B1 $\times$ 4	–
12	Apolo	Vi8 $\times$ 6	G-SUB $\times$ 4	Yi10P $\times$ 2
13	La Mirona	A15 $\times$ 8	KS21 $\times$ 8	JBL PRX 521Mi $\times$ x
14	Acapulco	Actor DX $\times$ 2	Actor DX $\times$ 2	Premium PR:O 12M $\times$ 2
15	REM	Doppia II 9050 $\times$ 4	DBS 18-2i $\times$ 4	DVS 10P $\times$ 2
16	Escenario Santander	VERA 36 $\times$ 10	B30 $\times$ 8	–
17	Trinchera	X-Array XN $\times$ 4	XDS $\times$ 4	Eliminator
18	Antzokia	METRIX $\times$ 8	METRIX SUB $\times$ 4	Point 15 $\times$ 2
19	Dabadaba	T20 $\times$ 2	B18 $\times$ 4	–
20	Hangar	RSX208 $\times$ 8	RSX218 $\times$ 2	Red Line 12 $\times$ 2
21	Berlin Club	PS15 $\times$ 8	LS18 $\times$ 4	–
22	Sanagustin Kulturgunea	XLE 191 $\times$ 12	X-SUB $\times$ 4	–

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Table D.2 (continued)

#	Venue	Main $\times$ qty	Sub $\times$ qty	FF $\times$ qty
23	Sala X	ARCS $\times$ 10	SB18 $\times$ 4	112P $\times$ 2
24	Auditori Muni- cipal Maria Rosa Juncosa	Ti10L $\times$ 10	Yi-SUB $\times$ 4	E8 $\times$ 4
25	Sala Ramon Romagosa	KW153 $\times$ 2	—	—
26	Plaça del Castell	Geo S1210 $\times$ 6	RS15 $\times$ 6	PS10 $\times$ 2
27	Apolo	KSL $\times$ 6	SL $\times$ 4	24S $\times$ 2
28	Moon	LX-V8 $\times$ 12	LX-218S $\times$ 4	GR12 $\times$ 2
29	Mamba!	Geo S1210 $\times$ 12	CD18 $\times$ 4	PS10 $\times$ 3
30	Custom	ARCS $\times$ 6	SB218 $\times$ 4	RCF HDL 6-A $\times$ 2
31	Teatro Eslava	ESR215 $\times$ 2	VHD2.18J $\times$ 2	ESD5 $\times$ 4
32	Capitol	KF-730 $\times$ 12	SB-1000z $\times$ 4	SM-200 IH $\times$ 2
33	Teatro Albeniz	V10P $\times$ 2	J-SUB $\times$ 2	—
34	Jimmy Jazz	Vera20 $\times$ 4	B17i $\times$ 4	C5 $\times$ 5
35	Sala Oasis	C29 $\times$ 6	B30 $\times$ 6	—
36	Auditori Muni- cipal Maria Rosa Juncosa	E12 $\times$ 4	Yi-SUB $\times$ 4	E8 $\times$ 2
37	Museu de la Ter- racota	Q7 $\times$ 2	Q-SUB $\times$ 4	—
38	Bodega Cooper- ativa de Falset	Event 28a $\times$ 6	Event 218a $\times$ 4	Altea- 408a $\times$ 2
39	Fòrum Romà D'Empúries	KARA II $\times$ 18	SB-28 $\times$ 8	KIVA II $\times$ 4
40	Fòrum de Bar- celona	K2 $\times$ 18	—	—

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